

# Supplementary Material for “Innovation-Normalized Detection in Compound-Gaussian Clutter: Exact Laws, Conformal Thresholds and Certified Integration under Pulsed Interference”

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All numbers below are generated from the saved outcomes in the project repository (`research/r7c`). Ten protocols were frozen and hash-recorded before their outcomes were computed: the main IPIX study (`PROTOCOL.md`), single-pulse detection (`detection/PROTOCOL_DETECTION.md`), onset (`detection/PROTOCOL_ONSET.md`), baselines (`baselines/PROTOCOL_BASELINES.md`), the 77 GHz radar (`jku/PROTOCOL_JKU.md`), classical detectors, gradual emergence and ACI contamination (`r10/PROTOCOL_R10.md`), order-statistic normalization with real interference (`r11/PROTOCOL_R11.md`), low false-alarm calibration, ANMF baselines, the real IPIX target and certified integration under pulsed interference (`r12/PROTOCOL_R12.md`), the guarded scale, clip level, certified trimming and pulse blanking (`r14/PROTOCOL_R14.md`), and the confirmation on NetRAD S-band sea clutter (`r15/PROTOCOL_R15.md`). A further protocol (`sdrdsp/PROTOCOL_SDRDSP.md`) was frozen before the SDRDSP data were obtained.

Study labels R7–R15 name these protocols in the repository (R1–R6 were development iterations on synthetic data); they are used only here. Deviations from the frozen protocols are listed in Section 2.

## 1 Data exposure

Table 1 states which data each stage used, so that the strength of each validation claim can be judged. Only the hold-out series, the 77 GHz data and NetRAD were not used while the methods were being developed.

## 2 Protocol deviations

- Main IPIX study: before any run, an encoding error in the protocol file was repaired byte-exactly and the hash rewritten; no outcome had been computed. A GPU backend was added for the likelihood evaluation of the noise-aware fit; it gives identical fitted parameters to the CPU path.
- Second radar: the descriptive hit rule summed range bins 20–247 instead of 20–250, because the fixed preprocessing removes bins above 247. The general-form gain curve was chosen after the pre-specified matched form failed to capture the magnitude of the measured gain, so it is post hoc.
- Order-statistic study (R11): the protocol’s rationale counted  $j + 1$  target-corrupted history innovations at look  $j$  after onset; the correct count,  $j$ , was adopted after the data were seen. The data follow the corrected count, and the main text uses only the corrected statement.
- Final study (R12): before the real-data runs, the numerical search for the Fisher’s  $g$  threshold was re-bracketed; the law is unchanged.

## 3 Unmet expectations

Each expected value was written down before the outcome was computed. Of the 48 outcomes predicted for the last three studies, twelve were not met.

- Final IPIX study (R12; 12 of 18 met): calibration of the saturating clipped statistic; the Tyler law (expected within twofold); the CA power detector’s white-noise law (expected off more than threefold); the cost of clipping (expected at most 1.5 dB); false alarms of non-coherent integration under real FMCW interference (expected at least fivefold in both runs); and the gain of AR gap reconstruction (expected at least 1 dB).
- Improvement study (R14; 12 of 15 met): the guard’s onset cost, the certificate at  $10^{-3}$  and the cost of trimming (Section 21).

Table 1: Data exposure by stage. “Seen” means that outcomes on these data were computed while methods, orders or comparisons could still change.

| Data                                  | Used for                                                                                      | Status of the reported results                                                                                                                                        |
|---------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPIX, 14 sessions, rotation 1         | Method development (R1–R7): method list, model orders, comparisons                            | Seen; not reported                                                                                                                                                    |
| IPIX, same sessions, rotations 2–3    | Main study, detection, onset, baselines, classical detectors, order-statistic scores (R7–R11) | Frozen-protocol evaluation, computed once per protocol; the recordings were seen in rotation 1, so this is not an untouched confirmation                              |
| IPIX, day transfer                    | Transfer and ACI analyses                                                                     | Frozen-protocol evaluation on the same recordings                                                                                                                     |
| IPIX #269, #287                       | Hold-out                                                                                      | Hashed and unopened until the main analysis was complete; evaluated once (small test sets)                                                                            |
| JKU 77 GHz, reference run             | Second-radar study                                                                            | Not used in development; frozen protocol; the general-form gain curve is post hoc                                                                                     |
| JKU 77 GHz, interference runs A and C | Real-interference test (R11); certified integration (R12)                                     | R11: only chirp-hit statistics inspected before its protocol. R12: R11 outcomes on these runs had been seen                                                           |
| IPIX, rotations 2–3                   | Guarded scale, clip level, trimming, blanking (R14)                                           | Frozen-protocol evaluation, computed once; same recordings as R7–R12                                                                                                  |
| IPIX, primary target cells            | Real-target study (R12)                                                                       | Excluded from the R7–R11 clutter studies; a development check (V4, before R7) had run IN-ARCP on the primary cell and found it blind. Frozen protocol, evaluated once |
| NetRAD S-band, node 3, 14 recordings  | Second-radar sea-clutter confirmation (R15)                                                   | Untouched: protocol frozen before any statistic of the samples was computed; evaluated once                                                                           |
| SDRDSP (pending)                      | Further sea-clutter confirmation                                                              | Protocol frozen and hashed before the data were obtained                                                                                                              |

- NetRAD confirmation (R15; 12 of 15 met): calibration of six statistics at  $10^{-4}$ , the pivotality ranking against CA16, and a rise of false alarms in segments with flagged pulses (Section 26).
- Main IPIX study ( $m = 16$ ,  $\alpha = 0.1$ ; radius relative to IN-ARCP): the noise-aware gain was expected to concentrate in noise-affected sessions. It did for NA-AR(1) (0.915 noisy against 0.989 clean) but not for NA-AR(2) and NA-AR(4) (0.871 and 0.866 noisy against 0.739 and 0.709 clean), whose gain comes mainly from the AR clutter dynamics (expectation Q1, mixed).
- Onset study: the gain of IN-ARCP over the long-window power detector was expected to shrink below 3 dB by an 8-pulse dwell. It persisted: 8.4 dB [5.2, 11.1] at  $P_d = 0.5$  (expectation O3, refuted).
- 77 GHz radar: the noise-aware scores were expected to be flatter than IN-ARCP across CNR classes. The AR(4) version was (mean largest class deviation 0.055 against 0.066); the AR(1) version was not (0.067; expectation J2, mixed).
- Classical detectors (R10). The R10 output below prints expectation R1 as FAILED. It is met in substance. For a target that rises over  $L$  pulses before the first look, IN-ARCP’s first-look gain over the long-window power detector (CAlloc in the paper) falls from 9.86 to 8.65 to 4.65 dB for  $L = 1, 4, 8$ . At  $L = 16$ , CAlloc reaches  $P_d = 0.5$  at 9.1 dB while IN-ARCP stays below it up to 25 dB, so the gain is below  $-15.9$  dB, well under the 5 dB bound. PAMF-H was expected to need at least 3 dB less SCR than integrated IN-ARCP over an 8-pulse dwell at  $P_d = 0.5$ ; it needed 1.0 dB [0.3, 1.7] less (4.0 dB at  $P_d = 0.9$ ). Gated ACI was expected to keep the clean false-alarm rate within  $[0.8, 1.2]\alpha$  at 5% target contamination; it gave  $0.55\alpha$  ( $\gamma = 0.02$ ) and  $0.68\alpha$  ( $\gamma = 0.005$ ).
- Order-statistic study (R11; 2 of 11 met): quintile false-alarm spread of the median OS score at most 2.0 (2.06); its cost for abrupt targets at most 1.5 dB (1.99 dB [0.89, 3.39]; Corollary 3 gives 0.69 dB); a first-look gain of at least 3 dB over IN-ARCP for 8-pulse ramps (1.46 dB); the integrated OS score closing half of integrated IN-ARCP’s gap to PAMF-H at  $P_d = 0.9$  (it was 2.5 dB worse); OS-law coverage error at most 0.015 (0.021); on the dense real-interference run, the OS score halving IN-ARCP’s masking (drops 0.39 and 0.35), zeroing cutting the hit-pulse

false-alarm rate by half (39%), and the OS score after zeroing within 0.5 dB of IN-ARCP (0.71 dB worse); on the reference run, IN-ARCP beating CA16 (CA16 was 0.78 dB better, the loss Proposition 2 predicts for  $c < 1$ ).

## 4 Hold-out series

Two further McMaster series (#269 and #287, the widely used “high” and “low” sea-state files, one provider-preprocessed VV cell each, from the same 1993 campaign) were hashed and left unopened until the main analysis was complete, and then evaluated once (Table 2). They confirm the direction of the main result, with larger gains. They also show a case in which the classical Gaussian threshold over-covers while conformal IN-ARCP stays near nominal. Their test sets are small (about 165 episodes per unit), so the hold-out is a directional check. Mondrian calibration returns infinite regions at  $1 - \alpha = 0.99$  there, because each bin receives fewer calibration episodes than the rank requires.

Table 2: Hold-out series #269 and #287 ( $1 - \alpha = 0.90$ ; six units per column).

| Method                                | $m = 8$      |                | $m = 16$     |                |
|---------------------------------------|--------------|----------------|--------------|----------------|
|                                       | Radius ratio | Coverage (min) | Radius ratio | Coverage (min) |
| Noise-aware AR(4) (proposed)          | 0.454        | 0.890 (0.816)  | 0.498        | 0.893 (0.847)  |
| Noise-aware AR(2) (proposed)          | 0.541        | 0.887 (0.828)  | 0.533        | 0.895 (0.859)  |
| AR(4), innovation-normalized          | 0.546        | 0.908 (0.847)  | 0.515        | 0.906 (0.883)  |
| AR(2), innovation-normalized          | 0.582        | 0.905 (0.840)  | 0.572        | 0.906 (0.871)  |
| Noise-aware AR(1)                     | 0.921        | 0.898 (0.840)  | 0.921        | 0.903 (0.859)  |
| IN-ARCP, AR(1)                        | 1.000        | 0.906 (0.853)  | 1.000        | 0.910 (0.859)  |
| Gaussian plug-in of IN-ARCP           | 1.178        | 0.949 (0.896)  | 1.104        | 0.945 (0.895)  |
| Mondrian IN-ARCP                      | 1.000        | 0.895 (0.804)  | 0.981        | 0.896 (0.810)  |
| Learned-scale CP                      | 0.629        | 0.893 (0.804)  | 0.662        | 0.919 (0.791)  |
| Invariant MLP CP                      | 1.809        | 0.884 (0.736)  | 2.158        | 0.903 (0.761)  |
| Raw-RMS normalized CP                 | 1.321        | 0.880 (0.706)  | 1.243        | 0.887 (0.706)  |
| Unnormalized CP                       | 1.142        | 0.868 (0.607)  | 1.161        | 0.859 (0.589)  |
| Gaussian plug-in of noise-aware AR(4) | 0.411        | 0.866 (0.816)  | 0.472        | 0.856 (0.755)  |

## 5 Summary of the laws

Table 3 collects the laws of Section IV of the paper and marks the results that are new.

## 6 Synthetic verification

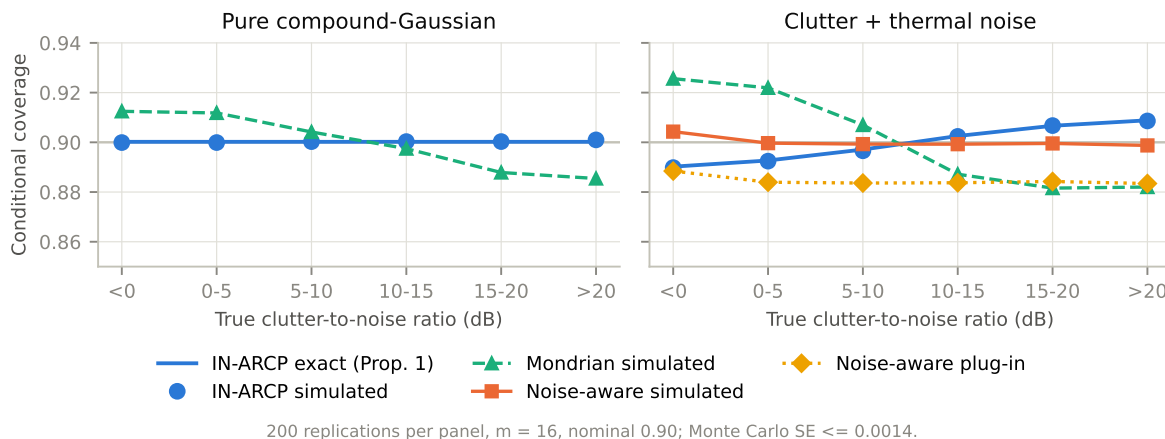


Figure 1: Synthetic verification: coverage conditional on the true clutter-to-noise ratio. The line is the exact law of Proposition 1; markers are simulation.

Table 3: Laws of whitened-innovation detection, with the equation numbers of the paper (AR(1) speckle, true coefficient, compound-Gaussian clutter;  $S_w = S/(1 - |\rho|^2)$ ,  $w(\omega) = P/S_c(\omega)$ ,  $\psi$  the P-ANMF threshold). New results are marked \*.

| Detector                 | $P_{fa}$                              | $P_d$ (Swerling 1)                |
|--------------------------|---------------------------------------|-----------------------------------|
| Look at onset            | $(1 + q^2/m)^{-m}$                    | $(1 + \frac{q^2}{m(1+S_w)})^{-m}$ |
| Look $\ell$ after onset* | same                                  | (5); horizon (6)                  |
| Guard of $\Delta$ pulses | same                                  | (5) at $\ell - \Delta$            |
| OS scale, onset          | $\prod_{i < k} \frac{m-i}{m-i+q^2}$   | same at $\frac{q^2}{1+S_w}$       |
| OS scale, after onset*   | same                                  | (7); exact by quadrature          |
| $K$ -pulse integration   | (8)                                   | (9)                               |
| P-ANMF, known $\omega$   | $(1 - \psi)^{N-1}$                    | (10): $SCR \times Nw(\omega)$     |
| P-ANMF, max over bins    | Fisher's $g$                          | inclusion-exclusion               |
| Certified clip/binary*   | $\leq \alpha$ , any amplitude         | by calibration                    |
| Conformal threshold      | $\text{Beta}(n+1-k_\alpha, k_\alpha)$ | Beta moment                       |

## 7 Proof of Proposition 2 and the latent-texture Mondrian result

Equation, proposition and theorem numbers without the prefix S refer to the paper.

*Proposition 2.* If  $e \sim \mathcal{CN}(0, v)$  and a Swerling-1 target of power  $SP$  is added, a threshold test on  $|e|^2/v$  at false-alarm probability  $\alpha$  detects with probability  $\alpha^{1/(1+SP/v)}$ , while the power-only test on  $|Y|^2/P$  detects with  $\alpha^{1/(1+S)}$ ; equal  $P_d$  therefore requires SCRs in the ratio  $G = P/v$ . For the center  $rH_m$  under (1) with AR(1) speckle,  $v = \nu\{c(1 + |r|^2 - 2\Re \bar{r}\rho) + 1 + |r|^2\}$ , which gives  $G_{IN}$ . For the optimal predictor from the infinite past,  $v$  is the one-step prediction-error variance of the AR(1)-plus-white-noise process,  $\exp\{\int \log f\}$  by the Kolmogorov-Szegő formula (Brockwell and Davis, *Time Series: Theory and Methods*, 1991). With  $f(\omega) = \nu\{c(1 - |\rho|^2) + |1 - \rho e^{-j\omega}|^2\}/|1 - \rho e^{-j\omega}|^2$ , the denominator contributes zero, and the numerator factorizes as  $\nu V|1 - \zeta e^{-j\omega}|^2$  with  $V(1 + |\zeta|^2) = \xi$ ,  $V\zeta = \rho$  and  $|\zeta| < 1$ , which gives  $v = \nu V$  and  $G_{opt}$ . Because the optimal predictor's error variance is at most that of any linear predictor, including 0 and  $rH_m$ ,  $G_{opt} \geq \max\{1, G_{IN}\}$ .

*Latent-texture Mondrian calibration.* Mondrian (group-conditional) conformal prediction calibrates separately within bins of an observed statistic and is valid conditional on the observed bin; Dewolf et al. report that estimated taxonomies mix up the true classes. For a latent texture the consequence can be stated exactly.

**Proposition S1.** Let  $\nu = 0$  and  $X = \sigma Z$ , with  $\sigma > 0$  distributed as  $G$ , non-degenerate and independent of  $Z \sim \mathcal{CN}(0, R)$ . Let  $S(Z)$  be a scale-invariant score with a continuous distribution function  $F_S$  that is strictly increasing on its support, let  $T(H) > 0$  be continuous and homogeneous of degree one, and calibrate within fixed bins  $B_b = [e_{b-1}, e_b)$ ,  $0 = e_0 < \dots < e_K = \infty$ ,  $K \geq 2$ , of  $T(H)$  with population quantiles  $q_b$ . Let  $\mathcal{C}(\sigma)$  be the coverage conditional on the texture. (i) If  $S(Z)$  is independent of  $T(Z_H)$ , then  $\mathcal{C}(\sigma) \equiv 1 - \alpha$ . (ii) If  $S(Z)$  given  $T(Z_H) = t$  is strictly stochastically decreasing in  $t$ , then  $\lim_{\sigma \rightarrow \infty} \mathcal{C}(\sigma) = F_S(q_K) < 1 - \alpha < F_S(q_1) = \lim_{\sigma \rightarrow 0} \mathcal{C}(\sigma)$ , whereas the unbinned threshold gives a constant  $\mathcal{C}(\sigma)$ . Condition (ii) holds exactly for IN-ARCP at the true coefficient with  $T = s_\rho$ .

*Proof.* Write  $t = T(Z_H)$ , so that the observed statistic is  $T(X_H) = \sigma t$  and  $S(X) = S(Z)$ . For bin  $b$  let  $w_b(t) = \mathbb{P}\{\sigma t \in B_b\}$ . Since  $\sigma$  is independent of  $Z$ , for every  $x$

$$\mathbb{P}\{S \leq x, \sigma t \in B_b\} = \mathbb{E}[\varphi_x(t) w_b(t)], \quad (S1)$$

where  $\varphi_x(t) = \mathbb{P}\{S \leq x \mid t\}$ . (i) If  $S$  is independent of  $t$ , then  $\varphi_x \equiv F_S(x)$ , every bin-conditional law of  $S$  equals  $F_S$ , all  $q_b$  equal  $F_S^{-1}(1 - \alpha)$  and  $\mathcal{C}(\sigma) = 1 - \alpha$ .

(ii) Let  $x_0 = F_S^{-1}(1 - \alpha)$ . For the top bin,  $w_K(t) = \mathbb{P}\{\sigma \geq e_{K-1}/t\}$  is nondecreasing in  $t$ . Because  $T$  is continuous, positive and homogeneous and  $Z_H$  has full support,  $t$  has full support on  $(0, \infty)$ ; as  $e_{K-1}/t$  then ranges over  $(0, \infty)$  and  $G$  is non-degenerate,  $w_K(t)$  is not almost surely constant. By assumption  $\varphi_{x_0}$  is strictly increasing. Let  $t'$  be an independent copy of  $t$ ,  $D_\varphi = \varphi_{x_0}(t) - \varphi_{x_0}(t')$  and  $\Delta_w = w_K(t) - w_K(t')$ . Then

$$\text{Cov}\{\varphi_{x_0}(t), w_K(t)\} = \frac{1}{2} \mathbb{E}[D_\varphi \Delta_w] > 0,$$

because  $D_\varphi \Delta_w \geq 0$ , with strict inequality on the event  $\{w_K(t) \neq w_K(t')\}$ , which has positive probability. By (S1),  $\mathbb{P}\{S \leq x_0 \mid \sigma t \in B_K\} = \mathbb{E}[\varphi_{x_0} w_K]/\mathbb{E}[w_K] > \mathbb{E}[\varphi_{x_0}] = 1 - \alpha$ . The bin-conditional distribution function of  $S$  is continuous, so there is  $x_1 < x_0$  at which it still exceeds  $1 - \alpha$ ; hence  $q_K \leq x_1 < x_0$  and,  $F_S$  being strictly increasing,  $F_S(q_K) < 1 - \alpha$ . For the bottom bin,  $w_1(t) = \mathbb{P}\{\sigma < e_1/t\}$  is nonincreasing, the covariance is negative, and the same steps give  $q_1 > x_0$  and  $F_S(q_1) > 1 - \alpha$ .

The coverage at texture  $\sigma$  is  $\mathcal{C}(\sigma) = \sum_b \mathbb{P}\{S \leq q_b, \sigma t \in B_b\}$ . As  $\sigma \rightarrow \infty$ ,  $\sigma t \rightarrow \infty$  almost surely, so  $\mathbf{1}\{\sigma t \in B_K\} \rightarrow 1$  and, by dominated convergence,  $\mathcal{C}(\sigma) \rightarrow F_S(q_K)$ ; as  $\sigma \rightarrow 0$ ,  $\mathcal{C}(\sigma) \rightarrow F_S(q_1)$ . An unbinned threshold  $q$  gives  $\mathcal{C}(\sigma) = F_S(q)$  for every  $\sigma$ .

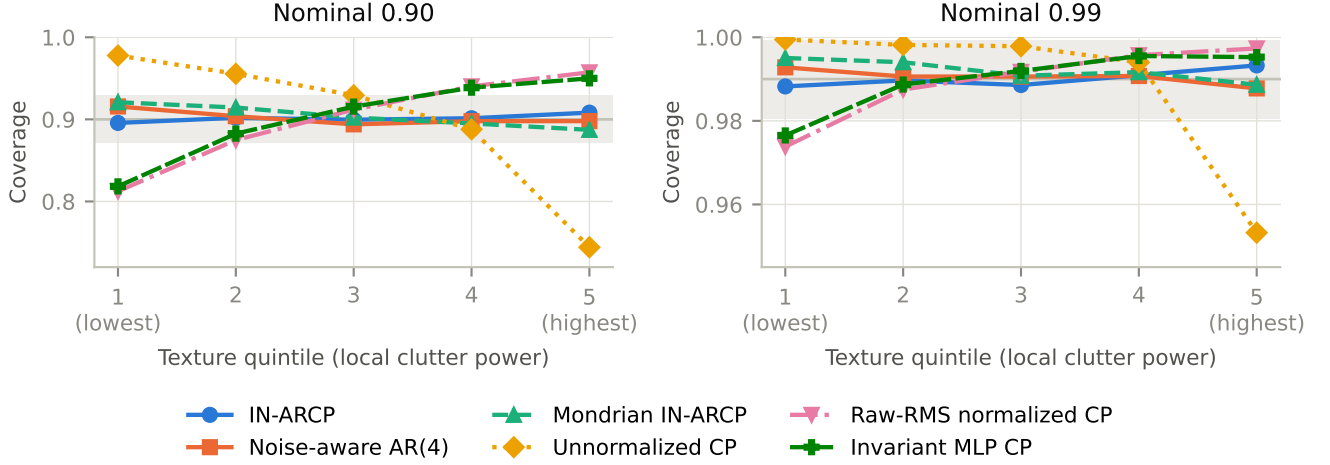
Finally, for IN-ARCP at  $r = \rho$  and  $T = s_\rho$ ,  $S = |E|/t$  with the innovation  $E$  independent of  $Z_H$ , so  $\varphi_x(t) = \mathbb{P}\{|E| \leq xt\}$  is strictly increasing in  $t$ , which is condition (ii).  $\blacksquare$

## 8 Zero-noise limit of the noise-aware score

**Proposition S2.** Let  $R$  be  $AR(1)$  with coefficient  $\rho$  and fix a nonzero history  $h$ . As  $\nu \downarrow 0$ ,  $\nu\hat{c}(h) \rightarrow h^H R_H^{-1} h/m$ ,  $\mu(h) \rightarrow \rho h_m$  and  $s(h) \rightarrow s_\rho(h)$ , so the NA score converges to the IN-ARCP score at  $r = \rho$ .

*Proof sketch.* With  $\gamma = \nu c$  the history log-likelihood converges locally uniformly on  $\gamma > 0$  to  $-m \log \gamma - \gamma^{-1} h^H R_H^{-1} h$ , whose unique maximizer is  $h^H R_H^{-1} h/m$ ; coercivity at 0 and  $\infty$  gives convergence of the maximizers. The Markov property gives  $\mu \rightarrow \rho h_m$ , and  $(1 - |\rho|^2) R_H^{-1} = W_\rho^H W_\rho$ , with  $W_\rho$  the bidiagonal whitening matrix, identifies the limiting predictive variance with  $s_\rho(h)^2$ .

## 9 Texture-conditional coverage and disc widths on IPIX



Shaded band: expected maximum quintile deviation from sampling noise alone. IPIX, rotations 2-3,  $m = 16$ .

Figure 2: Coverage by texture quintile on IPIX (rotations 2-3,  $m = 16$ ). Texture is proxied by the local clutter power ( $\pm 1024$  pulses, excluding the episode).

Table 4: IPIX width and coverage ( $m = 16$ ; rotations 2-3; 56 units). Radius ratio: geometric mean over units of mean disc radius relative to IN-ARCP on the same test episodes, with 95% day-cluster bootstrap interval. Max. dev.: mean over units of the largest absolute texture-quintile coverage deviation from nominal. At equal model order, NA-AR(4) has 0.891 [0.872, 0.905] times the radius of the innovation-normalized AR(4) disc.

| Method                                 | $1 - \alpha = 0.90$   |          |           | $1 - \alpha = 0.99$   |          |           |
|----------------------------------------|-----------------------|----------|-----------|-----------------------|----------|-----------|
|                                        | Radius ratio [95% CI] | Coverage | Max. dev. | Radius ratio [95% CI] | Coverage | Max. dev. |
| Noise-aware AR(4) (proposed)           | 0.753 [0.714, 0.815]  | 0.902    | 0.043     | 0.743 [0.706, 0.798]  | 0.991    | 0.011     |
| Noise-aware AR(2) (proposed)           | 0.777 [0.748, 0.824]  | 0.901    | 0.044     | 0.769 [0.745, 0.806]  | 0.991    | 0.011     |
| AR(4), innovation-normalized           | 0.845 [0.805, 0.905]  | 0.901    | 0.032     | 0.861 [0.822, 0.918]  | 0.990    | 0.011     |
| AR(2), innovation-normalized           | 0.895 [0.855, 0.953]  | 0.901    | 0.034     | 0.898 [0.857, 0.953]  | 0.990    | 0.010     |
| Noise-aware AR(1)                      | 0.966 [0.930, 0.994]  | 0.901    | 0.042     | 0.952 [0.907, 0.989]  | 0.990    | 0.011     |
| IN-ARCP, AR(1)                         | 1                     | 0.901    | 0.036     | 1                     | 0.990    | 0.011     |
| Gaussian plug-in of IN-ARCP            | 0.995 [0.988, 1.002]  | 0.900    | 0.034     | 0.993 [0.979, 1.005]  | 0.990    | 0.011     |
| Mondrian IN-ARCP                       | 0.980 [0.975, 0.984]  | 0.904    | 0.043     | 0.998 [0.988, 1.006]  | 0.992    | 0.011     |
| Learned-scale CP                       | 0.921 [0.875, 0.990]  | 0.900    | 0.039     | 1.072 [1.019, 1.147]  | 0.991    | 0.012     |
| Invariant MLP CP                       | 1.337 [1.288, 1.439]  | 0.901    | 0.096     | 1.585 [1.489, 1.780]  | 0.990    | 0.019     |
| Raw-RMS normalized CP                  | 1.368 [1.320, 1.449]  | 0.899    | 0.103     | 1.609 [1.533, 1.782]  | 0.989    | 0.021     |
| Unnormalized CP                        | 1.048 [1.010, 1.085]  | 0.899    | 0.165     | 1.475 [1.261, 1.720]  | 0.989    | 0.039     |
| Gaussian plug-in of noise-aware AR(4)  | 0.726 [0.687, 0.787]  | 0.885    | 0.049     | 0.698 [0.659, 0.758]  | 0.985    | 0.016     |
| Sampling-noise reference for max. dev. |                       |          | 0.029     |                       |          | 0.009     |

Table 5: Day transfer ( $m = 16$ ,  $1 - \alpha = 0.90$ ): fit and calibrate on other days, test on each of six held-out days.

| Method                                | Mean coverage | Worst held-out day | Radius ratio |
|---------------------------------------|---------------|--------------------|--------------|
| Noise-aware AR(4) (proposed)          | 0.898         | 0.863              | 0.936        |
| Noise-aware AR(2) (proposed)          | 0.898         | 0.883              | 0.956        |
| AR(4), innovation-normalized          | 0.901         | 0.894              | 0.993        |
| AR(2), innovation-normalized          | 0.899         | 0.891              | 1.024        |
| Noise-aware AR(1)                     | 0.900         | 0.882              | 1.037        |
| IN-ARCP, AR(1)                        | 0.903         | 0.882              | 1.000        |
| Gaussian plug-in of IN-ARCP           | 0.906         | 0.893              | 1.006        |
| Mondrian IN-ARCP                      | 0.907         | 0.881              | 0.984        |
| Learned-scale CP                      | 0.898         | 0.774              | 0.967        |
| Invariant MLP CP                      | 0.880         | 0.622              | 1.200        |
| Raw-RMS normalized CP                 | 0.906         | 0.681              | 1.466        |
| Unnormalized CP                       | 0.909         | 0.716              | 1.470        |
| Gaussian plug-in of noise-aware AR(4) | 0.876         | 0.842              | 0.893        |

## 10 Day transfer and texture-proxy sensitivity

Texture-proxy sensitivity (pre-specified windows; mean maximum quintile deviation at  $1 - \alpha = 0.90$ ,  $m = 16$ ): with  $\pm 512$  pulses IN-ARCP 0.035, NA-AR(4) 0.043, unnormalized 0.181, neural center 0.103; with  $\pm 2048$  pulses IN-ARCP 0.033, unnormalized 0.141. The ranking is the same as with  $\pm 1024$ .

## 11 Additional figures for the IPIX study

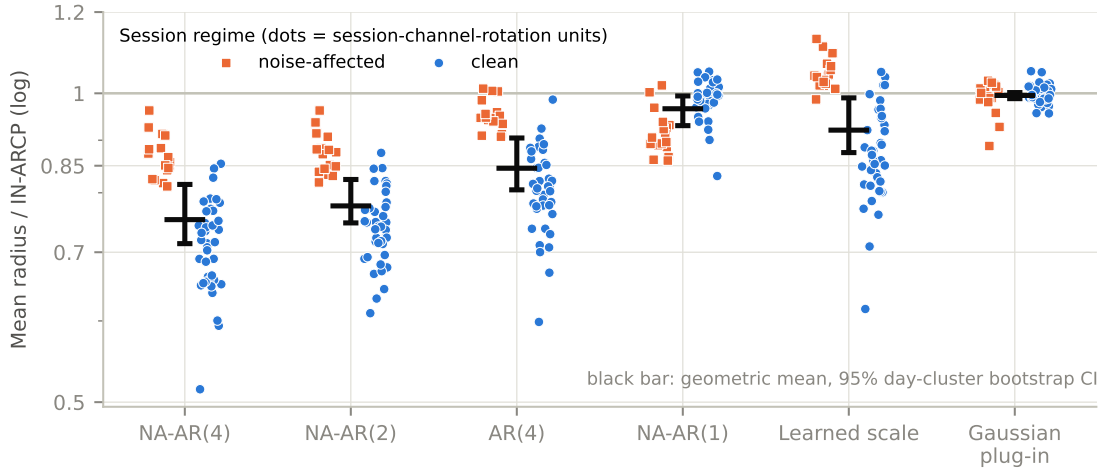


Figure 3: Per-unit radius relative to IN-ARCP at  $1 - \alpha = 0.90$ ,  $m = 16$ , by session regime (noise-affected: fitted noise power above 1% of mean power).

## 12 Detection study: all levels and methods

Pooled detection probability by SCR, measured false-alarm rates, exact-law predictions and SCR gains with 95% day-cluster intervals (output of `detection/analyze_detection.py`):

units: 56; days: 6; test episodes: 74980

```

=== 1-alpha = 0.9 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CAlloc 0.1030 0.166 0.219 0.307 0.433 0.579 0.711 0.814 0.888 0.934 0.963 0.979 0.988 0.993
CA16 0.1002 0.242 0.312 0.406 0.516 0.631 0.736 0.823 0.887 0.931 0.959 0.977 0.987 0.992
U 0.1011 0.446 0.549 0.651 0.745 0.823 0.883 0.926 0.955 0.974 0.985 0.992 0.995 0.997
RMS 0.1007 0.442 0.541 0.642 0.734 0.813 0.877 0.922 0.953 0.972 0.984 0.991 0.995 0.997
MLP 0.0990 0.447 0.548 0.650 0.743 0.823 0.884 0.927 0.956 0.974 0.985 0.992 0.995 0.997
LS 0.0995 0.534 0.624 0.710 0.784 0.847 0.898 0.935 0.960 0.977 0.987 0.993 0.996 0.997
MON 0.0961 0.540 0.634 0.720 0.794 0.858 0.907 0.941 0.965 0.979 0.988 0.993 0.996 0.998
GI 0.1004 0.542 0.635 0.722 0.795 0.859 0.908 0.942 0.965 0.980 0.989 0.993 0.996 0.998
IN1 0.0987 0.540 0.633 0.720 0.793 0.857 0.906 0.941 0.964 0.979 0.988 0.993 0.996 0.998

```

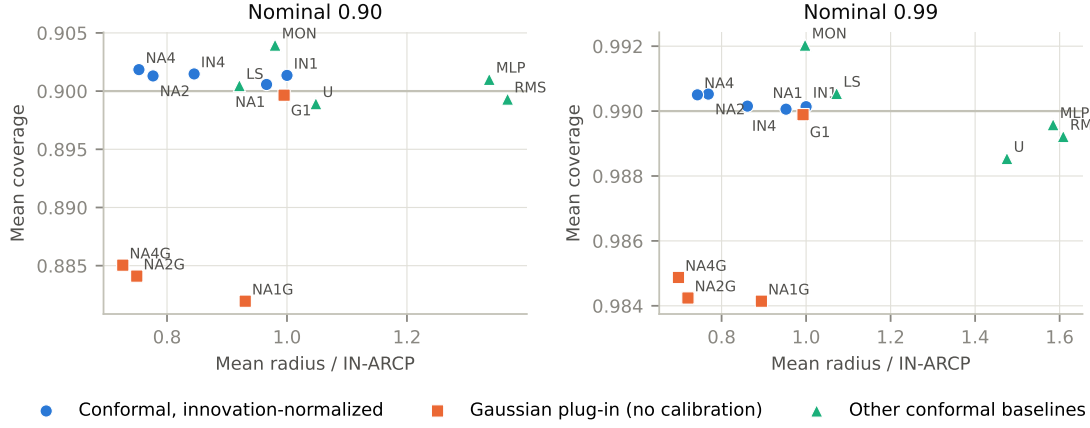


Figure 4: Mean coverage against mean radius (confirmatory,  $m = 16$ ). Gaussian plug-ins of the richer noise-aware models are shorter but under-cover; conformal calibration restores nominal coverage. Labels:  $NAp$  and  $INp$ , noise-aware and innovation-normalized  $AR(p)$  ( $IN1$  is  $IN$ -ARCP);  $G1$  and  $NApG$ , Gaussian plug-ins;  $MON$ , Mondrian;  $LS$ , learned scale;  $U$ , unnormalized;  $RMS$ , raw-RMS;  $MLP$ , neural center.

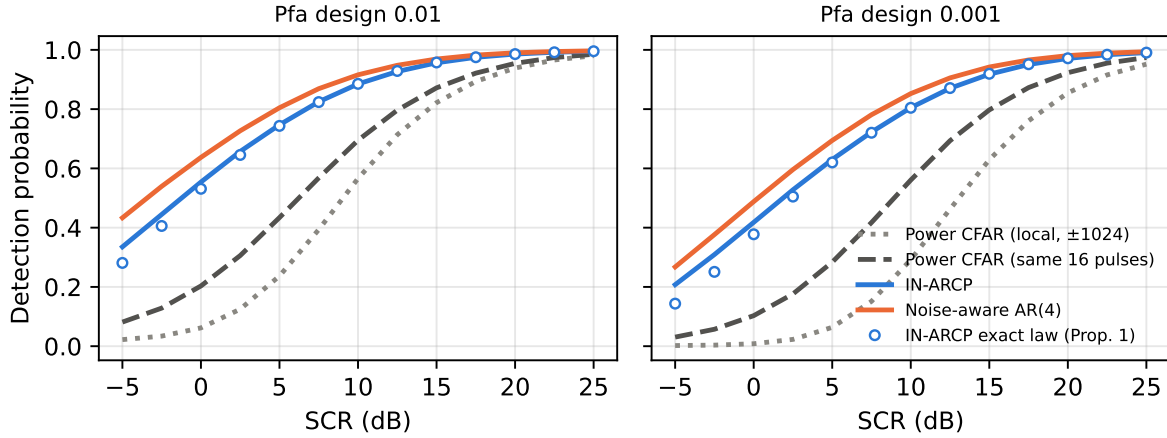


Figure 5: Detection of injected Swerling-1 targets on IPIX at design false-alarm probabilities 0.01 and 0.001.

|                                                                                                          |          |         |         |                      |          |                     |       |       |       |       |       |       |       |       |
|----------------------------------------------------------------------------------------------------------|----------|---------|---------|----------------------|----------|---------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| IN2                                                                                                      | 0.0991   | 0.553   | 0.643   | 0.728                | 0.801    | 0.862               | 0.909 | 0.943 | 0.966 | 0.980 | 0.989 | 0.993 | 0.996 | 0.998 |
| IN4                                                                                                      | 0.0985   | 0.574   | 0.661   | 0.741                | 0.811    | 0.869               | 0.914 | 0.946 | 0.967 | 0.981 | 0.989 | 0.994 | 0.996 | 0.998 |
| NA1                                                                                                      | 0.0994   | 0.553   | 0.648   | 0.735                | 0.810    | 0.872               | 0.917 | 0.948 | 0.969 | 0.982 | 0.990 | 0.994 | 0.997 | 0.998 |
| NA2                                                                                                      | 0.0987   | 0.609   | 0.692   | 0.767                | 0.834    | 0.887               | 0.926 | 0.954 | 0.972 | 0.984 | 0.991 | 0.995 | 0.997 | 0.998 |
| NA4                                                                                                      | 0.0982   | 0.614   | 0.696   | 0.770                | 0.836    | 0.888               | 0.927 | 0.954 | 0.973 | 0.984 | 0.991 | 0.995 | 0.997 | 0.998 |
| NA4G                                                                                                     | 0.1150   | 0.630   | 0.710   | 0.782                | 0.845    | 0.895               | 0.932 | 0.958 | 0.975 | 0.986 | 0.992 | 0.995 | 0.997 | 0.998 |
| IN1 exact-law prediction                                                                                 | 0.507    | 0.614   | 0.710   | 0.782                | 0.845    | 0.895               | 0.932 | 0.958 | 0.975 | 0.986 | 0.992 | 0.995 | 0.997 | 0.998 |
| exact-law MAE, IN1 Pd: per unit x SCR 0.0140; pooled curve 0.0053                                        |          |         |         |                      |          |                     |       |       |       |       |       |       |       |       |
| -- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --            |          |         |         |                      |          |                     |       |       |       |       |       |       |       |       |
| IN1                                                                                                      | SCR      | -3.0 dB | vs CA16 | 5.06 [ 3.25, 6.46]   | vs Caloc | 6.64 [ 4.50, 8.41]  |       |       |       |       |       |       |       |       |
| IN4                                                                                                      | SCR      | -3.1 dB | vs CA16 | 5.19 [ 3.41, 6.53]   | vs Caloc | 6.76 [ 4.69, 8.46]  |       |       |       |       |       |       |       |       |
| NA4                                                                                                      | SCR      | -3.4 dB | vs CA16 | 5.41 [ 3.85, 6.61]   | vs Caloc | 6.99 [ 5.10, 8.52]  |       |       |       |       |       |       |       |       |
| G1                                                                                                       | SCR      | -3.1 dB | vs CA16 | 5.10 [ 3.35, 6.44]   | vs Caloc | 6.67 [ 4.65, 8.39]  |       |       |       |       |       |       |       |       |
| U                                                                                                        | SCR      | -2.6 dB | vs CA16 | 4.61 [ 2.79, 6.04]   | vs Caloc | 6.19 [ 4.15, 8.03]  |       |       |       |       |       |       |       |       |
| MLP                                                                                                      | SCR      | -2.6 dB | vs CA16 | 4.66 [ 2.72, 6.11]   | vs Caloc | 6.23 [ 3.87, 8.10]  |       |       |       |       |       |       |       |       |
| LS                                                                                                       | SCR      | -2.8 dB | vs CA16 | 4.83 [ 2.83, 6.34]   | vs Caloc | 6.40 [ 4.06, 8.30]  |       |       |       |       |       |       |       |       |
| -- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --            |          |         |         |                      |          |                     |       |       |       |       |       |       |       |       |
| IN1                                                                                                      | SCR      | 5.2 dB  | vs CA16 | 7.98 [ 5.03, 10.43]  | vs Caloc | 7.93 [ 5.02, 10.37] |       |       |       |       |       |       |       |       |
| IN4                                                                                                      | SCR      | 4.3 dB  | vs CA16 | 8.86 [ 5.48, 11.49]  | vs Caloc | 8.80 [ 5.45, 11.44] |       |       |       |       |       |       |       |       |
| NA4                                                                                                      | SCR      | 3.1 dB  | vs CA16 | 10.05 [ 6.52, 12.77] | vs Caloc | 9.99 [ 6.51, 12.72] |       |       |       |       |       |       |       |       |
| G1                                                                                                       | SCR      | 5.1 dB  | vs CA16 | 8.02 [ 5.09, 10.42]  | vs Caloc | 7.97 [ 5.11, 10.38] |       |       |       |       |       |       |       |       |
| U                                                                                                        | SCR      | 7.1 dB  | vs CA16 | 6.09 [ 3.33, 8.23]   | vs Caloc | 6.04 [ 3.30, 8.21]  |       |       |       |       |       |       |       |       |
| MLP                                                                                                      | SCR      | 7.0 dB  | vs CA16 | 6.12 [ 3.70, 7.88]   | vs Caloc | 6.07 [ 3.70, 7.81]  |       |       |       |       |       |       |       |       |
| LS                                                                                                       | SCR      | 5.4 dB  | vs CA16 | 7.78 [ 4.32, 10.40]  | vs Caloc | 7.72 [ 4.35, 10.37] |       |       |       |       |       |       |       |       |
| === 1-alpha = 0.99 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) === |          |         |         |                      |          |                     |       |       |       |       |       |       |       |       |
| method                                                                                                   | Pfa_meas | -5.0    | -2.5    | 0.0                  | 2.5      | 5.0                 | 7.5   | 10.0  | 12.5  | 15.0  | 17.5  | 20.0  | 22.5  | 25.0  |
| Caloc                                                                                                    | 0.0110   | 0.022   | 0.034   | 0.062                | 0.125    | 0.237               | 0.394 | 0.566 | 0.714 | 0.822 | 0.893 | 0.939 | 0.966 | 0.980 |
| CA16                                                                                                     | 0.0103   | 0.081   | 0.129   | 0.203                | 0.306    | 0.434               | 0.568 | 0.695 | 0.796 | 0.872 | 0.922 | 0.954 | 0.974 | 0.985 |
| U                                                                                                        | 0.0115   | 0.131   | 0.208   | 0.308                | 0.428    | 0.558               | 0.680 | 0.782 | 0.860 | 0.913 | 0.948 | 0.970 | 0.983 | 0.990 |
| RMS                                                                                                      | 0.0108   | 0.176   | 0.271   | 0.383                | 0.503    | 0.625               | 0.733 | 0.822 | 0.887 | 0.931 | 0.960 | 0.977 | 0.987 | 0.993 |
| MLP                                                                                                      | 0.0104   | 0.176   | 0.272   | 0.388                | 0.510    | 0.632               | 0.742 | 0.831 | 0.893 | 0.935 | 0.962 | 0.978 | 0.988 | 0.993 |
| LS                                                                                                       | 0.0095   | 0.288   | 0.391   | 0.495                | 0.597    | 0.694               | 0.776 | 0.846 | 0.900 | 0.938 | 0.963 | 0.979 | 0.988 | 0.993 |
| MON                                                                                                      | 0.0080   | 0.326   | 0.434   | 0.546                | 0.651    | 0.743               | 0.822 | 0.883 | 0.927 | 0.956 | 0.974 | 0.985 | 0.992 | 0.995 |
| G1                                                                                                       | 0.0101   | 0.337   | 0.448   | 0.558                | 0.661    | 0.751               | 0.828 | 0.887 | 0.929 | 0.957 | 0.975 | 0.986 | 0.992 | 0.995 |
| IN1                                                                                                      | 0.0099   | 0.335   | 0.445   | 0.554                | 0.657    | 0.747               | 0.825 | 0.885 | 0.927 | 0.956 | 0.975 | 0.986 | 0.992 | 0.995 |
| IN2                                                                                                      | 0.0102   | 0.357   | 0.464   | 0.569                | 0.670    | 0.758               | 0.832 | 0.889 | 0.931 | 0.959 | 0.976 | 0.986 | 0.992 | 0.996 |
| IN4                                                                                                      | 0.0098   | 0.377   | 0.482   | 0.586                | 0.681    | 0.765               | 0.836 | 0.892 | 0.932 | 0.960 | 0.977 | 0.987 | 0.992 | 0.996 |
| NA1                                                                                                      | 0.0099   | 0.354   | 0.466   | 0.577                | 0.681    | 0.773               | 0.848 | 0.903 | 0.940 | 0.965 | 0.980 | 0.989 | 0.994 | 0.996 |
| NA2                                                                                                      | 0.0095   | 0.423   | 0.531   | 0.632                | 0.723    | 0.802               | 0.868 | 0.915 | 0.948 | 0.969 | 0.982 | 0.990 | 0.994 | 0.997 |

```

NA4      0.0095    0.433    0.539    0.637    0.727    0.805    0.869    0.916    0.948    0.969    0.982    0.990    0.994    0.997
NA4G     0.0151    0.465    0.566    0.659    0.745    0.819    0.878    0.921    0.952    0.971    0.984    0.991    0.995    0.997
IN1 exact-law prediction 0.281 0.406 0.531 0.645 0.743 0.824 0.885 0.929 0.957 0.975 0.986 0.992 0.995
exact-law MAE, IN1 Pd: per unit x SCR 0.0210; pooled curve 0.0105
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR -0.9 dB | vs CA16 6.96 [ 4.29, 9.04] | vs Caloc 9.81 [ 6.41,12.61]
IN4 SCR -1.3 dB | vs CA16 7.37 [ 4.54, 9.52] | vs Caloc 10.22 [ 6.67,13.26]
NA4 SCR -1.9 dB | vs CA16 8.03 [ 5.45,10.02] | vs Caloc 10.88 [ 7.56,13.66]
G1 SCR -1.0 dB | vs CA16 7.05 [ 4.40, 9.11] | vs Caloc 9.90 [ 6.48,12.71]
U SCR 3.8 dB | vs CA16 2.30 [ 0.78, 3.44] | vs Caloc 5.15 [ 3.34, 6.61]
MLP SCR 2.2 dB | vs CA16 3.88 [ 1.89, 5.34] | vs Caloc 6.73 [ 3.89, 8.70]
LS SCR 0.4 dB | vs CA16 5.72 [ 2.75, 7.96] | vs Caloc 8.57 [ 4.89,11.49]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 8.7 dB | vs CA16 7.62 [ 4.55,10.01] | vs Caloc 8.97 [ 5.58,11.67]
IN4 SCR 8.0 dB | vs CA16 8.35 [ 4.95,10.96] | vs Caloc 9.70 [ 5.90,12.64]
NA4 SCR 6.5 dB | vs CA16 9.81 [ 6.27,12.45] | vs Caloc 11.16 [ 7.28,14.14]
G1 SCR 8.7 dB | vs CA16 7.67 [ 4.71,10.02] | vs Caloc 9.02 [ 5.70,11.73]
U SCR 13.5 dB | vs CA16 2.80 [ 1.01, 4.34] | vs Caloc 4.15 [ 2.23, 5.73]
MLP SCR 12.0 dB | vs CA16 4.36 [ 2.30, 5.85] | vs Caloc 5.71 [ 3.23, 7.35]
LS SCR 10.2 dB | vs CA16 6.14 [ 2.75, 8.69] | vs Caloc 7.49 [ 3.78,10.46]

=== 1-alpha = 0.999 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
Caloc 0.0008 0.002 0.004 0.008 0.023 0.064 0.151 0.292 0.463 0.629 0.761 0.855 0.915 0.952
CA16 0.0007 0.031 0.057 0.103 0.176 0.284 0.417 0.561 0.693 0.797 0.873 0.923 0.955 0.974
U 0.0014 0.023 0.048 0.092 0.159 0.251 0.362 0.485 0.606 0.710 0.796 0.863 0.911 0.944
RMS 0.0007 0.057 0.108 0.184 0.287 0.411 0.539 0.665 0.772 0.853 0.910 0.946 0.969 0.982
MLP 0.0009 0.057 0.110 0.190 0.297 0.426 0.559 0.683 0.788 0.865 0.918 0.952 0.972 0.984
LS 0.0010 0.124 0.198 0.290 0.390 0.494 0.596 0.693 0.778 0.849 0.902 0.940 0.964 0.979
MDN 0.0000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
G1 0.0013 0.226 0.333 0.446 0.560 0.665 0.756 0.834 0.893 0.934 0.961 0.977 0.987 0.993
IN1 0.0011 0.208 0.309 0.418 0.528 0.630 0.723 0.805 0.870 0.918 0.951 0.971 0.984 0.991
IN2 0.0009 0.209 0.309 0.418 0.526 0.629 0.722 0.803 0.868 0.916 0.950 0.970 0.983 0.990
IN4 0.0008 0.230 0.332 0.440 0.543 0.641 0.729 0.804 0.866 0.913 0.947 0.969 0.982 0.990
NA1 0.0012 0.219 0.325 0.440 0.554 0.661 0.758 0.837 0.897 0.937 0.963 0.979 0.988 0.993
NA2 0.0009 0.272 0.380 0.491 0.598 0.695 0.783 0.853 0.906 0.943 0.966 0.981 0.989 0.994
NA4 0.0007 0.267 0.377 0.488 0.596 0.694 0.781 0.852 0.905 0.942 0.966 0.981 0.989 0.994
NA4G 0.0028 0.370 0.477 0.581 0.674 0.760 0.833 0.890 0.931 0.958 0.976 0.986 0.992 0.996
IN1 exact-law prediction 0.144 0.251 0.378 0.505 0.620 0.720 0.804 0.871 0.919 0.951 0.972 0.984 0.991
exact-law MAE, IN1 Pd: per unit x SCR 0.0266; pooled curve 0.0155
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 1.9 dB | vs CA16 6.84 [ 4.08, 8.94] | vs Caloc 10.95 [ 7.28,13.95]
IN4 SCR 1.7 dB | vs CA16 7.07 [ 4.04, 9.44] | vs Caloc 11.19 [ 7.42,14.41]
NA4 SCR 0.5 dB | vs CA16 8.27 [ 5.76,10.32] | vs Caloc 12.39 [ 8.88,15.43]
G1 SCR 1.2 dB | vs CA16 7.52 [ 5.19, 9.39] | vs Caloc 11.64 [ 8.28,14.48]
U SCR 10.4 dB | vs CA16 -1.64 [-4.25,-0.30] | vs Caloc 2.48 [ 0.60, 3.74]
MLP SCR 6.2 dB | vs CA16 2.55 [ 1.07, 3.82] | vs Caloc 6.67 [ 3.97, 8.33]
LS SCR 5.1 dB | vs CA16 3.69 [ 1.34, 5.58] | vs Caloc 7.81 [ 4.35,10.22]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 11.5 dB | vs CA16 7.22 [ 4.16, 9.55] | vs Caloc 9.57 [ 5.97,12.67]
IN4 SCR 11.2 dB | vs CA16 7.48 [ 4.09,10.22] | vs Caloc 9.90 [ 6.03,13.47]
NA4 SCR 9.7 dB | vs CA16 8.96 [ 6.01,11.33] | vs Caloc 11.37 [ 7.84,14.47]
G1 SCR 10.8 dB | vs CA16 7.88 [ 5.23,10.04] | vs Caloc 10.22 [ 6.92,13.23]
U SCR 18.6 dB | vs CA16 0.08 [-1.16, 0.98] | vs Caloc 2.35 [ 0.65, 3.73]
MLP SCR 15.9 dB | vs CA16 2.85 [ 1.26, 4.16] | vs Caloc 5.32 [ 3.01, 6.76]
LS SCR 14.6 dB | vs CA16 4.10 [ 1.22, 6.30] | vs Caloc 6.51 [ 2.89, 9.48]

```

## 13 Persistent targets after onset

Frozen onset study (detection/PROTOCOL\_ONSET.md): per-look detection probability, cumulative detection within  $K$  looks, and clean cumulative false-alarm probabilities:

```

units 56; segments 74980

=== per-look Pfa design 0.01 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0099 0.0189 0.0357 0.0683
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0101 0.0192 0.0369 0.0706
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0098 0.0191 0.0369 0.0721
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0095 0.0182 0.0363 0.0701
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0103 0.0170 0.0302 0.0540
clean cumulative false-alarm prob, K=1,2,4,8: Caloc 0.0113 0.0147 0.0209 0.0308
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002
G1 0.887 0.715 0.645 0.497 0.068 0.011 0.005 0.003
IN4 0.891 0.725 0.563 0.256 0.022 0.003 0.001 0.001
NA4 0.914 0.883 0.705 0.571 0.431 0.287 0.201 0.178
CA16 0.691 0.601 0.455 0.254 0.100 0.039 0.017 0.009
Caloc 0.562 0.564 0.564 0.563 0.565 0.565 0.563 0.565
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001
G1 0.887 0.025 0.014 0.008 0.004 0.003 0.002 0.002
IN4 0.891 0.049 0.004 0.001 0.001 0.001 0.001 0.001
NA4 0.914 0.666 0.153 0.015 0.006 0.004 0.003 0.003
CA16 0.691 0.610 0.460 0.186 0.033 0.013 0.006 0.004
Caloc 0.562 0.563 0.564 0.564 0.564 0.565 0.565 0.565
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002
G1 0.887 0.942 0.909 0.817 0.127 0.014 0.005 0.002
IN4 0.891 0.927 0.763 0.087 0.009 0.002 0.001 0.001
NA4 0.914 0.945 0.931 0.831 0.675 0.500 0.380 0.351
CA16 0.691 0.591 0.454 0.268 0.090 0.040 0.015 0.008
Caloc 0.562 0.564 0.564 0.563 0.564 0.564 0.565 0.562
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs Caloc [95% day-cluster CI] --
IN1 K=1: 9.82 [ 6.40,12.68] | K=2: 9.92 [ 6.58,12.67] | K=4: 9.19 [ 5.99,11.87] | K=8: 8.42 [ 5.22,11.05]
NA4 K=1: 10.93 [ 7.46,13.74] | K=2: 10.34 [ 7.31,12.81] | K=4: 9.82 [ 7.17,11.99] | K=8: 9.40 [ 7.19,11.28]
IN4 K=1: 10.23 [ 6.71,13.18] | K=2: 9.88 [ 6.44,12.62] | K=4: 9.14 [ 5.82,11.82] | K=8: 8.28 [ 4.83,11.05]
G1 K=1: 9.90 [ 6.58,12.74] | K=2: 9.97 [ 6.75,12.63] | K=4: 9.27 [ 6.23,11.85] | K=8: 8.53 [ 5.56,11.06]
CA16 K=1: 2.76 [ 1.81, 4.06] | K=2: 2.94 [ 1.88, 4.41] | K=4: 2.85 [ 1.67, 4.49] | K=8: 2.45 [ 1.06, 4.25]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs Caloc [95% day-cluster CI] --
IN1 K=1: 8.95 [ 5.56,11.69] | K=2: 10.00 [ 6.21,12.95] | K=4: 9.96 [ 5.96,13.07] | K=8: 9.45 [ 5.43,12.73]
NA4 K=1: 11.09 [ 7.22,14.10] | K=2: 12.33 [ 7.84,15.68] | K=4: 12.94 [ 8.38,16.44] | K=8: 13.26 [ 8.80,16.82]
IN4 K=1: 9.65 [ 5.85,12.64] | K=2: 10.99 [ 6.43,14.37] | K=4: 10.81 [ 6.09,14.37] | K=8: 10.15 [ 5.18,13.94]
G1 K=1: 9.01 [ 5.66,11.73] | K=2: 10.07 [ 6.39,13.05] | K=4: 10.05 [ 6.23,13.22] | K=8: 9.56 [ 5.56,12.83]
CA16 K=1: 1.41 [ 0.86, 2.14] | K=2: 1.43 [ 0.76, 2.32] | K=4: 1.20 [ 0.38, 2.32] | K=8: 0.53 [-0.47, 1.81]

=== per-look Pfa design 0.001 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0011 0.0017 0.0033 0.0063
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0013 0.0021 0.0040 0.0079
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0008 0.0014 0.0027 0.0055
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0007 0.0014 0.0027 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0007 0.0015 0.0025 0.0051

```

```

clean cumulative false-alarm prob, K=1,2,4,8: CAlloc 0.0010 0.0012 0.0018 0.0029
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1  0.804 0.587 0.347 0.025 0.001 0.000 0.000 0.000
G1    0.833 0.626 0.455 0.014 0.002 0.000 0.000 0.000
IN4   0.802 0.559 0.133 0.010 0.000 0.000 0.000 0.000
NA4   0.852 0.790 0.516 0.312 0.173 0.124 0.114 0.110
CA16  0.555 0.368 0.125 0.025 0.006 0.002 0.001 0.000
CAlloc 0.290 0.290 0.289 0.290 0.289 0.291 0.289 0.289
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1  0.804 0.003 0.000 0.000 0.000 0.000 0.000 0.000
G1    0.833 0.008 0.002 0.001 0.000 0.000 0.000 0.000
IN4   0.802 0.002 0.000 0.000 0.000 0.000 0.000 0.000
NA4   0.852 0.448 0.023 0.001 0.000 0.000 0.000 0.000
CA16  0.555 0.364 0.066 0.004 0.001 0.000 0.000 0.000
CAlloc 0.290 0.289 0.291 0.290 0.289 0.289 0.290 0.290
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1  0.804 0.878 0.629 0.057 0.001 0.000 0.000 0.000
G1    0.833 0.904 0.801 0.025 0.002 0.001 0.000 0.000
IN4   0.802 0.829 0.278 0.001 0.000 0.000 0.000 0.000
NA4   0.852 0.906 0.849 0.540 0.337 0.273 0.259 0.252
CA16  0.555 0.372 0.117 0.032 0.004 0.002 0.000 0.000
CAlloc 0.290 0.290 0.291 0.290 0.289 0.292 0.290 0.291
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1  K=1: 10.93 [ 7.28,14.02] | K=2: 11.99 [ 7.93,15.35] | K=4: 11.48 [ 7.26,15.09] | K=8: 10.88 [ 6.53,14.62]
NA4  K=1: 12.40 [ 9.09,15.50] | K=2: 13.19 [ 9.42,16.38] | K=4: 12.98 [ 9.27,15.90] | K=8: 12.73 [ 9.29,15.59]
IN4  K=1: 11.17 [ 7.43,14.42] | K=2: 12.12 [ 7.77,15.88] | K=4: 11.49 [ 7.08,15.34] | K=8: 10.87 [ 6.35,14.85]
G1   K=1: 11.63 [ 8.41,14.50] | K=2: 12.89 [ 9.11,16.16] | K=4: 12.54 [ 8.65,15.96] | K=8: 11.97 [ 8.03,15.55]
CA16 K=1: 4.06 [ 2.75, 5.93] | K=2: 4.00 [ 2.44, 6.10] | K=4: 3.57 [ 1.85, 5.85] | K=8: 3.00 [ 1.14, 5.43]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1  K=1: 9.54 [ 5.98,12.63] | K=2: 10.64 [ 6.86,13.92] | K=4: 10.38 [ 6.25,13.85] | K=8: 9.80 [ 5.45,13.28]
NA4  K=1: 11.30 [ 7.70,14.40] | K=2: 12.63 [ 8.56,16.01] | K=4: 13.09 [ 9.01,16.52] | K=8: 13.06 [ 9.04,16.55]
IN4  K=1: 9.86 [ 5.87,13.40] | K=2: 11.02 [ 6.57,14.94] | K=4: 10.61 [ 5.84,14.63] | K=8: 10.00 [ 5.16,14.13]
G1   K=1: 10.17 [ 6.99,13.18] | K=2: 11.42 [ 7.90,14.55] | K=4: 11.26 [ 7.42,14.46] | K=8: 10.68 [ 6.87,14.03]
CA16 K=1: 2.45 [ 1.56, 3.81] | K=2: 2.39 [ 1.22, 4.07] | K=4: 2.00 [ 0.67, 3.77] | K=8: 1.40 [-0.11, 3.33]

```

Exploratory: comparison at equal dwell false-alarm probability (0.01), with max and sum-of-squares dwell statistics:

EXPLORATORY: units 56; dwell Pfa design 0.01; persistent Swerling-1 targets, random Doppler

```

-- K = 1 looks --
IN1  max: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
IN1  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
NA4  max: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
NA4  sum: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
CAlloc max: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
CAlloc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 9.93 [ 6.46, 12.81] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.99 [ 7.58, 13.81] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 9.93 [ 6.37, 12.81] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.99 [ 7.61, 13.81] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB

-- K = 2 looks --
IN1  max: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.0 dB | Pd 0.9 7.1 dB
IN1  sum: measured dwell Pfa 0.0097 | SCR for Pd 0.5 -2.2 dB | Pd 0.9 6.8 dB
NA4  max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -2.5 dB | Pd 0.9 4.8 dB
NA4  sum: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
CAlloc max: measured dwell Pfa 0.0110 | SCR for Pd 0.5 8.1 dB | Pd 0.9 16.8 dB
CAlloc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 8.7 dB | Pd 0.9 17.2 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.72 [ 6.85, 13.68] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 10.12 [ 6.29, 13.09] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.15 [ 7.71, 13.85] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 12.46 [ 7.90, 15.85] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.91 [ 7.32, 13.74] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 10.40 [ 6.73, 13.29] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.42 [ 8.19, 13.92] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 13.13 [ 8.71, 16.43] dB

-- K = 4 looks --
IN1  max: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 7.4 dB
IN1  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -2.6 dB | Pd 0.9 6.6 dB
NA4  max: measured dwell Pfa 0.0087 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
NA4  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -3.3 dB | Pd 0.9 2.7 dB
CAlloc max: measured dwell Pfa 0.0109 | SCR for Pd 0.5 7.6 dB | Pd 0.9 16.2 dB
CAlloc sum: measured dwell Pfa 0.0111 | SCR for Pd 0.5 8.3 dB | Pd 0.9 16.8 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.29 [ 6.26, 13.55] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 9.47 [ 5.33, 12.64] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.08 [ 7.57, 13.82] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 12.70 [ 7.98, 16.26] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.94 [ 7.48, 13.70] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 10.22 [ 6.66, 13.08] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.67 [ 8.74, 13.95] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 14.17 [ 9.70, 17.61] dB

-- K = 8 looks --
IN1  max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -1.3 dB | Pd 0.9 8.3 dB
IN1  sum: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 7.0 dB
NA4  max: measured dwell Pfa 0.0091 | SCR for Pd 0.5 -2.9 dB | Pd 0.9 4.3 dB
NA4  sum: measured dwell Pfa 0.0100 | SCR for Pd 0.5 -3.8 dB | Pd 0.9 2.4 dB
CAlloc max: measured dwell Pfa 0.0103 | SCR for Pd 0.5 7.4 dB | Pd 0.9 16.0 dB
CAlloc sum: measured dwell Pfa 0.0105 | SCR for Pd 0.5 8.0 dB | Pd 0.9 16.4 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 9.31 [ 4.87, 12.82] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 8.14 [ 3.83, 11.48] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.85 [ 7.29, 13.54] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 12.13 [ 7.50, 15.64] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.70 [ 7.35, 13.40] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 9.35 [ 6.02, 12.16] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.74 [ 9.17, 13.75] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 14.01 [ 9.97, 17.21] dB

```

Exploratory: exact-law prediction of the post-onset decay:

```

Exact-law prediction vs measured per-look Pd after onset (IN-ARCP, SCR 10 dB, Pfa 0.01); pooled over 56 units (exploratory)
random predicted 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
measured 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002
matched predicted 0.885 0.023 0.013 0.007 0.004 0.002 0.002 0.001
measured 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001
opposite predicted 0.885 0.941 0.906 0.793 0.167 0.014 0.004 0.002
measured 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002

```

## 14 Numerical checks of the detection theory

Monte Carlo checks of Corollary 2, Proposition 2 (closed forms) and Proposition 3 (output of `detection/theory_checks.py`):

```
T1/T2 exact detection law vs Monte Carlo (Swirling-1 target in Y only; 400k draws)
rho=0.91-0.18j c=None r=0.91-0.18j SCR=1.0: MC 0.5358 exact 0.5364 closed form 0.5364
rho=0.91-0.18j c=10.0 r=0.88-0.18j SCR=3.0: MC 0.6381 exact 0.6374
rho=0.70 c=1.0 r=0.65 SCR=10.0: MC 0.4275 exact 0.4263

T3 noise-limited whitening gain: empirical (c+1)/E|Y-pred|^2 vs closed forms (m=16 history, 200k draws)
|rho|=0.7 c= 0.1: IN emp -1.46 dB closed -1.46 | oracle emp 0.03 dB closed(m=inf) 0.03 | clutter-only bound 2.92
|rho|=0.7 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 0.66 dB closed(m=inf) 0.67 | clutter-only bound 2.92
|rho|=0.7 c= 10.0: IN emp 2.24 dB closed 2.23 | oracle emp 2.29 dB closed(m=inf) 2.27 | clutter-only bound 2.92
|rho|=0.7 c= 100.0: IN emp 2.85 dB closed 2.84 | oracle emp 2.85 dB closed(m=inf) 2.84 | clutter-only bound 2.92
|rho|=0.93 c= 0.1: IN emp -2.32 dB closed -2.32 | oracle emp 0.12 dB closed(m=inf) 0.12 | clutter-only bound 8.69
|rho|=0.93 c= 1.0: IN emp 0.02 dB closed 0.00 | oracle emp 1.66 dB closed(m=inf) 1.65 | clutter-only bound 8.69
|rho|=0.93 c= 10.0: IN emp 5.34 dB closed 5.34 | oracle emp 5.76 dB closed(m=inf) 5.76 | clutter-only bound 8.69
|rho|=0.93 c= 100.0: IN emp 8.17 dB closed 8.18 | oracle emp 8.18 dB closed(m=inf) 8.19 | clutter-only bound 8.69
|rho|=0.98 c= 0.1: IN emp -2.50 dB closed -2.52 | oracle emp 0.20 dB closed(m=inf) 0.21 | clutter-only bound 14.02
|rho|=0.98 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 2.22 dB closed(m=inf) 2.22 | clutter-only bound 14.02
|rho|=0.98 c= 10.0: IN emp 6.69 dB closed 6.69 | oracle emp 7.78 dB closed(m=inf) 7.78 | clutter-only bound 14.02
|rho|=0.98 c= 100.0: IN emp 12.31 dB closed 12.32 | oracle emp 12.43 dB closed(m=inf) 12.44 | clutter-only bound 14.02

T4 persistent strong target: limiting score and Pd (IN-ARCP, r fitted = 0.93e*{-0.2j})
sqrt(m/(m-1)) = 1.033; thresholds q(0.1)=1.574 q(0.01)=2.310
Doppler w=+0.0: limiting score 0.937 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=-0.2: limiting score 0.613 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=+1.0: limiting score 1.029 | SCR 0 dB: Pd(0.01) 0.000 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
```

## 15 Baselines: parametric texture and adaptive conformal inference

B1/B2 within-session units: 56; transfer days: 6

```
=== 1-alpha = 0.9 ===
B1 NA4-K / NA4 radius (GM): 1.0092 [1.0064, 1.0115] | coverage NA4 0.9019 NA4-K 0.9015 | max dev NA4 0.043 NA4-K 0.043
B2 within ACI g=0.005: coverage 0.9010 vs split 0.9013 | radius ratio (finite) 0.999 [0.997,1.000] | infinite-region fraction 0.0000 | max dev 0.033 vs 0.036
B2 within ACI g=0.02: coverage 0.9007 vs split 0.9013 | radius ratio (finite) 0.999 [0.995,1.002] | infinite-region fraction 0.0000 | max dev 0.029 vs 0.036
B2 transfer IN1 : mean coverage 0.9029 worst day 0.8822 | radius vs split 1.000 | max dev 0.024 | per day 0.905 0.894 0.915 0.918 0.904 0.882
B2 transfer ACIO.005 : mean coverage 0.9032 worst day 0.8889 | radius vs split 1.001 | max dev 0.020 | per day 0.904 0.897 0.912 0.914 0.903 0.889
B2 transfer ACIO.02 : mean coverage 0.9024 worst day 0.8965 | radius vs split 1.000 | max dev 0.015 | per day 0.903 0.900 0.906 0.908 0.902 0.897

=== 1-alpha = 0.99 ===
B1 NA4-K / NA4 radius (GM): 1.0078 [1.0034, 1.0104] | coverage NA4 0.9905 NA4-K 0.9906 | max dev NA4 0.011 NA4-K 0.011
B2 within ACI g=0.005: coverage 0.9903 vs split 0.9901 | radius ratio (finite) 1.001 [0.996,1.007] | infinite-region fraction 0.0000 | max dev 0.010 vs 0.011
B2 within ACI g=0.02: coverage 0.9901 vs split 0.9901 | radius ratio (finite) 1.006 [0.998,1.015] | infinite-region fraction 0.0115 | max dev 0.009 vs 0.011
B2 transfer IN1 : mean coverage 0.9921 worst day 0.9875 | radius vs split 1.000 | max dev 0.006 | per day 0.987 0.992 0.994 0.996 0.994 0.989
B2 transfer ACIO.005 : mean coverage 0.9918 worst day 0.9889 | radius vs split 0.997 | max dev 0.005 | per day 0.989 0.991 0.993 0.995 0.993 0.990
B2 transfer ACIO.02 : mean coverage 0.9912 worst day 0.9902 | radius vs split 0.995 | max dev 0.004 | per day 0.990 0.990 0.992 0.993 0.991 0.990
```

## 16 Classical detectors, gradual emergence and ACI contamination (R10)

Frozen protocol `r10/PROTOCOL_R10.md`; stated expectations and verdicts in `r10/RESULTS_R10.md`. Output of `r10/analyze_r10.py`:

```
units 56; days 6; test segments 74980

=== Part C: dwell detectors at equal dwell Pfa (alpha = 0.01 unless stated) ===
-- measured dwell Pfa on clean test segments, alpha = 0.01 (K' = 1 2 4 8) --
IN1sum 0.0099 0.0097 0.0099 0.0102
NA4sum 0.0095 0.0094 0.0099 0.0100
CALocsum 0.0108 0.0108 0.0111 0.0105
PAMFH 0.0098 0.0109 0.0101 0.0096
NPAMF 0.0098 0.0109 0.0098 0.0101
MTDhann nan 0.0095 0.0095 0.0106
MTDrect nan 0.0095 0.0096 0.0097
-- measured dwell Pfa on clean test segments, alpha = 0.001 (K' = 1 2 4 8) --
IN1sum 0.0011 0.0008 0.0008 0.0009
NA4sum 0.0007 0.0011 0.0009 0.0010
CALocsum 0.0009 0.0010 0.0011 0.0012
PAMFH 0.0008 0.0010 0.0009 0.0010
NPAMF 0.0008 0.0009 0.0009 0.0012
MTDhann nan 0.0005 0.0009 0.0008
MTDrect nan 0.0005 0.0010 0.0009
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -2.2/ 6.7 -2.5/ 6.5 -2.7/ 6.9
NA4sum -1.9/ 6.6 -2.7/ 4.1 -3.3/ 2.7 -3.7/ 2.3
CALocsum 9.0/ 17.7 8.7/ 17.2 8.4/ 16.8 8.0/ 16.4
PAMFH -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.7 -3.7/ 3.0
NPAMF -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.6 -3.8/ 2.6
MTDhann - 8.4/ 20.7 4.5/ 19.2 2.3/ 17.9
MTDrect - 8.4/ 20.7 7.7/ 19.9 6.3/ 18.9
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.001; mean over units --
IN1sum 2.0/ 11.5 0.3/ 10.0 0.4/ 10.3 1.0/ 11.3
NA4sum 0.5/ 9.8 -1.3/ 7.5 -1.3/ 6.8 -1.4/ 7.2
CALocsum 13.1/ 21.0 12.8/ 20.5 12.6/ 20.2 12.2/ 20.1
PAMFH 1.8/ 11.2 -0.4/ 8.9 -1.2/ 7.6 -2.0/ 7.0
NPAMF 1.8/ 11.2 -0.6/ 8.6 -2.0/ 6.4 -3.0/ 5.0
MTDhann - 14.8/ 22.4 11.9/ 22.5 10.6/ 22.2
MTDrect - 14.8/ 22.4 14.9/ 22.6 13.8/ 22.1
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler matched, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 0.3/ 10.0 2.1/ 11.6 4.0/ 13.4
NA4sum -1.9/ 6.6 -2.0/ 6.7 -1.5/ 7.5 -0.1/ 9.2
CALocsum 9.0/ 17.7 8.8/ 17.5 8.5/ 17.2 8.2/ 16.8
PAMFH -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.1 0.6/ 10.1
NPAMF -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.0 0.2/ 9.5
MTDhann - 18.3/ 22.1 18.3/ 21.6 19.0/ 20.2
MTDrect - 18.3/ 22.1 18.3/ 21.4 17.8/ 21.7
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler opposite, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -3.0/ 3.6 -3.6/ 1.9 -3.8/ 1.3
NA4sum -1.9/ 6.6 -3.1/ 1.9 -3.7/ 0.3 -4.1/ -0.8
CALocsum 9.0/ 17.7 8.5/ 16.9 8.3/ 16.6 7.9/ 16.3
PAMFH -1.3/ 8.0 -3.0/ 2.2 -3.6/ 0.1 -4.1/ -1.1
NPAMF -1.3/ 8.0 -3.0/ 2.2 -3.6/ -0.1 -4.2/ -1.3
```

```

MTDhann      -      4.5/ 15.1      0.3/  9.4      -0.5/  8.5
MTDrect      -      4.5/ 15.1      4.0/ 14.6      2.6/ 13.0
-- gains (dB, positive = first detector needs less SCR) at Pd = 0.5, random Doppler, alpha 0.01; 95% day bootstrap --
K'=1 Pd=0.5 IN1sum vs PAMFH : -0.43 [ -0.59, -0.24] LOD0 -0.48..-0.37
K'=1 Pd=0.9 IN1sum vs PAMFH : -0.72 [ -1.07, -0.26] LOD0 -0.86..-0.44
K'=1 Pd=0.5 NA4sum vs PAMFH : 0.66 [ 0.44, 0.91] LOD0 0.59.. 0.75
K'=1 Pd=0.9 NA4sum vs PAMFH : 1.43 [ 1.37, 1.51] LOD0 1.41.. 1.45
K'=1 Pd=0.5 IN1sum vs Calocsum: 9.86 [ 6.31, 12.74] LOD0 8.99..11.60
K'=1 Pd=0.9 IN1sum vs Calocsum: 9.00 [ 5.58, 11.67] LOD0 8.07..10.65
K'=1 Pd=0.5 PAMFH vs Calocsum: 10.29 [ 6.79, 13.25] LOD0 9.39..12.07
K'=1 Pd=0.9 PAMFH vs Calocsum: 9.72 [ 6.08, 12.63] LOD0 8.51..11.52
K'=1 Pd=0.5 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=1 Pd=0.9 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.5 IN1sum vs PAMFH : -0.20 [ -0.30, 0.00] LOD0 -0.24..-0.15
K'=2 Pd=0.9 IN1sum vs PAMFH : -1.68 [ -2.39, -0.77] LOD0 -1.99..-1.10
K'=2 Pd=0.5 NA4sum vs PAMFH : 0.32 [ 0.07, 0.67] LOD0 0.17.. 0.47
K'=2 Pd=0.9 NA4sum vs PAMFH : 1.00 [ 0.84, 1.19] LOD0 0.93.. 1.13
K'=2 Pd=0.5 IN1sum vs MTDhann : 10.59 [ 9.46, 12.27] LOD0 10.19..11.13
K'=2 Pd=0.9 IN1sum vs MTDhann : 14.19 [ 12.11, 16.48] LOD0 13.70..15.24
K'=2 Pd=0.5 NA4sum vs MTDhann : 11.11 [ 10.34, 12.46] LOD0 10.81..11.45
K'=2 Pd=0.9 NA4sum vs MTDhann : 16.88 [ 14.09, 19.13] LOD0 16.31..18.18
K'=2 Pd=0.5 PAMFH vs MTDhann : 10.79 [ 9.75, 12.25] LOD0 10.43..11.28
K'=2 Pd=0.9 PAMFH vs MTDhann : 15.92 [ 12.95, 18.05] LOD0 15.25..17.25
K'=2 Pd=0.5 IN1sum vs Calocsum: 10.84 [ 7.30, 13.75] LOD0 9.98..12.63
K'=2 Pd=0.9 IN1sum vs Calocsum: 10.48 [ 6.68, 13.38] LOD0 9.42..12.33
K'=2 Pd=0.5 PAMFH vs Calocsum: 11.04 [ 7.55, 13.80] LOD0 10.13..12.79
K'=2 Pd=0.9 PAMFH vs Calocsum: 12.16 [ 7.57, 15.49] LOD0 10.53..14.32
K'=2 Pd=0.5 MTDhann vs Calocsum: 0.26 [ -2.49, 2.31] LOD0 -0.84.. 1.50
K'=2 Pd=0.9 MTDhann vs Calocsum: -3.47 [ -5.49, -2.08] LOD0 -4.46..-2.73
K'=2 Pd=0.5 PAMFH vs NPAMF : 0.00 [ -0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.9 PAMFH vs NPAMF : 0.01 [ -0.03, 0.04] LOD0 -0.00.. 0.02
K'=4 Pd=0.5 IN1sum vs PAMFH : -0.54 [ -0.88, -0.19] LOD0 -0.63..-0.37
K'=4 Pd=0.9 IN1sum vs PAMFH : -2.78 [ -3.49, -1.87] LOD0 -3.11..-2.23
K'=4 Pd=0.5 NA4sum vs PAMFH : 0.22 [ 0.04, 0.47] LOD0 0.10.. 0.29
K'=4 Pd=0.9 NA4sum vs PAMFH : 1.00 [ 0.81, 1.27] LOD0 0.91.. 1.11
K'=4 Pd=0.5 IN1sum vs MTDhann : 6.98 [ 5.95, 8.43] LOD0 6.74.. 7.86
K'=4 Pd=0.9 IN1sum vs MTDhann : 12.84 [ 11.08, 14.97] LOD0 12.34..13.68
K'=4 Pd=0.5 NA4sum vs MTDhann : 7.74 [ 6.53, 9.23] LOD0 7.35.. 8.73
K'=4 Pd=0.9 NA4sum vs MTDhann : 16.63 [ 14.09, 18.91] LOD0 15.98..17.77
K'=4 Pd=0.5 PAMFH vs MTDhann : 7.52 [ 6.45, 8.90] LOD0 7.25.. 8.45
K'=4 Pd=0.9 PAMFH vs MTDhann : 15.68 [ 13.18, 17.78] LOD0 15.12..16.79
K'=4 Pd=0.5 IN1sum vs Calocsum: 10.88 [ 7.44, 13.64] LOD0 10.04..12.60
K'=4 Pd=0.9 IN1sum vs Calocsum: 10.32 [ 6.71, 13.19] LOD0 9.41..12.08
K'=4 Pd=0.5 PAMFH vs Calocsum: 11.42 [ 8.29, 13.88] LOD0 10.63..12.98
K'=4 Pd=0.9 PAMFH vs Calocsum: 13.10 [ 8.63, 16.42] LOD0 11.64..15.19
K'=4 Pd=0.5 MTDhann vs Calocsum: 3.90 [ -0.24, 6.88] LOD0 2.19.. 5.72
K'=4 Pd=0.9 MTDhann vs Calocsum: -2.39 [ -4.54, -0.89] LOD0 -3.36..-1.60
K'=4 Pd=0.5 PAMFH vs NPAMF : -0.04 [ -0.11, 0.01] LOD0 -0.06..-0.00
K'=4 Pd=0.9 PAMFH vs NPAMF : -0.13 [ -0.21, -0.08] LOD0 -0.16..-0.11
K'=8 Pd=0.5 IN1sum vs PAMFH : -1.01 [ -1.67, -0.34] LOD0 -1.18..-0.70
K'=8 Pd=0.9 IN1sum vs PAMFH : -3.95 [ -4.43, -3.32] LOD0 -4.17..-3.57
K'=8 Pd=0.5 NA4sum vs PAMFH : 0.08 [ -0.14, 0.30] LOD0 -0.04.. 0.14
K'=8 Pd=0.9 NA4sum vs PAMFH : 0.70 [ 0.49, 1.03] LOD0 0.62.. 0.77
K'=8 Pd=0.5 IN1sum vs MTDhann : 4.95 [ 4.19, 5.97] LOD0 4.75.. 5.58
K'=8 Pd=0.9 IN1sum vs MTDhann : 10.96 [ 9.11, 12.61] LOD0 10.54..11.83
K'=8 Pd=0.5 NA4sum vs MTDhann : 6.04 [ 4.85, 7.58] LOD0 5.41.. 6.80
K'=8 Pd=0.9 NA4sum vs MTDhann : 15.60 [ 13.19, 17.56] LOD0 14.90..16.71
K'=8 Pd=0.5 PAMFH vs MTDhann : 5.96 [ 4.85, 7.48] LOD0 5.45.. 6.71
K'=8 Pd=0.9 PAMFH vs MTDhann : 14.90 [ 12.56, 16.68] LOD0 14.14..15.99
K'=8 Pd=0.5 IN1sum vs Calocsum: 10.65 [ 7.32, 13.37] LOD0 9.82..12.34
K'=8 Pd=0.9 IN1sum vs Calocsum: 9.41 [ 5.89, 12.28] LOD0 8.54..11.12
K'=8 Pd=0.5 PAMFH vs Calocsum: 11.66 [ 8.91, 13.74] LOD0 10.94..13.03
K'=8 Pd=0.9 PAMFH vs Calocsum: 13.36 [ 9.35, 16.42] LOD0 12.11..15.29
K'=8 Pd=0.5 MTDhann vs Calocsum: 5.70 [ 1.71, 8.63] LOD0 4.24.. 7.58
K'=8 Pd=0.9 MTDhann vs Calocsum: -1.54 [ -3.31, -0.10] LOD0 -2.03..-0.71
K'=8 Pd=0.5 PAMFH vs NPAMF : -0.15 [ -0.27, -0.04] LOD0 -0.18..-0.09
K'=8 Pd=0.9 PAMFH vs NPAMF : -0.44 [ -0.55, -0.30] LOD0 -0.48..-0.40

```

C1 check: max |Pd(PAMF-H, K'=1) - Pd(IN4 look 0)| over SCR = 0.0000 (expectation < 0.005)

C2: PAMF-H needs 1.01 dB less SCR than IN1-sum at K'=8 (expectation >= 3 dB): FAILED

C3: MTD-CA needs 5.70 dB less SCR than Caloc-sum at K'=8 (expectation >= 5 dB): OK

C4: PAMF-H matched-Doppler SCR is 4.25 dB worse than random (expectation >= 3 dB): OK

=== Part R: gradual emergence (random Doppler, alpha 0.01) ===

-- placement pre: per-look Pd at SCR 10 dB (looks 0.7), mean over units --

```

L= 1 IN1      0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4      0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 Caloc    0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
L= 4 IN1      0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002
L= 4 NA4      0.790 0.821 0.643 0.473 0.281 0.203 0.183 0.173
L= 4 Caloc    0.559 0.556 0.557 0.557 0.559 0.558 0.559 0.557
L= 8 IN1      0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001
L= 8 NA4      0.678 0.684 0.404 0.237 0.194 0.179 0.171 0.166
L= 8 Caloc    0.557 0.559 0.558 0.556 0.558 0.557 0.558 0.557
L=16 IN1      0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001
L=16 NA4      0.204 0.203 0.184 0.175 0.168 0.165 0.162 0.160
L=16 Caloc    0.558 0.558 0.557 0.557 0.558 0.558 0.560 0.559

```

-- placement pre: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --

```

L= 1 IN1      look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4      look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4      look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16     look0 6.2 | cum8 2.8 | first look 0.27
L= 1 CA1oc    look0 9.0 | cum8 5.4 | first look 0.63
L= 4 IN1      look0 0.4 | cum8 -2.5 | first look 0.16
L= 4 IN4      look0 0.1 | cum8 -1.9 | first look 0.12
L= 4 NA4      look0 -1.7 | cum8 -4.0 | first look 0.22
L= 4 CA16     look0 7.7 | cum8 4.2 | first look 0.32
L= 4 CA1oc    look0 9.0 | cum8 5.4 | first look 0.63
L= 8 IN1      look0 4.4 | cum8 0.3 | first look 0.21
L= 8 IN4      look0 9.3 | cum8 7.4 | first look 0.21
L= 8 NA4      look0 -0.9 | cum8 -4.0 | first look 0.27
L= 8 CA16     look0 11.7 | cum8 8.3 | first look 0.42
L= 8 CA1oc    look0 9.1 | cum8 5.4 | first look 0.63
L=16 IN1      look0 nan | cum8 nan | first look 1.49
L=16 IN4      look0 nan | cum8 nan | first look 2.29
L=16 NA4      look0 18.7 | cum8 13.0 | first look 0.67
L=16 CA16     look0 nan | cum8 nan | first look 0.88
L=16 CA1oc    look0 9.1 | cum8 5.4 | first look 0.63

```

-- placement pre: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --

```

L= 1 IN1sum   -2.7 NA4sum -3.7 Calocsum 8.0 PAMFH -3.8 NPAMF -3.8 MTDhann 2.3 MTDirect 6.3
L= 4 IN1sum   -2.1 NA4sum -3.6 Calocsum 8.0 PAMFH -3.5 NPAMF -3.6 MTDhann 2.3 MTDirect 6.2
L= 8 IN1sum   0.3 NA4sum -3.5 Calocsum 8.0 PAMFH -3.1 NPAMF -3.5 MTDhann 2.3 MTDirect 6.3
L=16 IN1sum   nan NA4sum 9.3 Calocsum 8.0 PAMFH -1.8 NPAMF -3.0 MTDhann 2.3 MTDirect 6.2

```

-- placement in: per-look Pd at SCR 10 dB (looks 0.7), mean over units --

```

L= 1 IN1      0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4      0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 Caloc    0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557

```

```

L= 4 IN1      0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015
L= 4 NA4      0.558 0.726 0.786 0.792 0.821 0.643 0.471 0.280
L= 4 Caloc    0.036 0.179 0.388 0.569 0.557 0.558 0.558 0.557
L= 8 IN1      0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656
L= 8 NA4      0.304 0.529 0.630 0.677 0.695 0.701 0.693 0.677
L= 8 Caloc    0.015 0.037 0.092 0.180 0.285 0.387 0.479 0.557
L=16 IN1      0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484
L=16 NA4      0.104 0.307 0.436 0.513 0.555 0.579 0.585 0.583
L=16 Caloc    0.012 0.016 0.023 0.036 0.059 0.091 0.133 0.179
-- placement in: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --
L= 1 IN1      look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4      look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4      look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16     look0  6.2 | cum8  2.8 | first look 0.27
L= 1 Caloc    look0  9.0 | cum8  5.4 | first look 0.63
L= 4 IN1      look0 11.0 | cum8 -2.8 | first look 0.91
L= 4 IN4      look0 10.2 | cum8 -2.5 | first look 0.82
L= 4 NA4      look0  8.9 | cum8 -3.8 | first look 0.77
L= 4 CA16     look0 18.2 | cum8  3.7 | first look 1.70
L= 4 Caloc    look0 21.0 | cum8  6.0 | first look 2.46
L= 8 IN1      look0 16.2 | cum8 -2.0 | first look 1.94
L= 8 IN4      look0 15.4 | cum8 -1.6 | first look 1.73
L= 8 NA4      look0 14.3 | cum8 -3.0 | first look 1.60
L= 8 CA16     look0 23.6 | cum8  5.6 | first look 3.23
L= 8 Caloc    look0 24.8 | cum8  7.5 | first look 4.35
L=16 IN1      look0 19.9 | cum8  3.1 | first look 2.84
L=16 IN4      look0 18.9 | cum8  2.9 | first look 2.58
L=16 NA4      look0 17.7 | cum8 -0.2 | first look 2.54
L=16 CA16     look0  nan | cum8 11.6 | first look 4.01
L=16 Caloc    look0  nan | cum8 13.6 | first look 4.99
-- placement in: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --
L= 1 INisum   -2.7 NA4sum -3.7 Calocsum  8.0 PAMFH -3.7 NPAMF -3.8 MTDhann  2.3 MTDrect  6.3
L= 4 INisum   -2.1 NA4sum -3.2 Calocsum  9.3 PAMFH -3.0 NPAMF -3.1 MTDhann  2.9 MTDrect  7.7
L= 8 INisum   -0.3 NA4sum -2.3 Calocsum 12.0 PAMFH -1.8 NPAMF -1.7 MTDhann  6.8 MTDrect 10.7
L=16 INisum   5.6 NA4sum  1.7 Calocsum 18.0 PAMFH  2.8 NPAMF  3.2 MTDhann 12.9 MTDrect 16.8
R-pre single-look gain IN1 vs Caloc, L= 1:  9.86 [ 6.31, 12.78] LOD0  8.99..11.60
R-pre single-look gain IN1 vs Caloc, L= 4:  8.65 [ 5.11, 11.50] LOD0  7.82..10.40
R-pre single-look gain IN1 vs Caloc, L= 8:  4.65 [ 0.90,  7.62] LOD0  3.71.. 6.54
R-pre single-look gain IN1 vs Caloc, L=16:  nan [  nan,  nan] LOD0  nan..  nan
R1: gain decreasing in L: False; below 5 dB at L=16: False -> FAILED
R3: IN1 first-detection look (R-in) by L: [0.12, 0.91, 1.94, 2.84] -> OK

=== Part A: ACI with target-contaminated feedback (IN1, alpha 0.01) ===
pi=0      split Pfa/a 0.99 Pd  nan | naive0.005 Pfa/a 0.97 Pd  nan | gated0.005 Pfa/a 0.97 Pd  nan | naive0.02 Pfa/a 0.99 Pd  nan | gated0.02 Pfa/a 0.99 Pd  nan
pi=0.01   split Pfa/a 0.99 Pd 0.893 | naive0.005 Pfa/a 0.68 Pd 0.880 | gated0.005 Pfa/a 0.89 Pd 0.888 | naive0.02 Pfa/a 0.52 Pd 0.664 | gated0.02 Pfa/a 0.82 Pd 0.850
pi=0.05   split Pfa/a 0.99 Pd 0.890 | naive0.005 Pfa/a 0.24 Pd 0.426 | gated0.005 Pfa/a 0.68 Pd 0.834 | naive0.02 Pfa/a 0.14 Pd 0.275 | gated0.02 Pfa/a 0.55 Pd 0.635
A1 (gamma 0.02): naive clean Pfa/alpha 0.14 (< 0.8?) and Pd 0.275 vs split 0.890 -> OK
A2 (gamma 0.02): gated clean Pfa/alpha 0.55 within [0.8, 1.2]? -> FAILED

```

Exact-law prediction of IN-ARCP's per-look detection probability under gradual emergence (endpoint R2):

```

pre|L1      pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
           meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009
pre|L4      pred 0.789 0.702 0.541 0.101 0.015 0.006 0.003 0.002
           meas 0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002 | MAE 0.009
pre|L8      pred 0.648 0.434 0.049 0.012 0.005 0.003 0.002 0.001
           meas 0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001 | MAE 0.008
pre|L16     pred 0.015 0.007 0.004 0.002 0.002 0.001 0.001 0.001
           meas 0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001 | MAE 0.001
in|L1      pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
           meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009
in|L4      pred 0.429 0.665 0.750 0.789 0.702 0.541 0.101 0.015
           meas 0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015 | MAE 0.021
in|L8      pred 0.155 0.407 0.535 0.604 0.643 0.662 0.665 0.648
           meas 0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656 | MAE 0.033
in|L16     pred 0.038 0.168 0.296 0.379 0.432 0.461 0.472 0.462
           meas 0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484 | MAE 0.037
R2: exact-law MAE per unit x look x L x placement = 0.016 (expectation <= 0.03): OK

```

## 17 Order-statistic normalization and real interference (R11)

Frozen protocol r11/PROTOCOL.R11.md; expectations and verdicts in r11/RESULTS.R11.md (2 of 11 met). Monte Carlo check of Corollary 3 (r11/theory\_os.py):

```

Pfa T=1.0: law 0.29412 MC 0.29393 z=-0.26
Pfa T=3.0: law 0.03612 MC 0.03613 z=+0.03
Pfa T=8.0: law 0.00067 MC 0.00066 z=-0.26
Pd S=1.0: law 0.5160 MC 0.5156
Pd S=10.0: law 0.9266 MC 0.9269
OS(k=12) loss vs CA in homogeneous clutter at Pfa 0.01, Pd 0.5: 0.30 dB
0 contaminated history samples (random positions): Pd at S=10 OS 0.926 RMS 0.931
2 contaminated history samples (random positions): Pd at S=10 OS 0.835 RMS 0.000
4 contaminated history samples (random positions): Pd at S=10 OS 0.000 RMS 0.000
5 contaminated history samples (random positions): Pd at S=10 OS 0.000 RMS 0.000

```

Output of r11/analyze\_r11.py (Part O: IPIX; Part J: 77 GHz with real interference; N: no zeroing, Z: fast-time zeroing; strata: number of interference-hit chirps among the 16 history chirps):

```

=== Part O: IPIX, 56 units, 74980 test segments ===
-- O1: look-0 Pfa (alpha 0.01) and texture-quintile Pfa/alpha (low -> high), spread --
IN1      Pfa 0.0099 | 1.16 1.05 1.12 0.93 0.66 | spread 1.76
OS1_8    Pfa 0.0108 | 1.48 1.19 1.12 0.90 0.72 | spread 2.06
OS1_12   Pfa 0.0102 | 1.25 1.10 1.09 0.87 0.79 | spread 1.59
IN4      Pfa 0.0098 | 1.07 1.01 0.92 1.00 0.91 | spread 1.17
OS4      Pfa 0.0100 | 1.28 1.02 0.97 0.90 0.85 | spread 1.50
NA4      Pfa 0.0095 | 0.72 0.94 0.94 0.95 1.20 | spread 1.67
Caloc    Pfa 0.0108 | 0.69 0.75 0.98 1.12 1.84 | spread 2.68
-- dwell Pfa at K=8 (design 0.01): INisum 0.0102 OS1_8sum 0.0108 PAMFH 0.0096 PAMFOS 0.0102 Calocsum 0.0105
-- O2: abrupt target (L=1, look 0) SCR for Pd 0.5; difference vs IN1 --
IN1      SCR -0.82 dB | minus IN1  0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
OS1_8    SCR  1.17 dB | minus IN1  1.99 [ 0.89,  3.39] LOD0 1.50.. 2.56
OS1_12   SCR -0.49 dB | minus IN1  0.33 [ 0.17,  0.64] LOD0 0.21.. 0.42
IN4      SCR -1.25 dB | minus IN1 -0.43 [-0.60, -0.24] LOD0 -0.48..-0.37
OS4      SCR  0.01 dB | minus IN1  0.84 [ 0.20,  1.70] LOD0 0.58.. 1.25
NA4      SCR -1.92 dB | minus IN1 -1.10 [-1.25, -0.95] LOD0 -1.14..-1.06
Caloc    SCR  9.04 dB | minus IN1  9.86 [ 6.39, 12.73] LOD0 8.99..11.60
-- O3: persistent target (L=1), per-look Pd at SCR 10 dB, looks 0..7 --

```

```

IN1      0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
OS1_8    0.801 0.754 0.744 0.730 0.714 0.698 0.677 0.650
OS1_12   0.871 0.811 0.796 0.774 0.740 0.056 0.006 0.003
IN4      0.891 0.725 0.563 0.257 0.021 0.003 0.001 0.001
OS4      0.844 0.825 0.804 0.748 0.699 0.669 0.624 0.009
NA4      0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
CAlloc   0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
-- O4: ramps; first-look SCR for Pd .5 | SCR for Pd .5 within 8 looks | dwell SCR (K=8) --
pre L= 1: IN1 -0.8| -3.1 OS1_8 1.2| -1.2 OS1_12 -0.5| -2.8 OS4 0.0| -1.9 NA4 -1.9| -4.1 CAlloc 9.0| 5.4
dwell Pd.5/.9: INisum -2.7/ 6.9 OS1_8sum -0.4/ 9.2 PAMFH -3.7/ 3.0 PAMFOS -2.9/ 4.5 CAllocsum 8.0/ 16.4
pre L= 4: IN1 0.4| -2.5 OS1_8 1.8| -0.4 OS1_12 0.1| -2.2 OS4 0.5| -1.1 NA4 -1.7| -4.0 CAlloc 9.0| 5.4
dwell Pd.5/.9: INisum -2.1/ 16.7 OS1_8sum 0.7/ 11.9 PAMFH -3.5/ 13.2 PAMFOS -2.6/ 12.2 CAllocsum 8.0/ 16.4
pre L= 8: IN1 4.4| 0.3 OS1_8 2.9| 0.8 OS1_12 2.9| -0.1 OS4 2.7| 1.2 NA4 -0.9| -4.0 CAlloc 9.1| 5.4
dwell Pd.5/.9: INisum 0.3/ 19.3 OS1_8sum 2.6/ 15.5 PAMFH -3.1/ 15.1 PAMFOS -2.0/ 19.5 CAllocsum 8.0/ 16.3
pre L=16: IN1 nan| nan OS1_8 nan| nan OS1_12 nan| nan OS4 nan| nan NA4 18.7| 13.0 CAlloc 9.1| 5.4
dwell Pd.5/.9: INisum nan/ nan OS1_8sum nan/ nan PAMFH -1.8/ 19.3 PAMFOS 3.1/ nan CAllocsum 8.0/ 16.4
in L= 1: IN1 -0.8| -3.1 OS1_8 1.2| -1.2 OS1_12 -0.5| -2.8 OS4 0.0| -1.9 NA4 -1.9| -4.1 CAlloc 9.0| 5.4
dwell Pd.5/.9: INisum -2.7/ 6.9 OS1_8sum -0.4/ 9.2 PAMFH -3.7/ 3.0 PAMFOS -2.9/ 4.5 CAllocsum 8.0/ 16.4
in L= 4: IN1 11.0| -2.8 OS1_8 11.8| -0.6 OS1_12 11.4| -2.4 OS4 11.7| -1.3 NA4 8.9| -3.8 CAlloc 21.0| 6.0
dwell Pd.5/.9: INisum -2.1/ 9.6 OS1_8sum 0.7/ 11.1 PAMFH -3.0/ 9.2 PAMFOS -2.2/ 10.6 CAllocsum 9.3/ 17.7
in L= 8: IN1 16.2| -2.0 OS1_8 15.1| 0.7 OS1_12 16.0| -1.5 OS4 15.0| -0.1 NA4 14.3| -3.0 CAlloc 24.8| 7.5
dwell Pd.5/.9: INisum -0.3/ 12.1 OS1_8sum 3.0/ 13.5 PAMFH -1.8/ 13.6 PAMFOS -0.8/ 15.0 CAllocsum 12.0/ 20.4
in L=16: IN1 19.9| 3.1 OS1_8 20.8| 5.8 OS1_12 20.2| 3.8 OS4 19.8| 5.0 NA4 17.7| -0.2 CAlloc nan| 13.6
dwell Pd.5/.9: INisum 5.6/ 18.2 OS1_8sum 8.3/ 18.1 PAMFH 2.8/ 19.6 PAMFOS 4.3/ 20.3 CAllocsum 18.0/ 23.9
R-pre L=8 first-look SCR, IN1 minus OS1_8: 1.46 [ -0.31, 2.95] LODD 0.80.. 1.99
dwell K=8 Pd=0.5: SCR(INisum) - SCR(PAMFH) = 1.01 [ 0.36, 1.67] LODD 0.70.. 1.18
dwell K=8 Pd=0.5: SCR(OS1_8sum) - SCR(PAMFH) = 3.31 [ 0.79, 6.23] LODD 1.95.. 4.20
dwell K=8 Pd=0.5: SCR(PAMFOS) - SCR(PAMFH) = 0.80 [ 0.09, 1.69] LODD 0.46.. 1.13
dwell K=8 Pd=0.5: SCR(OS1_8sum) - SCR(INisum) = 2.30 [ 0.43, 4.59] LODD 1.26.. 3.08
dwell K=8 Pd=0.9: SCR(INisum) - SCR(PAMFH) = 3.95 [ 3.32, 4.43] LODD 3.57.. 4.17
dwell K=8 Pd=0.9: SCR(OS1_8sum) - SCR(PAMFH) = 6.46 [ 5.34, 7.67] LODD 6.00.. 6.68
dwell K=8 Pd=0.9: SCR(PAMFOS) - SCR(PAMFH) = 1.49 [ 0.92, 2.20] LODD 1.27.. 1.77
dwell K=8 Pd=0.9: SCR(OS1_8sum) - SCR(INisum) = 2.52 [ 1.26, 4.13] LODD 2.04.. 3.09
-- O7: OS exact law vs measured coverage (MAE per unit) --
OS1_8 1-a=0.9: law 0.9052 measured 0.9008 MAE 0.0214
OS1_8 1-a=0.99: law 0.9921 measured 0.9892 MAE 0.0051
OS1_12 1-a=0.9: law 0.8972 measured 0.9009 MAE 0.0121
OS1_12 1-a=0.99: law 0.9899 measured 0.9898 MAE 0.0032
OS4 1-a=0.9: law 0.9050 measured 0.9009 MAE 0.0125
OS4 1-a=0.99: law 0.9924 measured 0.9900 MAE 0.0035

VERDICTS Part 0
O1 spread 2.06 <= 2.0: FAILED
O2 OS1_8 - IN1 = 1.99 dB in [0, 1.5]: FAILED
O3 OS1_8 look-4 Pd 0.714 >= 0.5: OK
O4 R-pre L=8 IN1 - OS1_8 first-look SCR = 1.46 dB >= 3: FAILED
O5 fraction of INisum's Pd.9 gap to PAMF-H closed by OS1_8sum = -0.64 >= 0.5: FAILED
O6 PAMF-OS - PAMF-H at Pd .5 = 0.80 dB within 1: OK
O7 OS1_8 law MAE at 0.90 = 0.0214 <= 0.015: FAILED

=== Part J: JKU, 8 station x Rx units ===
-- run ref: chirp hit rate 0.000; zeroed fraction N/Z 0.0000/0.0492
[N] episodes non-hit Y by history hits {'O': 443520, '1-2': 0, '3-5': 0, '>=6': 0}; hit-Y 0
[N] IN1 Pfa by stratum 0.0101 nan nan nan | Y-hit nan | Pd010 0.499 nan nan nan | SCR50 10.0 nan nan nan
[N] OS1_8 Pfa by stratum 0.0102 nan nan nan | Y-hit nan | Pd010 0.452 nan nan nan | SCR50 10.8 nan nan nan
[N] CA16 Pfa by stratum 0.0103 nan nan nan | Y-hit nan | Pd010 0.550 nan nan nan | SCR50 9.3 nan nan nan
[N] OS16_8 Pfa by stratum 0.0103 nan nan nan | Y-hit nan | Pd010 0.496 nan nan nan | SCR50 10.1 nan nan nan
[Z] episodes non-hit Y by history hits {'O': 443520, '1-2': 0, '3-5': 0, '>=6': 0}; hit-Y 0
[Z] IN1 Pfa by stratum 0.0102 nan nan nan | Y-hit nan | Pd010 0.480 nan nan nan | SCR50 10.3 nan nan nan
[Z] OS1_8 Pfa by stratum 0.0101 nan nan nan | Y-hit nan | Pd010 0.435 nan nan nan | SCR50 11.0 nan nan nan
[Z] CA16 Pfa by stratum 0.0104 nan nan nan | Y-hit nan | Pd010 0.524 nan nan nan | SCR50 9.7 nan nan nan
[Z] OS16_8 Pfa by stratum 0.0106 nan nan nan | Y-hit nan | Pd010 0.472 nan nan nan | SCR50 10.4 nan nan nan
-- run A: chirp hit rate 0.464; zeroed fraction N/Z 0.0000/0.0761
[N] episodes non-hit Y by history hits {'O': 337680, '1-2': 140880, '3-5': 11520, '>=6': 227760}; hit-Y 626160
[N] IN1 Pfa by stratum 0.0098 0.0093 0.0070 0.0010 | Y-hit 0.0176 | Pd010 0.350 0.287 0.271 0.112 | SCR50 12.4 13.4 13.7 17.7
[N] OS1_8 Pfa by stratum 0.0101 0.0103 0.0080 0.0020 | Y-hit 0.0179 | Pd010 0.310 0.252 0.239 0.100 | SCR50 13.1 14.1 14.3 18.4
[N] CA16 Pfa by stratum 0.0107 0.0095 0.0071 0.0005 | Y-hit 0.0217 | Pd010 0.388 0.334 0.314 0.131 | SCR50 11.7 12.5 12.8 16.3
[N] OS16_8 Pfa by stratum 0.0122 0.0112 0.0105 0.0010 | Y-hit 0.0305 | Pd010 0.349 0.301 0.291 0.126 | SCR50 12.4 13.1 13.3 16.7
[Z] episodes non-hit Y by history hits {'O': 337680, '1-2': 140880, '3-5': 11520, '>=6': 227760}; hit-Y 626160
[Z] IN1 Pfa by stratum 0.0106 0.0101 0.0113 0.0103 | Y-hit 0.0107 | Pd010 0.467 0.455 0.452 0.429 | SCR50 10.5 10.7 10.7 11.2
[Z] OS1_8 Pfa by stratum 0.0105 0.0097 0.0102 0.0102 | Y-hit 0.0100 | Pd010 0.422 0.407 0.402 0.387 | SCR50 11.2 11.4 11.5 11.9
[Z] CA16 Pfa by stratum 0.0107 0.0102 0.0088 0.0110 | Y-hit 0.0104 | Pd010 0.494 0.490 0.489 0.494 | SCR50 10.1 10.2 10.2 10.1
[Z] OS16_8 Pfa by stratum 0.0108 0.0103 0.0088 0.0112 | Y-hit 0.0108 | Pd010 0.448 0.444 0.446 0.447 | SCR50 10.8 10.9 10.9 10.9
-- run C: chirp hit rate 0.052; zeroed fraction N/Z 0.0000/0.0712
[N] episodes non-hit Y by history hits {'O': 1023600, '1-2': 100080, '3-5': 112320, '>=6': 42960}; hit-Y 65040
[N] IN1 Pfa by stratum 0.0107 0.0132 0.0080 0.0043 | Y-hit 0.0288 | Pd010 0.464 0.311 0.243 0.218 | SCR50 10.6 13.2 14.4 15.0
[N] OS1_8 Pfa by stratum 0.0108 0.0147 0.0084 0.0056 | Y-hit 0.0268 | Pd010 0.419 0.285 0.221 0.187 | SCR50 11.3 13.8 15.1 15.8
[N] CA16 Pfa by stratum 0.0112 0.0123 0.0061 0.0033 | Y-hit 0.0434 | Pd010 0.515 0.377 0.310 0.266 | SCR50 9.8 10.2 10.0 13.7
[N] OS16_8 Pfa by stratum 0.0111 0.0147 0.0068 0.0042 | Y-hit 0.0397 | Pd010 0.466 0.345 0.275 0.233 | SCR50 10.6 12.5 13.7 14.4
[Z] episodes non-hit Y by history hits {'O': 1023600, '1-2': 100080, '3-5': 112320, '>=6': 42960}; hit-Y 65040
[Z] IN1 Pfa by stratum 0.0105 0.0100 0.0085 0.0113 | Y-hit 0.0106 | Pd010 0.471 0.438 0.411 0.431 | SCR50 10.5 11.0 11.4 11.1
[Z] OS1_8 Pfa by stratum 0.0105 0.0103 0.0091 0.0119 | Y-hit 0.0103 | Pd010 0.427 0.394 0.366 0.391 | SCR50 11.2 11.7 12.2 11.8
[Z] CA16 Pfa by stratum 0.0105 0.0102 0.0100 0.0123 | Y-hit 0.0105 | Pd010 0.514 0.495 0.480 0.494 | SCR50 9.8 10.1 10.3 10.1
[Z] OS16_8 Pfa by stratum 0.0104 0.0102 0.0099 0.0126 | Y-hit 0.0120 | Pd010 0.466 0.448 0.436 0.443 | SCR50 10.5 10.9 11.1 10.9

VERDICTS Part J
J1 run A >=6-hit stratum Pd010 drop vs ref: IN1 0.387 (>=0.2?), OS1_8 0.353 (<= half?) -> FAILED [n]=6 stratum: 227760]
J2 run A: IN1 hit-Y Pfa N 0.0176 -> Z 0.0107; reduction 0.39 (>=0.5?) -> FAILED
J2 run C: IN1 hit-Y Pfa N 0.0288 -> Z 0.0106; reduction 0.63 (>=0.5?) -> OK
J3 run A overall SCR50: Z+OS1_8 - Z+IN1 = 0.71 [0.66,0.76] dB (<= +0.5?) -> FAILED
overall SCR50 (non-hit Y, all strata): N-ref: IN1 10.0 OS1_8 10.8 CA16 9.3 OS16_8 10.1 | N-A: IN1 15.7 OS1_8 16.4 CA16 14.6 OS16_8 15.1 | N-C: IN1 11.4 OS1_8 12.1 CA16 10.5
OS16_8 11.3 | Z-ref: IN1 10.3 OS1_8 11.0 CA16 9.7 OS16_8 10.4 | Z-A: IN1 10.9 OS1_8 11.6 CA16 10.1 OS16_8 10.9 | Z-C: IN1 10.6 OS1_8 11.3 CA16 9.9 OS16_8 10.6
J4 ref: CA16 - IN1 SCR50 = -0.78 dB (>0?) -> FAILED

```

## 18 Audit of existing outcomes: conditional false-alarm rates, dependence, leave-one-day-out, fit diagnostics

Output of r10/audit\_existing.py (descriptive and sensitivity analyses requested in review; no new runs):

```

=== 1. IPIX false-alarm rate by texture quintile (confirmatory, m = 16, +/-1024 proxy) ===
-- design alpha = 0.1: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.04 0.98 1.00 0.99 0.92 | spread 1.1x | exceedances 1557 1466 1501 1483 1373 of 74980 episodes
IN4 1.00 0.99 0.97 1.00 0.97 | spread 1.0x | exceedances 1495 1474 1456 1501 1455 of 74980 episodes
NA4 0.84 0.96 1.06 1.02 1.02 | spread 1.3x | exceedances 1252 1436 1591 1540 1538 of 74980 episodes
U 0.22 0.44 0.71 1.12 2.56 | spread 11.5x | exceedances 325 662 1057 1680 3846 of 74980 episodes
RMS 1.88 1.25 0.88 0.60 0.43 | spread 4.4x | exceedances 2813 1876 1321 899 636 of 74980 episodes
MLP 1.82 1.17 0.84 0.61 0.50 | spread 3.6x | exceedances 2732 1761 1264 927 749 of 74980 episodes
MON 0.79 0.86 0.98 1.05 1.13 | spread 1.4x | exceedances 1177 1277 1464 1579 1689 of 74980 episodes
-- design alpha = 0.01: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.18 1.03 1.14 0.91 0.67 | spread 1.7x | exceedances 175 155 172 137 101 of 74980 episodes

```

```

IN4 1.09 0.99 0.93 0.99 0.93 | spread 1.2x | exceedances 163 149 141 150 140 of 74980 episodes
NA4 0.72 0.94 0.94 0.93 1.22 | spread 1.7x | exceedances 107 139 142 139 184 of 74980 episodes
U 0.05 0.18 0.21 0.60 4.68 | spread 87.5x | exceedances 8 27 32 89 710 of 74980 episodes
RMS 2.62 1.25 0.83 0.43 0.27 | spread 9.8x | exceedances 390 187 124 64 40 of 74980 episodes
MLP 2.35 1.13 0.81 0.45 0.48 | spread 5.2x | exceedances 353 167 121 67 71 of 74980 episodes
MON 0.50 0.60 0.92 0.83 1.14 | spread 2.3x | exceedances 74 90 139 125 171 of 74980 episodes
-- maximum quintile deviation: observed vs binomial reference vs dependence-aware (time-shift) reference, alpha = 0.1 --
IN1 observed 0.036 | binomial ref 0.029 | time-shift ref 0.033
IN4 observed 0.032 | binomial ref 0.029 | time-shift ref 0.033
NA4 observed 0.043 | binomial ref 0.029 | time-shift ref 0.039
U observed 0.165 | binomial ref 0.029 | time-shift ref 0.062

=== 2. Dependence of IN1 exceedance indicators (alpha = 0.1), mean over units ===
lag-1 (256 pulses) autocorrelation within bin: -0.007 (range -0.067 to 0.046)
correlation across bins at the same time: 0.045

=== 3. Leave-one-day-out range of headline effects (estimate on all 6 days; min..max over the 6 LODD fits) ===
NA-AR(4)/IN-AR(4) radius ratio (0.90): 0.891 LODD 0.884..0.895
IN1 quintile Pfa spread (alpha 0.01): 1.75x LODD 1.72..1.83x
single-pulse gain IN1 vs CA16 Pd=0.5: 6.96 dB LODD 5.97.. 8.16
single-pulse gain IN1 vs CALoc Pd=0.5: 9.81 dB LODD 8.94..11.54
single-pulse gain NA4 vs IN1 Pd=0.5: 1.07 dB LODD 1.03.. 1.11
single-pulse gain NA4 vs IN4 Pd=0.5: 0.66 dB LODD 0.58.. 0.75
single-pulse gain IN1 vs CA16 Pd=0.9: 7.62 dB LODD 6.59.. 9.01
single-pulse gain IN1 vs CALoc Pd=0.9: 8.97 dB LODD 8.10..10.62
single-pulse gain NA4 vs IN1 Pd=0.9: 2.19 dB LODD 1.88.. 2.39
single-pulse gain NA4 vs IN4 Pd=0.9: 1.46 dB LODD 1.43.. 1.52
eight-look dwell gain IN1 vs CALoc-sum Pd=0.5: 10.70 dB LODD 9.86..12.38
eight-look dwell gain NA4 vs CALoc-sum Pd=0.5: 11.74 dB LODD 11.01..13.00
per-day values of the single-pulse IN1 vs CALoc gain (Pd 0.5): 1993-11-07 3.5, 1993-11-08 8.4, 1993-11-09 13.8, 1993-11-10 13.3, 1993-11-11 12.2, 1993-11-18 11.4

=== 4. Fit diagnostics (all confirmatory units, training thirds) ===
|r| of IN-ARCP: median 0.960, max 0.980; clamp |r| <= 0.98 binding in 5 of 56 units
NA-AR(2) Nelder-Mead converged in 56 of 56 units; NA-AR(4) in 16 of 56 (median iterations 1500, cap 1500)
whole-unit time (all 13 procedures, GPU): median 71 s

=== 5. Second radar: Pfa/alpha by CNR class (mean over units with data), exact law, counts ===
class (dB) <-5 -5..0 0..5 5..10 10..20 >=20
test episodes 721095 265416 229845 68556 31870 338
IN1 a=0.1 1.03 1.03 1.00 0.83 0.38 0.00 units: 24 24 24 24 24 2
NA4 a=0.1 0.83 1.36 1.19 0.88 0.54 0.00 units: 24 24 24 24 24 2
CA16 a=0.1 1.07 1.05 0.96 0.58 0.15 0.00 units: 24 24 24 24 24 2
law a=0.1 1.04 1.00 0.95 0.86 0.68 0.34 (exact law, 3-decimal coverage from jku_summary)
IN1 a=0.01 1.07 1.06 0.95 0.52 0.08 0.00 units: 24 24 24 24 24 2
NA4 a=0.01 0.69 1.68 1.27 0.66 0.22 0.00 units: 24 24 24 24 24 2
CA16 a=0.01 1.14 1.05 0.78 0.19 0.02 0.00 units: 24 24 24 24 24 2
law a=0.01 1.10 1.00 0.90 0.70 0.50 0.20 (exact law, 3-decimal coverage from jku_summary)

```

Exploratory: sensitivity of NA-AR(4) to optimizer convergence (restart from the capped solution for up to 4,500 further Nelder–Mead iterations; `r10/explore_na4_convergence.py`):

```

units 56; converged at cap 1500: 16; converged after restart (+4500): 55
log-likelihood gain per training episode: median 3.13e-06, max 1.54e-04
1-alpha=0.9: radius ratio long/orig geo-mean 0.9989 (range 0.9790..1.0132); coverage orig 0.9019 long 0.9020
1-alpha=0.99: radius ratio long/orig geo-mean 0.9997 (range 0.9492..1.0327); coverage orig 0.9905 long 0.9906

```

## 19 Proofs of the whitened-innovation laws

Throughout,  $z \sim \mathcal{CN}(0, I)$  denotes whitened clutter innovations (unit variance after dividing by  $\sigma_e$ ),  $g \sim \mathcal{CN}(0, 1)$  the Swerling-1 amplitude, and  $u$  the whitened target signature, so that the whitened episode is  $z + gu \sim \mathcal{CN}(0, I + uu^H)$ .  $E$ ,  $E_{\pm}$  are standard exponentials and  $\Gamma_k$  a Gamma( $k, 1$ ) variable, all independent.

### 19.1 Proposition (post-onset rank-one law)

Write the detection event as  $F > 0$  with  $F = x^H M x$ ,  $x = z + gu$  and  $M = \text{diag}(-\beta I_m, 1)$ ,  $\beta = q^2/m$ . As in Appendix A of the paper,  $F \stackrel{d}{=} \sum_k \lambda_k E_k$  with  $\lambda_k$  the eigenvalues of  $C^{1/2} M C^{1/2}$ ,  $C = I + uu^H$ , which are those of  $M C = M + (Mu)u^H$ . For a rank-one update of a diagonal matrix,

$$\det(\lambda I - M - Muu^H) = \det(\lambda I - M) \left( 1 - \sum_i \frac{M_{ii} |u_i|^2}{\lambda - M_{ii}} \right) = (\lambda - 1)(\lambda + \beta)^m \left( 1 - \frac{\mathcal{E}_0}{\lambda - 1} + \frac{\beta \mathcal{E}_H}{\lambda + \beta} \right),$$

with  $\mathcal{E}_0 = |u_Y|^2$  and  $\mathcal{E}_H = \|u_H\|^2$ . Hence  $-\beta$  is an eigenvalue of multiplicity  $m - 1$  and the other two solve  $(\lambda - 1)(\lambda + \beta) - \mathcal{E}_0(\lambda + \beta) + \beta \mathcal{E}_H(\lambda - 1) = 0$ , which is the quadratic of the proposition. Their product is  $-\beta(1 + \mathcal{E}_0 + \mathcal{E}_H) < 0$ , so  $\lambda_+ > 0 > \lambda_-$ . Then

$$\mathbb{P}\{F > 0\} = \mathbb{P}\{\lambda_+ E_+ > |\lambda_-| E_- + \beta \Gamma_{m-1}\} = \mathbb{E} e^{-(|\lambda_-| E_- + \beta \Gamma_{m-1})/\lambda_+} = \frac{\lambda_+}{\lambda_+ + |\lambda_-|} \left( 1 + \frac{\beta}{\lambda_+} \right)^{-(m-1)}.$$

For  $\mathcal{E}_H = 0$  the roots are  $1 + \mathcal{E}_0$  and  $-\beta$ , and the expression reduces to  $(1 + \beta/(1 + \mathcal{E}_0))^{-m}$ . The signature of a persistent target follows from the whitened amplitudes in (4): the onset innovation carries  $S_w$ , every later one  $S_w |d|^2$ .

### 19.2 Corollary (visibility horizon)

With  $\mathcal{E}_0 = S_w D$  and  $\mathcal{E}_H = S_w(1 + (\ell - 1)D)$ ,  $D = |d(\omega)|^2$ , the linear coefficient is  $B = 1 - \beta + S_w \chi$ ,  $\chi = D - \beta\{1 + (\ell - 1)D\}$ , and  $\lambda_{\pm} = \{B \pm (B^2 + 4\beta(1 + \mathcal{E}_0 + \mathcal{E}_H))^{1/2}\}/2$  with  $1 + \mathcal{E}_0 + \mathcal{E}_H = O(S_w)$ . If  $\chi > 0$ , then  $\lambda_+ \sim S_w \chi \rightarrow \infty$  while  $\lambda_-$  stays bounded, so  $P_d \rightarrow 1$ . If  $\chi < 0$ , then  $\lambda_+ \rightarrow \beta(1 + \ell D)/|\chi|$  stays bounded while  $\lambda_- \sim S_w \chi \rightarrow -\infty$ , so  $\lambda_+ / (\lambda_+ - \lambda_-) \rightarrow 0$  and  $P_d \rightarrow 0$ . The sign change  $\chi = 0$  is  $\ell = 1 + 1/\beta - 1/D$ .

### 19.3 Order-statistic scale

Conditional on  $G = |g|^2$ , the history powers  $|z_j + gu_j|^2$  are independent, exponential for clean innovations and non-central (with  $2|z_j + gu_j|^2 \sim \chi_2^2(2G|u_j|^2)$ ) for contaminated ones. The distribution function of their  $k$ -th smallest value  $X_{(k)}$  is the Poisson-binomial probability that at least  $k$  are below  $x$ , and  $P_d = \mathbb{E}_G \int \mathbb{P}\{|z_Y + gu_Y|^2 > q^2 x \mid G\} dF_{X_{(k)}|G}(x)$ . The code evaluates this by midpoint quadrature in the distribution function of  $G$  and a dynamic program for the Poisson-binomial law. If  $\ell \leq m - k$  contaminated powers diverge,  $X_{(k)}$  is the  $k$ -th smallest of the  $m - \ell$  clean ones, which is independent of  $z_Y$ , and the OS-CFAR law with  $m - \ell$  cells gives (7). If  $\ell > m - k$ ,  $X_{(k)}$  diverges with the target while  $|z_Y + gu_Y|^2/X_{(k)}$  tends to a constant below any practical threshold, as in Proposition 3.

### 19.4 Dwell integration

Under  $H_0$  the event is  $\Gamma_K > \tau\Gamma_m$ , and conditioning on  $\Gamma_m$  with the Poisson form of the Gamma tail gives  $\sum_{i \leq K} \mathbb{E}[e^{-\tau\Gamma_m} (\tau\Gamma_m)^i / i!] = \sum_{i \leq K} \binom{m+i-1}{i} \tau^i (1+\tau)^{-(m+i)}$ . With a Swerling-1 target the dwell form has one eigenvalue  $\delta = 1 + \mathcal{E}_D$  and  $K - 1$  unit ones. For  $v > 0$ ,

$$\mathbb{P}\{\delta E + \Gamma_{K-1} > v\} = \mathbb{P}\{\Gamma_{K-1} > v\} + e^{-v/\delta} (1 - 1/\delta)^{-(K-1)} \mathbb{P}\{\Gamma_{K-1} \leq v(1 - 1/\delta)\}.$$

Averaging over  $v = \tau\Gamma_m$  uses  $\mathbb{E}e^{-s\Gamma_m} = (1+s)^{-m}$  and  $\mathbb{E}[e^{-s\Gamma_m} \mathbb{P}\{\Gamma_{K-1} > \tau\Gamma_m \mid \Gamma_m\}] = \sum_{i \leq K-2} \binom{m+i-1}{i} \tau^i (1+s+\tau')^{-(m+i)}$  with  $s+\tau' = \tau$ , which gives (9).

### 19.5 Self-normalized whitened Doppler

With a known on-grid Doppler, the whitened dwell vector splits into its DFT coefficient on the target bin,  $|\cdot|^2 \sim (1 + NS_w |d|^2)E$ , and the orthogonal remainder  $\Gamma_{N-1}$ . The statistic  $X/(X + \Gamma_{N-1})$  exceeds  $t$  iff  $X(1-t) > t\Gamma_{N-1}$ , which has probability  $\{1 + t/((1-t)(1 + NS_w |d|^2))\}^{-(N-1)}$ ; at  $S_w = 0$  this is  $(1-t)^{N-1}$ . For the maximum over the  $N$  bins, let  $I_j$  be independent exponentials with means  $\mu_j$  and  $J$  a set of bins with  $|J|t < 1$ . Substituting  $x_j = tT + y_j$  for  $j \in J$ , where  $T = \sum_i I_i$ , the map has Jacobian  $1/(1 - |J|t)$  and turns the exponent into a sum of independent exponentials, so

$$\mathbb{P}\{I_j > tT \ \forall j \in J\} = \frac{1}{1 - |J|t} \prod_{i=1}^N \frac{1}{1 + \theta_J \mu_i}, \quad \theta_J = \frac{t \sum_{j \in J} \mu_j^{-1}}{1 - |J|t},$$

and zero if  $|J|t \geq 1$ . With all  $\mu_i = 1$  this is  $(1 - |J|t)^{N-1}$ , and inclusion-exclusion gives Fisher's law  $\sum_r (-1)^{r-1} \binom{N}{r} (1 - rt)^{N-1}$  (R. A. Fisher, Proc. R. Soc. Lond. A 125, 1929). With one bin of mean  $1 + NS_w |d|^2$ , inclusion-exclusion over sets that do or do not contain it gives  $P_d$  (`laws.pd_snd_max`).

### 19.6 Theorem (certified integration)

*Domination.* Let  $\mathcal{J}_H$  and  $\mathcal{J}_D$  be the corrupted history and dwell innovations. The  $k_H$ -th smallest of the history powers is a nondecreasing function of each power, so over all values of the corrupted entries it is smallest when they are zero; hence  $s_{k_H}^2 \geq \underline{s}^2$ , the value computed with those entries set to zero. Every uncorrupted dwell term is a clean power divided by  $s_{k_H}^2 \geq \underline{s}^2$ , and every clipped term is at most  $\kappa$ . Therefore  $T \leq W$  for all interference values. For binary integration, an uncorrupted exceedance of the actual statistic is an exceedance at  $\underline{s}^2$ , and a corrupted term counts at most one. Adding elements to  $\mathcal{J}$  can only lower  $\underline{s}^2$  and replace terms by their maximum, so  $W$  is nondecreasing in  $\mathcal{J}$ . *Calibration.* Let the test hit set  $\mathcal{J}$  be independent of the clutter and let  $\tilde{\mathcal{J}} \sim \mathcal{D}$  with  $\mathcal{J} \subseteq \tilde{\mathcal{J}}$  (a coupling that exists when  $\mathcal{D}$  dominates the law of  $\mathcal{J}$  in the inclusion order, by Strassen's theorem; a bound on the hit rate alone does not suffice). Then  $T \leq W(X, \mathcal{J}) \leq W(X, \tilde{\mathcal{J}})$ . The pairs  $(X_i, \mathcal{J}_i)$  and  $(X, \tilde{\mathcal{J}})$  are exchangeable, so the split-conformal rank argument gives  $\mathbb{P}\{W(X, \tilde{\mathcal{J}}) > \hat{q}\} \leq \alpha$ . No property of the interference values is used, so the bound holds for adversarial amplitudes chosen after seeing the clutter, as long as the hit positions are independent of the clutter. *Trimmed sums.* If a corrupted dwell term can take any value, the trimmed sum over the  $K - t$  smallest terms is unbounded once more than  $t$  dwell terms are corrupted, which a Bernoulli hit model allows with positive probability. The conformal quantile of the worst case is then infinite whenever that probability exceeds  $\alpha$ .

### 19.7 Conformal calibration size and loss

For continuous i.i.d. scores the  $k_\alpha$ -th order statistic of  $n$  calibration scores, mapped through the score distribution function, is  $\text{Beta}(k_\alpha, n + 1 - k_\alpha)$ ; the conditional false-alarm probability is one minus it, hence  $\text{Beta}(n + 1 - k_\alpha, k_\alpha)$ . Because  $k_\alpha$  is a ceiling,  $\mathbb{P}\{P_{fa} > 2\alpha\}$  is not monotone in  $n$ ; an exhaustive scan (`laws.n_required`) gives the smallest  $n$  and the  $n$  beyond which the condition always holds. With ties the rank argument still gives  $\mathbb{P}\{T > \hat{q}\} \leq \alpha$ , but the Beta law no longer applies. For  $P_d = P_{fa}^\gamma$ ,  $\mathbb{E}U^\gamma = B(n + 1 - k_\alpha + \gamma, k_\alpha)/B(n + 1 - k_\alpha, k_\alpha)$ , with  $B$  the Beta function.

## 20 R12: Monte Carlo checks, full results and verdicts

Frozen protocol `r12/PROTOCOL_R12.md` (sha256 a32706e7...); expectations and verdicts in `r12/RESULTS_R12.md` (12 of 18 met). Monte Carlo checks of most closed forms in Section IV (`r12/verify_laws.py`, 400,000 replications); the guarded law and an independent re-check are in Section 22. The V7 rows marked “law assumes Gaussian” test the Kraut–Scharf law with compound-Gaussian secondary data, where it fails 2.8–3.4-fold; they are excluded from the  $|z|$  summary because that law does not claim to hold there. The summary’s 65 checks are the rows with a  $z$ -score, less every row with a bracketed label (the two rows just named, and the four rows of the asymptotic fixed-point law, whose  $|z| < 1.6$ ); the V8 Beta-law rows and the V9 certificate rows report agreement or violation rates, not  $z$ -scores. The printed maximum 2.91 is rounded from 2.9125; each of the 65 lies within 3 standard errors. The calibration sizes printed under V8 came from a faulty search and are superseded by `r12/calibration_size.txt` (149, 1,497 and 14,978; every  $n$  beyond 313, 3,146 and 31,477):

```

V1-V3 IN-ARCP per-look score: onset, post-onset rank-one law, visibility horizon
Pd clutter Doppler, SCR 10 dB, look 0 theory 0.93148 MC 0.93112 z -0.46
Pd clutter Doppler, SCR 10 dB, look 1 theory 0.00057 MC 0.00062 z +0.71
Pd clutter Doppler, SCR 10 dB, look 2 theory 0.00056 MC 0.00057 z +0.08
Pd clutter Doppler, SCR 10 dB, look 4 theory 0.00056 MC 0.00051 z -0.65
Pd clutter Doppler, SCR 10 dB, look 8 theory 0.00055 MC 0.00062 z +0.98
visibility horizon (clutter): l* = -200.08
Pd clutter Doppler, SCR 25 dB, look 0 theory 0.99772 MC 0.99770 z -0.16
Pd clutter Doppler, SCR 25 dB, look 1 theory 0.00002 MC 0.00001 z -0.65
Pd clutter Doppler, SCR 25 dB, look 2 theory 0.00002 MC 0.00002 z +0.08
Pd clutter Doppler, SCR 25 dB, look 4 theory 0.00002 MC 0.00002 z +0.10
Pd clutter Doppler, SCR 25 dB, look 8 theory 0.00002 MC 0.00002 z +0.13
visibility horizon (clutter): l* = -200.08
Pd opposite Doppler, SCR 10 dB, look 0 theory 0.93148 MC 0.93064 z -1.06
Pd opposite Doppler, SCR 10 dB, look 1 theory 0.97857 MC 0.97844 z -0.28
Pd opposite Doppler, SCR 10 dB, look 2 theory 0.96169 MC 0.96205 z +0.59
Pd opposite Doppler, SCR 10 dB, look 4 theory 0.15389 MC 0.15473 z +0.74
Pd opposite Doppler, SCR 10 dB, look 8 theory 0.00045 MC 0.00043 z -0.25
visibility horizon (opposite): l* = 3.73
Pd opposite Doppler, SCR 25 dB, look 0 theory 0.99772 MC 0.99778 z +0.37
Pd opposite Doppler, SCR 25 dB, look 1 theory 0.99931 MC 0.99939 z +0.95
Pd opposite Doppler, SCR 25 dB, look 2 theory 0.99875 MC 0.99890 z +1.37
Pd opposite Doppler, SCR 25 dB, look 4 theory 0.01381 MC 0.01368 z -0.36
Pd opposite Doppler, SCR 25 dB, look 8 theory 0.00001 MC 0.00001 z -0.36
visibility horizon (opposite): l* = 3.73
random Doppler (closed form averaged over w) at 10 dB:
Pd random Doppler, 10 dB, look 0 theory 0.93148 MC 0.93113 z -0.63
Pd random Doppler, 10 dB, look 1 theory 0.74117 MC 0.74048 z -0.69
Pd random Doppler, 10 dB, look 2 theory 0.67605 MC 0.67458 z -1.41
Pd random Doppler, 10 dB, look 4 theory 0.08049 MC 0.08077 z +0.45
Pd random Doppler, 10 dB, look 8 theory 0.00080 MC 0.00081 z +0.31

V4 order-statistic score (k = 8, m = 16): Pfa law and post-onset law
Pfa OS (k=8) theory 0.01000 MC 0.01003 z +0.17
Pd OS exact, quarter Doppler, 10 dB, look 0 theory 0.92017 MC 0.92008 z -0.10
Pd OS exact, quarter Doppler, 10 dB, look 2 theory 0.94773 MC 0.94665 z -1.53
strong-target limit: 0.9471
Pd OS exact, quarter Doppler, 10 dB, look 6 theory 0.90930 MC 0.90896 z -0.38
strong-target limit: 0.9079
Pd OS exact, quarter Doppler, 10 dB, look 8 theory 0.83542 MC 0.83647 z +0.89
strong-target limit: 0.8339
Pd OS exact, quarter Doppler, 10 dB, look 9 theory 0.00136 MC 0.00107 z -2.48
strong-target limit: 0.0000
Pd OS exact, quarter Doppler, 25 dB, look 0 theory 0.99735 MC 0.99717 z -1.11
Pd OS exact, quarter Doppler, 25 dB, look 2 theory 0.99834 MC 0.99796 z -2.91
strong-target limit: 0.9983
Pd OS exact, quarter Doppler, 25 dB, look 6 theory 0.99695 MC 0.99662 z -1.88
strong-target limit: 0.9969
Pd OS exact, quarter Doppler, 25 dB, look 8 theory 0.99418 MC 0.99366 z -2.16
strong-target limit: 0.9941
Pd OS exact, quarter Doppler, 25 dB, look 9 theory 0.00004 MC 0.00005 z +0.25
strong-target limit: 0.0000

V5 dwell integration with a shared history (m = 16): Pfa and Pd laws
Pfa dwell K=2 theory 0.01000 MC 0.00996 z -0.16
Pd dwell K=2, 0 dB, quarter Doppler theory 0.73521 MC 0.73494 z -0.27
Pd dwell K=2, 10 dB, quarter Doppler theory 0.96800 MC 0.96865 z +1.65
Pfa dwell K=4 theory 0.01000 MC 0.00982 z -0.81
Pd dwell K=4, 0 dB, quarter Doppler theory 0.82829 MC 0.82673 z -1.84
Pd dwell K=4, 10 dB, quarter Doppler theory 0.98077 MC 0.98072 z -0.14
Pfa dwell K=8 theory 0.01000 MC 0.01006 z +0.27
Pd dwell K=8, 0 dB, quarter Doppler theory 0.87697 MC 0.87728 z +0.42
Pd dwell K=8, 10 dB, quarter Doppler theory 0.98671 MC 0.98684 z +0.51
power integration over 8 pulses at |rho| = 0.93: effective looks K_eff = 1.41

V6 self-normalized whitened Doppler (P-ANMF): Fisher's g and the NMF law
Pfa known bin K=8 theory 0.01000 MC 0.00995 z -0.16
Pfa max (Fisher g) K=8 theory 0.01000 MC 0.01001 z +0.03
Pd known bin K=8, -10 dB theory 0.74050 MC 0.74046 z -0.03
Pd max K=8, -10 dB theory 0.60215 MC 0.60083 z -0.85
Pd known bin K=8, 0 dB theory 0.96852 MC 0.96817 z -0.64
Pd max K=8, 0 dB theory 0.94670 MC 0.94600 z -0.99
Pfa known bin K=16 theory 0.01000 MC 0.00967 z -1.05
Pfa max (Fisher g) K=16 theory 0.01000 MC 0.00983 z -0.54
Pd known bin K=16, -10 dB theory 0.75351 MC 0.75314 z -0.27
Pd max K=16, -10 dB theory 0.60966 MC 0.61132 z +1.07
Pd known bin K=16, 0 dB theory 0.97047 MC 0.97081 z +0.63
Pd max K=16, 0 dB theory 0.94856 MC 0.94997 z +2.01
Pfa known bin K=64 theory 0.01000 MC 0.01024 z +0.38
Pfa max (Fisher g) K=64 theory 0.01000 MC 0.01056 z +0.89
Pd known bin K=64, -10 dB theory 0.66765 MC 0.66292 z -1.59
Pd max K=64, -10 dB theory 0.45636 MC 0.45604 z -0.10
Pd known bin K=64, 0 dB theory 0.95709 MC 0.95716 z +0.06
Pd max K=64, 0 dB theory 0.91784 MC 0.91952 z +0.97

V7 ANMF with the sample covariance (Kraut-Scharf law), Gaussian and compound-Gaussian data
Pfa ANMF-SCM N=8, Ks=24, Gaussian theory 0.01000 MC 0.00990 z -0.14
Pfa ANMF-FP N=8, Ks=24, Gaussian [asymptotic law] theory 0.01000 MC 0.00890 z -1.56
Pfa ANMF-SCM N=8, Ks=24, CG secondary+test [law assumes Gaussian] theory 0.01000 MC 0.03435 z +34.61
Pfa ANMF-FP N=8, Ks=24, CG secondary+test [asymptotic law] theory 0.01000 MC 0.00920 z -1.14
Pfa ANMF-SCM N=8, Ks=40, Gaussian theory 0.01000 MC 0.00980 z -0.28
Pfa ANMF-FP N=8, Ks=40, Gaussian [asymptotic law] theory 0.01000 MC 0.00955 z -0.64
Pfa ANMF-SCM N=8, Ks=40, CG secondary+test [law assumes Gaussian] theory 0.01000 MC 0.02795 z +25.51
Pfa ANMF-FP N=8, Ks=40, CG secondary+test [asymptotic law] theory 0.01000 MC 0.01105 z +1.49

```

```

V8 conformal calibration: conditional Pfa ~ Beta(n+1-k, k); expected Pd
n=1000, alpha=0.01: mean Pfa theory 0.00999 MC 0.01000; P(Pfa > 2 alpha) theory 0.0047 MC 0.0049
E[Pd] at SNR 10 dB (Pd = Pfa^(1/(1+SNR))): theory 0.65515 MC 0.65521
n=2000, alpha=0.001: mean Pfa theory 0.00100 MC 0.00100; P(Pfa > 2 alpha) theory 0.0914 MC 0.0909
E[Pd] at SNR 10 dB (Pd = Pfa^(1/(1+SNR))): theory 0.52206 MC 0.52211
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.01: n >= 153
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.001: n >= 1559
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.0001: n >= 15520

V9 certified integration under pulsed interference (m = 16, K = 8, median history, AR(1))
q: conformal quantile of the worst case over amplitudes, on clean episodes with hit positions drawn at
the design rate p. Test hits at rate p with Gaussian amplitudes (JNR re local power) or exact cancellation.
clip p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0106, clean-thr 0.0106, CA-NCI 0.0101; Pd(10 dB) certified 0.978, clean-thr 0.978
clip p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0034, clean-thr 0.0169, CA-NCI 0.1324; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0032, clean-thr 0.0167, CA-NCI 0.1170; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.02 cancelling : violations 0e+00; Pfa certified 0.0029, clean-thr 0.0144, CA-NCI 0.0528; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0316, CA-NCI 0.2266; Pd(10 dB) certified 0.938, clean-thr 0.975
clip p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0297, CA-NCI 0.2016; Pd(10 dB) certified 0.933, clean-thr 0.975
clip p=0.05 cancelling : violations 0e+00; Pfa certified 0.0002, clean-thr 0.0218, CA-NCI 0.0884; Pd(10 dB) certified 0.936, clean-thr 0.976
trim p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0103, clean-thr 0.0103, CA-NCI 0.0096; Pd(10 dB) certified 0.978, clean-thr 0.978
trim p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0286, CA-NCI 0.1332; Pd(10 dB) certified 0.000, clean-thr 0.977
trim p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0242, CA-NCI 0.1175; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.02 cancelling : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0166, CA-NCI 0.0529; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0821, CA-NCI 0.2260; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0659, CA-NCI 0.2016; Pd(10 dB) certified 0.000, clean-thr 0.975
trim p=0.05 cancelling : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0297, CA-NCI 0.0880; Pd(10 dB) certified 0.000, clean-thr 0.975
bin p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0043, clean-thr 0.0100; Pd(10 dB) certified 0.967, clean-thr 0.967
bin p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0028, clean-thr 0.0135, CA-NCI 0.1313; Pd(10 dB) certified 0.959, clean-thr 0.968
bin p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0026, clean-thr 0.0127, CA-NCI 0.1174; Pd(10 dB) certified 0.958, clean-thr 0.967
bin p=0.02 cancelling : violations 0e+00; Pfa certified 0.0020, clean-thr 0.0090, CA-NCI 0.0522; Pd(10 dB) certified 0.958, clean-thr 0.967
bin p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0464, CA-NCI 0.2271; Pd(10 dB) certified 0.928, clean-thr 0.965
bin p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0002, clean-thr 0.0396, CA-NCI 0.2045; Pd(10 dB) certified 0.927, clean-thr 0.965
bin p=0.05 cancelling : violations 0e+00; Pfa certified 0.0001, clean-thr 0.0200, CA-NCI 0.0891; Pd(10 dB) certified 0.929, clean-thr 0.965

```

Summary: 65 probability checks, max |z| = 2.91, 0 with |z| > 3; runtime 274 s

IPIX parts L (low false-alarm calibration), A (dwell detection with ANMF baselines), T (real target) and I (injected pulsed interference), with 95% day-cluster intervals (r12/analyze\_r12.py):

R12 IPIX analysis: 56 units

Part L: pooled Pfa / alpha [95% day-cluster interval]; texture-quintile spread (max/min)

```

alpha = 0.01
ANMF-FP(w) conf 1.04 [0.98, 1.07] spread 2.6
ANMF-FP(w) anal 1.62 [1.39, 1.82] spread 2.5
ANMF-SCM(w) conf 1.06 [0.99, 1.12] spread 2.3
ANMF-SCM(w) anal 2.02 [1.69, 2.28] spread 2.2
CA16 conf 1.00 [0.99, 1.01] spread 1.1
CA16 anal 0.80 [0.71, 0.85] spread 1.0
Cl1p-OS1 conf 1.05 [0.99, 1.08] spread 2.0
Cl1p-OS1 anal 4.84 [3.72, 6.72] spread 1.2
D-IN4 conf 1.02 [0.98, 1.07] spread 1.7
D-IN4 anal 2.02 [1.73, 2.27] spread 1.6
IN1 conf 1.01 [0.99, 1.02] spread 1.3
IN1 anal 0.96 [0.92, 1.02] spread 1.3
IN1 boot 1.63 [1.47, 1.75] spread 1.3
IN4 conf 1.00 [0.99, 1.01] spread 1.1
IN4 anal 1.03 [1.01, 1.05] spread 1.1
NA4 conf 1.00 [0.95, 1.08] spread 1.9
NA4 anal 1.56 [1.40, 1.70] spread 1.6
OS1_8 conf 1.04 [1.01, 1.08] spread 1.7
OS1_8 anal 2.15 [1.06, 3.57] spread 2.1
OSraw conf 1.01 [0.96, 1.04] spread 1.1
OSraw anal 2.07 [1.47, 2.55] spread 1.1
P-ANMF conf 1.04 [0.95, 1.11] spread 4.4
P-ANMF anal 4.36 [3.58, 5.29] spread 2.3
P-ANMF(bin) conf 1.10 [1.01, 1.19] spread 1.5
P-ANMF(bin) anal 1.78 [0.73, 2.61] spread 2.3
PAMF-H(w) conf 0.99 [0.96, 1.03] spread 2.2
PAMF-H(w) anal 1.97 [1.06, 2.70] spread 2.2

alpha = 0.001
ANMF-FP(w) conf 1.05 [0.96, 1.12] spread 2.1
ANMF-FP(w) anal 2.66 [1.76, 3.48] spread 2.2
ANMF-SCM(w) conf 1.04 [0.94, 1.14] spread 1.8
ANMF-SCM(w) anal 3.57 [2.38, 4.54] spread 2.0
CA16 conf 1.00 [0.94, 1.06] spread 1.1
CA16 anal 0.56 [0.41, 0.63] spread 1.3
Cl1p-OS1 conf 0.67 [0.34, 0.95] spread 12.1
Cl1p-OS1 anal 17.19 [11.68, 25.51] spread 1.3
D-IN4 conf 1.05 [0.86, 1.20] spread 1.8
D-IN4 anal 4.35 [3.16, 5.33] spread 2.1
IN1 conf 1.03 [1.01, 1.04] spread 1.5
IN1 anal 1.02 [0.91, 1.12] spread 1.4
IN1 boot 2.43 [2.15, 2.67] spread 1.4
IN4 conf 1.02 [0.96, 1.08] spread 1.1
IN4 anal 1.19 [1.01, 1.31] spread 1.1
NA4 conf 1.05 [1.00, 1.08] spread 3.0
NA4 anal 2.79 [2.38, 3.24] spread 2.3
OS1_8 conf 1.02 [0.89, 1.13] spread 1.9
OS1_8 anal 8.34 [2.00, 15.87] spread 4.0
OSraw conf 0.98 [0.91, 1.02] spread 1.9
OSraw anal 5.56 [3.15, 7.36] spread 1.4
P-ANMF conf 1.04 [0.90, 1.16] spread 9.9
P-ANMF anal 13.06 [10.39, 17.19] spread 4.5
P-ANMF(bin) conf 1.14 [0.95, 1.26] spread 1.7
P-ANMF(bin) anal 3.05 [1.01, 5.06] spread 2.4
PAMF-H(w) conf 1.00 [0.89, 1.08] spread 2.1
PAMF-H(w) anal 3.66 [1.66, 5.60] spread 2.4

alpha = 0.0001
ANMF-FP(w) conf 0.82 [0.67, 0.95] spread 2.0
ANMF-FP(w) anal 6.01 [2.87, 9.15] spread 1.8
ANMF-SCM(w) conf 0.85 [0.75, 1.01] spread 3.2
ANMF-SCM(w) anal 8.07 [3.98, 12.11] spread 1.8
CA16 conf 0.75 [0.68, 0.90] spread 1.3
CA16 anal 0.49 [0.26, 0.70] spread 1.6
Cl1p-OS1 conf 0.03 [0.00, 0.11] spread 6272109.2
Cl1p-OS1 anal 59.89 [40.97, 85.35] spread 1.5
D-IN4 conf 1.02 [0.59, 1.31] spread 2.2
D-IN4 anal 14.25 [7.84, 19.14] spread 2.0
IN1 conf 0.73 [0.54, 0.99] spread 2.7
IN1 anal 1.92 [1.26, 2.61] spread 2.0
IN1 boot 4.27 [3.44, 4.74] spread 1.3
IN4 conf 0.86 [0.64, 1.11] spread 1.9
IN4 anal 2.73 [1.48, 3.67] spread 1.3

```

|             |      |       |                |        |      |
|-------------|------|-------|----------------|--------|------|
| NA4         | conf | 1.04  | [0.84, 1.19]   | spread | 2.9  |
| NA4         | anal | 7.46  | [4.55, 9.73]   | spread | 3.2  |
| OS1_8       | conf | 0.92  | [0.76, 1.14]   | spread | 3.0  |
| OS1_8       | anal | 45.91 | [7.96, 90.46]  | spread | 7.1  |
| OSraw       | conf | 0.75  | [0.57, 1.07]   | spread | 3.3  |
| OSraw       | anal | 18.33 | [8.92, 25.59]  | spread | 1.8  |
| P-ANMF      | conf | 1.47  | [0.75, 2.57]   | spread | 23.2 |
| P-ANMF      | anal | 49.61 | [35.38, 73.34] | spread | 7.0  |
| P-ANMF(bin) | conf | 1.10  | [0.79, 1.31]   | spread | 2.0  |
| P-ANMF(bin) | anal | 5.47  | [1.35, 9.99]   | spread | 2.0  |
| PAMF-H(w)   | conf | 0.89  | [0.55, 1.22]   | spread | 4.0  |
| PAMF-H(w)   | anal | 9.88  | [4.13, 15.83]  | spread | 2.4  |

L1 conformal within band: False; offenders: [['Clip-OS1', 0.001, np.float64(0.67)], ['Clip-OS1', 0.0001, np.float64(0.03)]]  
L2 analytic whitened rules > 2x at 1e-4: ['IN4', 'OS1\_8', 'NA4', 'D-IN4', 'PAMF-H(w)', 'P-ANMF(bin)', 'P-ANMF', 'Clip-OS1']  
L3 False; L4 False; L5 True; L6 True

Part A: SCR (dB) for Pd = 0.5 / 0.9 from pooled curves (alpha = 0.01 and 0.001), measured Pfa

|       |           |       |       |       |       |     |        |
|-------|-----------|-------|-------|-------|-------|-----|--------|
| 0.01  | ANMF-FP   | SCR50 | 0.36  | SCR90 | 20.26 | Pfa | 0.0101 |
| 0.01  | ANMF-SCM  | SCR50 | -0.18 | SCR90 | 20.27 | Pfa | 0.0099 |
| 0.01  | Bin-OS1   | SCR50 | -1.20 | SCR90 | 14.49 | Pfa | 0.0065 |
| 0.01  | Clip-OS1  | SCR50 | -2.32 | SCR90 | 12.99 | Pfa | 0.0102 |
| 0.01  | D-IN1     | SCR50 | -4.36 | SCR90 | 8.34  | Pfa | 0.0103 |
| 0.01  | D-IN4     | SCR50 | -5.00 | SCR90 | 7.05  | Pfa | 0.0097 |
| 0.01  | Exc-D-IN1 | SCR50 | nan   | SCR90 | nan   | Pfa | 0.0099 |
| 0.01  | IN1look   | SCR50 | -1.22 | SCR90 | 10.98 | Pfa | 0.0099 |
| 0.01  | OSraw     | SCR50 | 9.41  | SCR90 | 19.46 | Pfa | 0.0108 |
| 0.01  | P-ANMF    | SCR50 | 0.20  | SCR90 | nan   | Pfa | 0.0102 |
| 0.01  | PAMF-H    | SCR50 | -5.00 | SCR90 | 5.13  | Pfa | 0.0096 |
| 0.001 | ANMF-FP   | SCR50 | 9.77  | SCR90 | nan   | Pfa | 0.0010 |
| 0.001 | ANMF-SCM  | SCR50 | 9.53  | SCR90 | nan   | Pfa | 0.0009 |
| 0.001 | Bin-OS1   | SCR50 | nan   | SCR90 | nan   | Pfa | 0.0002 |
| 0.001 | Clip-OS1  | SCR50 | nan   | SCR90 | nan   | Pfa | 0.0004 |
| 0.001 | D-IN1     | SCR50 | -0.14 | SCR90 | 12.44 | Pfa | 0.0009 |
| 0.001 | D-IN4     | SCR50 | -1.65 | SCR90 | 12.52 | Pfa | 0.0007 |
| 0.001 | Exc-D-IN1 | SCR50 | nan   | SCR90 | nan   | Pfa | 0.0007 |
| 0.001 | IN1look   | SCR50 | 1.86  | SCR90 | 14.12 | Pfa | 0.0011 |
| 0.001 | OSraw     | SCR50 | 14.35 | SCR90 | 24.54 | Pfa | 0.0007 |
| 0.001 | P-ANMF    | SCR50 | nan   | SCR90 | nan   | Pfa | 0.0011 |
| 0.001 | PAMF-H    | SCR50 | -4.25 | SCR90 | 9.88  | Pfa | 0.0010 |

A1 ANMF-FP minus PAMF-H: 5.36 dB [1.93, 8.83] (expect >= 1)  
A2 Clip-OS1 minus D-IN1: 2.04 dB [0.00, 2.90] (expect <= 1.5)  
A3 OSraw minus IN1 (first look): 10.62 dB [7.05, 12.69] (expect >= 5)

Part T: real target cell. Mean Pd over units (clutter Pfa), alpha = 1e-3 [1e-2]

|         |      |        |                       |            |
|---------|------|--------|-----------------------|------------|
| P-ANMF  | conf | N=8    | Pd 0.217 (Pfa 0.0011) | [Pd 0.352] |
| P-ANMF  | conf | N=16   | Pd 0.224 (Pfa 0.0013) | [Pd 0.362] |
| P-ANMF  | conf | N=32   | Pd 0.225 (Pfa 0.0013) | [Pd 0.388] |
| P-ANMF  | conf | N=64   | Pd 0.297 (Pfa 0.0013) | [Pd 0.469] |
| P-ANMF  | conf | N=128  | Pd 0.372 (Pfa 0.0012) | [Pd 0.538] |
| P-ANMF  | conf | N=256  | Pd 0.399 (Pfa 0.0015) | [Pd 0.573] |
| P-ANMF  | conf | N=512  | Pd 0.367 (Pfa 0.0020) | [Pd 0.567] |
| P-ANMF  | conf | N=1024 | Pd 0.409 (Pfa 0.0026) | [Pd 0.581] |
| P-ANMF  | anal | N=8    | Pd 0.366 (Pfa 0.0127) | [Pd 0.443] |
| P-ANMF  | anal | N=16   | Pd 0.447 (Pfa 0.0319) | [Pd 0.528] |
| P-ANMF  | anal | N=32   | Pd 0.563 (Pfa 0.0689) | [Pd 0.629] |
| P-ANMF  | anal | N=64   | Pd 0.669 (Pfa 0.1206) | [Pd 0.724] |
| P-ANMF  | anal | N=128  | Pd 0.747 (Pfa 0.1781) | [Pd 0.793] |
| P-ANMF  | anal | N=256  | Pd 0.794 (Pfa 0.2352) | [Pd 0.837] |
| P-ANMF  | anal | N=512  | Pd 0.829 (Pfa 0.2986) | [Pd 0.868] |
| P-ANMF  | anal | N=1024 | Pd 0.868 (Pfa 0.4038) | [Pd 0.905] |
| CA-NCI  | conf | N=8    | Pd 0.003 (Pfa 0.0011) | [Pd 0.014] |
| CA-NCI  | conf | N=16   | Pd 0.003 (Pfa 0.0011) | [Pd 0.015] |
| CA-NCI  | conf | N=32   | Pd 0.004 (Pfa 0.0012) | [Pd 0.015] |
| CA-NCI  | conf | N=64   | Pd 0.004 (Pfa 0.0013) | [Pd 0.016] |
| CA-NCI  | conf | N=128  | Pd 0.004 (Pfa 0.0014) | [Pd 0.016] |
| CA-NCI  | conf | N=256  | Pd 0.004 (Pfa 0.0017) | [Pd 0.016] |
| CA-NCI  | conf | N=512  | Pd 0.005 (Pfa 0.0018) | [Pd 0.016] |
| CA-NCI  | conf | N=1024 | Pd 0.007 (Pfa 0.0023) | [Pd 0.014] |
| ANMF-FP | conf | N=8    | Pd 0.129 (Pfa 0.0011) | [Pd 0.291] |
| ANMF-FP | conf | N=16   | Pd 0.179 (Pfa 0.0012) | [Pd 0.328] |

history-normalized on target: IN1 Pd 0.0004 (clutter Pfa 0.0010); Clip-OS1 Pd 0.0004 (clutter Pfa 0.0007)  
T1 True; T2 True (0.86 of units); T3 True (1.00 of units)  
T4 descriptive: P-ANMF N=1024 mean Pd at 1e-3: 0.409

Part I: injected pulsed interference. Pooled Pfa / alpha [upper day-cluster bound]; SCR50

|        |        |               |       |                     |       |       |
|--------|--------|---------------|-------|---------------------|-------|-------|
| p=0.02 | JNR=10 | D-IN1         | Pfa/a | 10.27 (upper 11.45) | SCR50 | -2.18 |
| p=0.02 | JNR=10 | PAMF-H        | Pfa/a | 10.76 (upper 12.67) | SCR50 | -5.00 |
| p=0.02 | JNR=10 | P-ANMF        | Pfa/a | 0.82 (upper 0.89)   | SCR50 | 3.03  |
| p=0.02 | JNR=10 | ANMF-FP       | Pfa/a | 1.00 (upper 1.18)   | SCR50 | 2.72  |
| p=0.02 | JNR=10 | D-IN4         | Pfa/a | 11.21 (upper 12.68) | SCR50 | -4.45 |
| p=0.02 | JNR=10 | Clip-OS1      | Pfa/a | 1.04 (upper 1.12)   | SCR50 | -2.19 |
| p=0.02 | JNR=10 | Clip-OS1-cert | Pfa/a | 0.52 (upper 0.60)   | SCR50 | -0.85 |
| p=0.02 | JNR=10 | Bin-OS1       | Pfa/a | 0.73 (upper 0.85)   | SCR50 | -1.08 |
| p=0.02 | JNR=10 | Bin-OS1-cert  | Pfa/a | 0.29 (upper 0.34)   | SCR50 | 0.17  |
| p=0.02 | JNR=10 | Exc-D-IN1     | Pfa/a | 1.22 (upper 1.29)   | SCR50 | nan   |
| p=0.02 | JNR=30 | D-IN1         | Pfa/a | 12.89 (upper 13.04) | SCR50 | -1.45 |
| p=0.02 | JNR=30 | PAMF-H        | Pfa/a | 13.70 (upper 14.14) | SCR50 | -5.00 |
| p=0.02 | JNR=30 | P-ANMF        | Pfa/a | 0.82 (upper 0.89)   | SCR50 | 5.75  |
| p=0.02 | JNR=30 | ANMF-FP       | Pfa/a | 1.07 (upper 1.22)   | SCR50 | 3.43  |
| p=0.02 | JNR=30 | D-IN4         | Pfa/a | 13.53 (upper 13.91) | SCR50 | -3.85 |
| p=0.02 | JNR=30 | Clip-OS1      | Pfa/a | 1.07 (upper 1.15)   | SCR50 | -2.24 |
| p=0.02 | JNR=30 | Clip-OS1-cert | Pfa/a | 0.55 (upper 0.61)   | SCR50 | -0.89 |
| p=0.02 | JNR=30 | Bin-OS1       | Pfa/a | 0.75 (upper 0.88)   | SCR50 | -1.10 |
| p=0.02 | JNR=30 | Bin-OS1-cert  | Pfa/a | 0.32 (upper 0.36)   | SCR50 | 0.15  |
| p=0.02 | JNR=30 | Exc-D-IN1     | Pfa/a | 1.09 (upper 1.17)   | SCR50 | nan   |
| p=0.05 | JNR=10 | D-IN1         | Pfa/a | 17.09 (upper 18.76) | SCR50 | 1.11  |
| p=0.05 | JNR=10 | PAMF-H        | Pfa/a | 17.65 (upper 20.62) | SCR50 | -2.39 |
| p=0.05 | JNR=10 | P-ANMF        | Pfa/a | 0.58 (upper 0.69)   | SCR50 | 6.14  |
| p=0.05 | JNR=10 | ANMF-FP       | Pfa/a | 1.01 (upper 1.18)   | SCR50 | 5.57  |
| p=0.05 | JNR=10 | D-IN4         | Pfa/a | 19.12 (upper 21.44) | SCR50 | -0.70 |
| p=0.05 | JNR=10 | Clip-OS1      | Pfa/a | 1.17 (upper 1.28)   | SCR50 | -1.98 |
| p=0.05 | JNR=10 | Clip-OS1-cert | Pfa/a | 0.10 (upper 0.11)   | SCR50 | 7.37  |
| p=0.05 | JNR=10 | Bin-OS1       | Pfa/a | 1.05 (upper 1.37)   | SCR50 | -0.87 |
| p=0.05 | JNR=10 | Bin-OS1-cert  | Pfa/a | 0.05 (upper 0.08)   | SCR50 | 8.42  |
| p=0.05 | JNR=10 | Exc-D-IN1     | Pfa/a | 1.55 (upper 1.83)   | SCR50 | nan   |
| p=0.05 | JNR=30 | D-IN1         | Pfa/a | 21.23 (upper 21.72) | SCR50 | 10.96 |
| p=0.05 | JNR=30 | PAMF-H        | Pfa/a | 22.12 (upper 23.12) | SCR50 | 4.33  |
| p=0.05 | JNR=30 | P-ANMF        | Pfa/a | 0.55 (upper 0.60)   | SCR50 | 18.56 |
| p=0.05 | JNR=30 | ANMF-FP       | Pfa/a | 1.21 (upper 1.34)   | SCR50 | 9.03  |
| p=0.05 | JNR=30 | D-IN4         | Pfa/a | 22.97 (upper 23.55) | SCR50 | 5.59  |
| p=0.05 | JNR=30 | Clip-OS1      | Pfa/a | 1.28 (upper 1.38)   | SCR50 | -2.09 |
| p=0.05 | JNR=30 | Clip-OS1-cert | Pfa/a | 0.11 (upper 0.14)   | SCR50 | 7.13  |
| p=0.05 | JNR=30 | Bin-OS1       | Pfa/a | 1.20 (upper 1.52)   | SCR50 | -0.99 |
| p=0.05 | JNR=30 | Bin-OS1-cert  | Pfa/a | 0.05 (upper 0.08)   | SCR50 | 8.17  |
| p=0.05 | JNR=30 | Exc-D-IN1     | Pfa/a | 1.37 (upper 1.50)   | SCR50 | nan   |

I1 True; I2 True; I3 True (Clip-OS1 clean SCR50 -2.32; cert at p=0.02: -0.85 / -0.89)

Verdicts: L1 NOT met, L2 met, L3 NOT met, L4 NOT met, L5 met, L6 met, A1 met, A2 NOT met, A3 met, T1 met, T2 met, T3 met, I1 met, I2 met, I3 met

77 GHz data with real interference (`r12/analyze_r12_jku.py`); N: no fast-time mitigation, Z: zeroing, ZAR: zeroing with AR(8) gap reconstruction; strata: number of hit chirps in the 24-chirp segment (0, 1–2, 3–5,  $\geq 6$ ):

R12 Part J: 8 (station, rx) units; hit rate (test frames): ref 0.000, A 0.454, C 0.046  
zeroed fraction (test frames): ref 0.049, A 0.076, C 0.070

```
run ref, mitigation N: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.04 | 1.04 0.00 0.00 0.00 ; SCR50 10.01 [ 10.01]
OS1_8    1.00 | 1.00 0.00 0.00 0.00 ; SCR50 10.80 [ 10.80]
D-IN1     1.03 | 1.03 0.00 0.00 0.00 ; SCR50  5.91 [  5.91]
Clip-OS1  1.00 | 1.00 0.00 0.00 0.00 ; SCR50  6.84 [  6.84]
Bin-OS1   0.67 | 0.67 0.00 0.00 0.00 ; SCR50  7.96 [  7.96]
Clip-OS1-cert 1.00 | 1.00 0.00 0.00 0.00 ; SCR50  6.84 [  6.84]
Bin-OS1-cert 0.67 | 0.67 0.00 0.00 0.00 ; SCR50  7.96 [  7.96]

run ref, mitigation Z: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.07 | 1.07 0.00 0.00 0.00 ; SCR50 10.33 [ 10.33]
OS1_8    0.98 | 0.98 0.00 0.00 0.00 ; SCR50 11.11 [ 11.11]
D-IN1     0.99 | 0.99 0.00 0.00 0.00 ; SCR50  6.24 [  6.24]
Clip-OS1  0.97 | 0.97 0.00 0.00 0.00 ; SCR50  7.11 [  7.11]
Bin-OS1   0.63 | 0.63 0.00 0.00 0.00 ; SCR50  8.32 [  8.32]
Clip-OS1-cert 0.97 | 0.97 0.00 0.00 0.00 ; SCR50  7.11 [  7.11]
Bin-OS1-cert 0.63 | 0.63 0.00 0.00 0.00 ; SCR50  8.32 [  8.32]

run ref, mitigation ZAR: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.05 | 1.05 0.00 0.00 0.00 ; SCR50 10.21 [ 10.21]
OS1_8    1.00 | 1.00 0.00 0.00 0.00 ; SCR50 10.97 [ 10.97]
D-IN1     1.01 | 1.01 0.00 0.00 0.00 ; SCR50  6.09 [  6.09]
Clip-OS1  0.99 | 0.99 0.00 0.00 0.00 ; SCR50  7.00 [  7.00]
Bin-OS1   0.66 | 0.66 0.00 0.00 0.00 ; SCR50  8.17 [  8.17]
Clip-OS1-cert 0.99 | 0.99 0.00 0.00 0.00 ; SCR50  7.00 [  7.00]
Bin-OS1-cert 0.66 | 0.66 0.00 0.00 0.00 ; SCR50  8.17 [  8.17]

run A, mitigation N: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.09 | 1.12 1.22 0.56 1.06 ; SCR50 15.29 [ 11.91]
OS1_8    1.16 | 1.05 1.21 0.83 1.20 ; SCR50 16.01 [ 12.70]
D-IN1     0.81 | 0.94 1.00 0.49 0.72 ; SCR50 11.14 [  7.68]
Clip-OS1  1.01 | 0.92 0.98 1.11 1.05 ; SCR50 11.95 [  8.66]
Bin-OS1   0.68 | 0.60 0.65 0.76 0.71 ; SCR50 13.08 [  9.83]
Clip-OS1-cert 0.11 | 0.40 0.03 0.00 0.00 ; SCR50  nan [ 13.74]
Bin-OS1-cert 0.06 | 0.22 0.00 0.00 0.00 ; SCR50  nan [ 14.93]

run A, mitigation Z: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.04 | 1.14 1.08 1.25 0.99 ; SCR50 11.07 [ 10.32]
OS1_8    0.98 | 1.04 1.04 1.32 0.94 ; SCR50 11.86 [ 11.14]
D-IN1     0.96 | 0.99 1.06 0.83 0.93 ; SCR50  6.96 [  6.11]
Clip-OS1  0.98 | 0.97 1.18 0.76 0.95 ; SCR50  7.86 [  7.05]
Bin-OS1   0.62 | 0.58 0.75 0.62 0.61 ; SCR50  9.05 [  8.33]
Clip-OS1-cert 0.11 | 0.42 0.02 0.00 0.00 ; SCR50  nan [ 12.24]
Bin-OS1-cert 0.06 | 0.22 0.00 0.00 0.00 ; SCR50  nan [ 13.63]

run A, mitigation ZAR: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.07 | 1.15 1.13 1.04 1.03 ; SCR50 10.89 [ 10.27]
OS1_8    1.04 | 1.08 1.12 1.11 1.00 ; SCR50 11.63 [ 11.03]
D-IN1     1.01 | 1.01 1.10 0.69 1.00 ; SCR50  6.74 [  6.06]
Clip-OS1  1.01 | 0.93 1.11 0.83 1.02 ; SCR50  7.65 [  7.03]
Bin-OS1   0.65 | 0.58 0.84 0.76 0.64 ; SCR50  8.82 [  8.26]
Clip-OS1-cert 0.11 | 0.41 0.02 0.00 0.00 ; SCR50  nan [ 12.18]
Bin-OS1-cert 0.06 | 0.22 0.01 0.00 0.00 ; SCR50  nan [ 13.52]

run C, mitigation N: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.13 | 1.08 1.31 1.57 0.90 ; SCR50 11.32 [ 10.60]
OS1_8    1.08 | 1.01 1.33 1.63 0.93 ; SCR50 12.08 [ 11.28]
D-IN1     1.42 | 1.19 3.56 1.99 0.89 ; SCR50  7.15 [  6.34]
Clip-OS1  1.29 | 1.11 2.38 2.00 1.13 ; SCR50  8.11 [  7.29]
Bin-OS1   0.90 | 0.75 1.73 1.39 0.95 ; SCR50  9.22 [  8.42]
Clip-OS1-cert 0.54 | 0.68 0.39 0.00 0.00 ; SCR50 14.52 [ 10.40]
Bin-OS1-cert 0.33 | 0.42 0.17 0.00 0.00 ; SCR50 15.78 [ 11.63]

run C, mitigation Z: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.04 | 1.05 0.97 1.04 0.98 ; SCR50 10.54 [ 10.36]
OS1_8    0.98 | 0.98 0.96 0.99 1.00 ; SCR50 11.32 [ 11.15]
D-IN1     1.00 | 1.00 0.99 0.94 1.03 ; SCR50  6.41 [  6.22]
Clip-OS1  1.02 | 1.03 1.03 0.98 1.02 ; SCR50  7.26 [  7.08]
Bin-OS1   0.64 | 0.63 0.63 0.56 0.79 ; SCR50  8.48 [  8.31]
Clip-OS1-cert 0.51 | 0.65 0.26 0.00 0.00 ; SCR50 13.94 [ 10.00]
Bin-OS1-cert 0.27 | 0.36 0.08 0.00 0.00 ; SCR50 15.19 [ 11.33]

run C, mitigation ZAR: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1      1.05 | 1.06 0.99 1.05 1.05 ; SCR50 10.41 [ 10.28]
OS1_8    1.01 | 1.01 1.03 1.03 1.03 ; SCR50 11.18 [ 11.04]
D-IN1     1.04 | 1.04 1.10 0.99 1.04 ; SCR50  6.25 [  6.10]
Clip-OS1  1.02 | 1.02 1.04 1.04 0.96 ; SCR50  7.15 [  7.01]
Bin-OS1   0.66 | 0.65 0.64 0.63 0.76 ; SCR50  8.32 [  8.18]
Clip-OS1-cert 0.50 | 0.65 0.26 0.00 0.00 ; SCR50 13.88 [  9.93]
Bin-OS1-cert 0.30 | 0.39 0.08 0.00 0.00 ; SCR50 15.08 [ 11.22]

J1 (Clip-OS1-cert Pfa <= 1.2 alpha in run C under N and Z): True  [N 0.0054, Z 0.0051]
J2 (D-IN1 under N Pfa >= 5 alpha in runs A and C): False  [A 0.0081, C 0.0142]
J3 (ZAR improves IN1 SCR50 over Z by >= 1 dB in run A): False  [0.18 dB]
```

## 21 Guarded scale, clip level, certified trimming, pulse blanking and bursts (R14)

Frozen protocol `r14/PROTOCOL_R14.md` (sha256 9f771091...), written after a synthetic dry run and before any IPIX outcome; expectations and verdicts in `r14/RESULTS_R14.md` (12 of 15 met). The settings are those of the final IPIX study, except for dense calibration: non-overlapping 24-pulse segments, about 14,600 per unit, so that  $\alpha = 10^{-3}$  is resolved. The study tested five changes:

- a guard of  $\Delta \in \{0, 8, 16\}$  pulses between the scale window and the tested pulse;

- clip levels  $\kappa \in \{4, 6, 9, 12, 18\}$ ;
- a certified trimmed sum, which discards the largest dwell terms; the number discarded is the smallest for which more corrupted dwell innovations than discarded terms have probability at most  $\alpha/5$  under the hit model (4 of eight at a 2% and 5 at a 5% hit rate at  $\alpha = 10^{-2}$ ; no such number exists for bursts);
- pulse blanking of isolated spikes, calibrated with masks drawn from the hit model and passed through the same rule (mask-matched calibration);
- bursts of eight consecutive pulses at the same mean hit rate.

The certified thresholds use the worst case of Theorem 1 with hits drawn either from the independent-hit model or from the burst model. The guard leaves the false-alarm law unchanged; the measured  $P_{fa}/\alpha$  with  $\Delta = 8$  is 1.03 at  $10^{-2}$  and 1.03 at  $10^{-3}$ . The real IPIX target stays undetected with every guard (exceedance 0.0005 at  $10^{-3}$  with  $\Delta = 8$ ), because it is present in every history.

*Unmet expectations of R14.*

- **Onset cost (G3).** The guard was expected to cost at most 0.5 dB at onset. It costs 0.9 dB with  $\Delta = 8$  and 1.2 dB with  $\Delta = 16$ , because the scale is older.
- **Certification at  $10^{-3}$  (K2).** The certificate at the clip level chosen by the saturation rule was expected to reach  $P_d = 0.5$  below 25 dB in all four interference conditions at  $10^{-3}$ . With 5% hits the certificate is vacuous at  $10^{-3}$  for every clip level.
- **Cost of trimming (T2).** Trimming was expected to cost more than clipping at both hit rates. At 5% it costs +5.1 dB against +5.9 dB for clipping ( $\kappa = 6$ ).

The SCR for  $P_d = 0.5$  is fragile under 5% hits, because the certified detection curve is nearly flat there, so the main text reports  $P_d$  at 10 dB.

Output of `r14/analyze_r14.py`. Columns of the interference table: condition,  $\alpha$ , statistic, pooled  $P_{fa}/\alpha$  [upper day-cluster bound], SCR for  $P_d = 0.5$ , its cost against clean clipping ( $\kappa = 6$ ), and  $P_d$  at 10 and 25 dB.

```
Part G: guarded history scale (56 units)
g= 0 alpha=0.01: clean Pfa/alpha 1.01 [0.98, 1.04]
g= 0 alpha=0.001: clean Pfa/alpha 1.02 [0.94, 1.10]
g= 0 quintile spread at 0.01: 1.44
g= 8 alpha=0.01: clean Pfa/alpha 1.03 [0.97, 1.07]
g= 8 alpha=0.001: clean Pfa/alpha 1.03 [0.84, 1.21]
g= 8 quintile spread at 0.01: 1.40
g=16 alpha=0.01: clean Pfa/alpha 1.00 [0.93, 1.03]
g=16 alpha=0.001: clean Pfa/alpha 1.03 [0.81, 1.15]
g=16 quintile spread at 0.01: 1.86
random g= 0: per-look Pd@10dB 0.88 0.71 0.64 0.50 0.08 0.01 0.00 0.00 0.00 | mean looks1-8 0.24 | SCR50 look0 -1.4 | SCR50 within8 <=-5
random g= 8: per-look Pd@10dB 0.87 0.82 0.82 0.82 0.82 0.82 0.82 0.82 0.82 | mean looks1-8 0.82 | SCR50 look0 -0.5 | SCR50 within8 <=-5
random g=16: per-look Pd@10dB 0.87 0.82 0.82 0.82 0.82 0.82 0.82 0.82 0.82 | mean looks1-8 0.82 | SCR50 look0 -0.1 | SCR50 within8 <=-5
opposite g= 0: per-look Pd@10dB 0.88 0.94 0.91 0.82 0.16 0.01 0.00 0.00 0.00 | mean looks1-8 0.36 | SCR50 look0 -1.4 | SCR50 within8 <=-5
opposite g= 8: per-look Pd@10dB 0.87 0.95 0.95 0.95 0.95 0.95 0.95 0.95 0.95 | mean looks1-8 0.95 | SCR50 look0 -0.5 | SCR50 within8 <=-5
opposite g=16: per-look Pd@10dB 0.87 0.94 0.94 0.94 0.94 0.94 0.94 0.94 0.94 | mean looks1-8 0.94 | SCR50 look0 -0.1 | SCR50 within8 <=-5
matched g= 0: per-look Pd@10dB 0.88 0.02 0.01 0.01 0.00 0.00 0.00 0.00 0.00 | mean looks1-8 0.01 | SCR50 look0 -1.4 | SCR50 within8 -2.1
matched g= 8: per-look Pd@10dB 0.87 0.07 0.07 0.07 0.07 0.07 0.07 0.07 0.07 | mean looks1-8 0.07 | SCR50 look0 -0.5 | SCR50 within8 -1.2
matched g=16: per-look Pd@10dB 0.87 0.07 0.07 0.07 0.07 0.07 0.07 0.07 0.07 | mean looks1-8 0.07 | SCR50 look0 -0.1 | SCR50 within8 -0.8
lav (exploratory, AR(1) no noise) random g= 0: 0.91 0.73 0.66 0.51 0.07 0.01 0.00 0.00 0.00
lav (exploratory, AR(1) no noise) random g= 8: 0.91 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85
lav (exploratory, AR(1) no noise) random g=16: 0.91 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85
ramp L= 8 g= 0: detection within 8 looks @10dB 0.745, SCR50 -0.8
ramp L= 8 g= 8: detection within 8 looks @10dB 0.908, SCR50 <=-5
ramp L= 8 g=16: detection within 8 looks @10dB 0.909, SCR50 <=-5
ramp L=16 g= 0: detection within 8 looks @10dB 0.032, SCR50 n.r.
ramp L=16 g= 8: detection within 8 looks @10dB 0.869, SCR50 -4.5
ramp L=16 g=16: detection within 8 looks @10dB 0.903, SCR50 <=-5
real target g= 0 alpha=0.001: exceedance 0.0003
real target g= 0 alpha=0.01: exceedance 0.0041
real target g= 8 alpha=0.001: exceedance 0.0005
real target g= 8 alpha=0.01: exceedance 0.0058
real target g=16 alpha=0.001: exceedance 0.0008
real target g=16 alpha=0.01: exceedance 0.0068

Parts K/T/X/B: dwell integration (56 units)
trim levels t (p, alpha, model): (0.02, 0.01, bern) -> 4; (0.02, 0.01, burst) -> vacuous; (0.02, 0.001, bern) -> 5; (0.02, 0.001, burst) -> vacuous; (0.05, 0.01, bern) -> 5;
(0.05, 0.01, burst) -> vacuous; (0.05, 0.001, bern) -> 6; (0.05, 0.001, burst) -> vacuous
kappa rule (mode over units): {0.01: 6.0, 0.001: 9.0} (distribution: {0.01: {4.0: 4, 6.0: 29, 9.0: 13, 12.0: 2, 18.0: 4, -1.0: 4}, 0.001: {6.0: 2, 9.0: 17, 12.0: 9, 18.0:
12, -1.0: 16}})
clean clipped (kappa 6) SCR50: alpha 0.01 -2.4, 0.001 14.5
clip kappa= 4 alpha=0.01: clean Pfa/alpha 1.04 [0.97, 1.08], clean SCR50 -2.0
clip kappa= 4 alpha=0.001: clean Pfa/alpha 0.09 [0.00, 0.23], clean SCR50 n.r.
clip kappa= 6 alpha=0.01: clean Pfa/alpha 1.05 [0.99, 1.08], clean SCR50 -2.4
clip kappa= 6 alpha=0.001: clean Pfa/alpha 0.67 [0.35, 0.94], clean SCR50 14.5
clip kappa= 9 alpha=0.01: clean Pfa/alpha 1.05 [0.98, 1.09], clean SCR50 -2.3
clip kappa= 9 alpha=0.001: clean Pfa/alpha 0.90 [0.58, 1.15], clean SCR50 2.5
clip kappa=12 alpha=0.01: clean Pfa/alpha 1.06 [1.00, 1.09], clean SCR50 -2.2
clip kappa=12 alpha=0.001: clean Pfa/alpha 0.95 [0.65, 1.20], clean SCR50 2.1
clip kappa=18 alpha=0.01: clean Pfa/alpha 1.06 [1.01, 1.09], clean SCR50 -2.0
clip kappa=18 alpha=0.001: clean Pfa/alpha 1.03 [0.79, 1.19], clean SCR50 1.9
blank alpha=0.01: clean Pfa/alpha 1.07 [1.03, 1.09], clean SCR50 -1.4
blank alpha=0.001: clean Pfa/alpha 1.06 [0.93, 1.14], clean SCR50 3.1

condition      alpha  statistic      Pfa/alpha [upper]  SCR50  cost vs clean clip6
bern p=0.02 10 dB 0.01 clean-thr clip6 1.08 [ 1.11] -2.3 +0.1 0.85 0.99
bern p=0.02 10 dB 0.01 cert(bern) clip4 0.50 [ 0.61] -0.0 +2.3 0.75 0.85
bern p=0.02 10 dB 0.01 cert(burst) clip4 0.00 [ 0.00] n.r. n.r. 0.00 0.00
bern p=0.02 10 dB 0.01 cert(bern) clip6 0.53 [ 0.61] -0.8 +1.6 0.81 0.95
bern p=0.02 10 dB 0.01 cert(burst) clip6 0.00 [ 0.00] n.r. n.r. 0.00 0.00
bern p=0.02 10 dB 0.01 cert(bern) clip9 0.52 [ 0.58] -0.8 +1.6 0.81 0.96
```

|                    |       |                     |              |      |       |      |      |
|--------------------|-------|---------------------|--------------|------|-------|------|------|
| bern p=0.02 10 dB  | 0.01  | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.01  | cert(bern) clip12   | 0.50 [ 0.54] | -0.5 | +1.8  | 0.81 | 0.98 |
| bern p=0.02 10 dB  | 0.01  | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.01  | cert(bern) clip18   | 0.50 [ 0.56] | 0.2  | +2.5  | 0.79 | 0.98 |
| bern p=0.02 10 dB  | 0.01  | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.01  | cert(bern) trim t=4 | 0.53 [ 0.57] | -0.3 | +2.1  | 0.79 | 0.97 |
| bern p=0.02 10 dB  | 0.01  | blank(bern masks)   | 1.05 [ 1.08] | -1.2 | +1.2  | 0.79 | 0.97 |
| bern p=0.02 10 dB  | 0.01  | blank(burst masks)  | 1.00 [ 1.02] | -1.0 | +1.3  | 0.79 | 0.97 |
| bern p=0.02 10 dB  | 0.001 | clean-thr clip6     | 0.65 [ 0.92] | 14.7 | +0.2  | 0.46 | 0.54 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) clip4    | 0.04 [ 0.10] | n.r. | n.r.  | 0.04 | 0.05 |
| bern p=0.02 10 dB  | 0.001 | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) clip6    | 0.07 [ 0.11] | n.r. | n.r.  | 0.12 | 0.16 |
| bern p=0.02 10 dB  | 0.001 | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) clip9    | 0.27 [ 0.34] | 8.9  | -5.6  | 0.52 | 0.63 |
| bern p=0.02 10 dB  | 0.001 | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) clip12   | 0.30 [ 0.41] | 4.5  | -10.0 | 0.63 | 0.76 |
| bern p=0.02 10 dB  | 0.001 | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) clip18   | 0.30 [ 0.40] | 4.3  | -10.2 | 0.64 | 0.79 |
| bern p=0.02 10 dB  | 0.001 | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 10 dB  | 0.001 | cert(bern) trim t=5 | 0.39 [ 0.44] | 4.0  | -10.5 | 0.68 | 0.94 |
| bern p=0.02 10 dB  | 0.001 | blank(bern masks)   | 1.09 [ 1.17] | 3.4  | -11.1 | 0.68 | 0.93 |
| bern p=0.02 10 dB  | 0.001 | blank(burst masks)  | 0.95 [ 1.10] | 3.7  | -10.8 | 0.68 | 0.93 |
| bern p=0.02 30 dB  | 0.01  | clean-thr clip6     | 1.10 [ 1.12] | -2.3 | +0.1  | 0.85 | 0.99 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) clip4    | 0.53 [ 0.63] | -0.1 | +2.3  | 0.75 | 0.85 |
| bern p=0.02 30 dB  | 0.01  | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) clip6    | 0.55 [ 0.61] | -0.8 | +1.5  | 0.81 | 0.95 |
| bern p=0.02 30 dB  | 0.01  | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) clip9    | 0.53 [ 0.58] | -0.8 | +1.6  | 0.81 | 0.96 |
| bern p=0.02 30 dB  | 0.01  | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) clip12   | 0.52 [ 0.56] | -0.6 | +1.8  | 0.81 | 0.98 |
| bern p=0.02 30 dB  | 0.01  | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) clip18   | 0.54 [ 0.59] | 0.1  | +2.5  | 0.79 | 0.98 |
| bern p=0.02 30 dB  | 0.01  | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.01  | cert(bern) trim t=4 | 0.57 [ 0.60] | -0.3 | +2.0  | 0.79 | 0.97 |
| bern p=0.02 30 dB  | 0.01  | blank(bern masks)   | 1.08 [ 1.11] | -1.2 | +1.2  | 0.79 | 0.97 |
| bern p=0.02 30 dB  | 0.01  | blank(burst masks)  | 1.01 [ 1.04] | -1.1 | +1.3  | 0.79 | 0.97 |
| bern p=0.02 30 dB  | 0.001 | clean-thr clip6     | 0.65 [ 0.90] | 14.7 | +0.1  | 0.46 | 0.54 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) clip4    | 0.04 [ 0.09] | n.r. | n.r.  | 0.04 | 0.05 |
| bern p=0.02 30 dB  | 0.001 | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) clip6    | 0.07 [ 0.11] | n.r. | n.r.  | 0.12 | 0.16 |
| bern p=0.02 30 dB  | 0.001 | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) clip9    | 0.26 [ 0.33] | 8.8  | -5.8  | 0.52 | 0.63 |
| bern p=0.02 30 dB  | 0.001 | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) clip12   | 0.30 [ 0.39] | 4.4  | -10.1 | 0.63 | 0.76 |
| bern p=0.02 30 dB  | 0.001 | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) clip18   | 0.31 [ 0.40] | 4.2  | -10.3 | 0.64 | 0.79 |
| bern p=0.02 30 dB  | 0.001 | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.02 30 dB  | 0.001 | cert(bern) trim t=5 | 0.43 [ 0.49] | 3.9  | -10.6 | 0.68 | 0.94 |
| bern p=0.02 30 dB  | 0.001 | blank(bern masks)   | 1.13 [ 1.18] | 3.4  | -11.1 | 0.68 | 0.93 |
| bern p=0.02 30 dB  | 0.001 | blank(burst masks)  | 0.99 [ 1.12] | 3.7  | -10.8 | 0.68 | 0.93 |
| bern p=0.05 10 dB  | 0.01  | clean-thr clip6     | 1.22 [ 1.31] | -2.1 | +0.3  | 0.85 | 0.98 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) clip4    | 0.02 [ 0.04] | n.r. | n.r.  | 0.06 | 0.09 |
| bern p=0.05 10 dB  | 0.01  | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) clip6    | 0.12 [ 0.14] | 3.7  | +6.1  | 0.64 | 0.75 |
| bern p=0.05 10 dB  | 0.01  | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) clip9    | 0.13 [ 0.17] | 2.4  | +4.8  | 0.71 | 0.87 |
| bern p=0.05 10 dB  | 0.01  | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) clip12   | 0.12 [ 0.16] | 2.3  | +4.7  | 0.72 | 0.90 |
| bern p=0.05 10 dB  | 0.01  | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) clip18   | 0.13 [ 0.16] | 2.8  | +5.2  | 0.72 | 0.93 |
| bern p=0.05 10 dB  | 0.01  | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.01  | cert(bern) trim t=5 | 0.15 [ 0.17] | 2.9  | +5.2  | 0.72 | 0.95 |
| bern p=0.05 10 dB  | 0.01  | blank(bern masks)   | 1.13 [ 1.21] | -0.8 | +1.6  | 0.78 | 0.96 |
| bern p=0.05 10 dB  | 0.01  | blank(burst masks)  | 1.07 [ 1.12] | -0.7 | +1.7  | 0.78 | 0.96 |
| bern p=0.05 10 dB  | 0.001 | clean-thr clip6     | 0.66 [ 0.94] | 15.1 | +0.6  | 0.45 | 0.54 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) clip4    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) clip6    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) clip9    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) clip12   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) clip18   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | cert(bern) trim t=6 | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 10 dB  | 0.001 | blank(bern masks)   | 1.67 [ 1.91] | 3.9  | -10.6 | 0.67 | 0.93 |
| bern p=0.05 10 dB  | 0.001 | blank(burst masks)  | 1.52 [ 1.66] | 4.3  | -10.2 | 0.66 | 0.93 |
| bern p=0.05 30 dB  | 0.01  | clean-thr clip6     | 1.28 [ 1.37] | -2.1 | +0.3  | 0.85 | 0.98 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) clip4    | 0.02 [ 0.05] | n.r. | n.r.  | 0.06 | 0.09 |
| bern p=0.05 30 dB  | 0.01  | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) clip6    | 0.12 [ 0.14] | 3.5  | +5.9  | 0.64 | 0.75 |
| bern p=0.05 30 dB  | 0.01  | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) clip9    | 0.14 [ 0.18] | 2.3  | +4.6  | 0.71 | 0.87 |
| bern p=0.05 30 dB  | 0.01  | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) clip12   | 0.13 [ 0.17] | 2.2  | +4.6  | 0.73 | 0.90 |
| bern p=0.05 30 dB  | 0.01  | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) clip18   | 0.17 [ 0.19] | 2.7  | +5.1  | 0.72 | 0.93 |
| bern p=0.05 30 dB  | 0.01  | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.01  | cert(bern) trim t=5 | 0.23 [ 0.26] | 2.8  | +5.1  | 0.72 | 0.95 |
| bern p=0.05 30 dB  | 0.01  | blank(bern masks)   | 1.21 [ 1.25] | -0.9 | +1.5  | 0.78 | 0.96 |
| bern p=0.05 30 dB  | 0.01  | blank(burst masks)  | 1.15 [ 1.18] | -0.7 | +1.6  | 0.78 | 0.96 |
| bern p=0.05 30 dB  | 0.001 | clean-thr clip6     | 0.69 [ 0.93] | 14.9 | +0.4  | 0.45 | 0.54 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) clip4    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) clip6    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) clip9    | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) clip12   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) clip18   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | cert(bern) trim t=6 | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| bern p=0.05 30 dB  | 0.001 | blank(bern masks)   | 2.25 [ 2.54] | 3.9  | -10.6 | 0.67 | 0.93 |
| bern p=0.05 30 dB  | 0.001 | blank(burst masks)  | 2.12 [ 2.39] | 4.3  | -10.2 | 0.66 | 0.93 |
| burst p=0.02 30 dB | 0.01  | clean-thr clip6     | 2.24 [ 2.33] | -2.2 | +0.1  | 0.84 | 0.97 |
| burst p=0.02 30 dB | 0.01  | cert(bern) clip4    | 1.11 [ 1.29] | 0.1  | +2.4  | 0.74 | 0.84 |
| burst p=0.02 30 dB | 0.01  | cert(burst) clip4   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.01  | cert(bern) clip6    | 1.46 [ 1.58] | -0.8 | +1.6  | 0.80 | 0.94 |
| burst p=0.02 30 dB | 0.01  | cert(burst) clip6   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.01  | cert(bern) clip9    | 1.75 [ 1.89] | -0.8 | +1.6  | 0.80 | 0.95 |
| burst p=0.02 30 dB | 0.01  | cert(burst) clip9   | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.01  | cert(bern) clip12   | 1.96 [ 2.11] | -0.5 | +1.8  | 0.80 | 0.97 |
| burst p=0.02 30 dB | 0.01  | cert(burst) clip12  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.01  | cert(bern) clip18   | 2.18 [ 2.32] | 0.2  | +2.5  | 0.78 | 0.96 |
| burst p=0.02 30 dB | 0.01  | cert(burst) clip18  | 0.00 [ 0.00] | n.r. | n.r.  | 0.00 | 0.00 |

|                    |       |                     |               |      |       |      |      |
|--------------------|-------|---------------------|---------------|------|-------|------|------|
| burst p=0.02 30 dB | 0.01  | cert(bern) trim t=4 | 2.42 [ 2.48]  | -0.4 | +2.0  | 0.78 | 0.96 |
| burst p=0.02 30 dB | 0.01  | blank(bern masks)   | 2.31 [ 2.36]  | -1.4 | +1.0  | 0.79 | 0.97 |
| burst p=0.02 30 dB | 0.01  | blank(burst masks)  | 2.25 [ 2.31]  | -1.2 | +1.1  | 0.79 | 0.96 |
| burst p=0.02 30 dB | 0.001 | clean-thr clip6     | 3.86 [ 5.34]  | 16.1 | +1.6  | 0.45 | 0.53 |
| burst p=0.02 30 dB | 0.001 | cert(bern) clip4    | 0.31 [ 0.81]  | n.r. | n.r.  | 0.04 | 0.05 |
| burst p=0.02 30 dB | 0.001 | cert(burst) clip4   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.001 | cert(bern) clip6    | 0.90 [ 1.61]  | n.r. | n.r.  | 0.12 | 0.15 |
| burst p=0.02 30 dB | 0.001 | cert(burst) clip6   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.001 | cert(bern) clip9    | 3.86 [ 4.99]  | 9.2  | -5.3  | 0.51 | 0.62 |
| burst p=0.02 30 dB | 0.001 | cert(burst) clip9   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.001 | cert(bern) clip12   | 5.36 [ 6.72]  | 4.6  | -9.9  | 0.62 | 0.75 |
| burst p=0.02 30 dB | 0.001 | cert(burst) clip12  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.001 | cert(bern) clip18   | 8.50 [10.92]  | 4.4  | -10.1 | 0.63 | 0.78 |
| burst p=0.02 30 dB | 0.001 | cert(burst) clip18  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.02 30 dB | 0.001 | cert(bern) trim t=5 | 14.54 [15.12] | 3.9  | -10.6 | 0.68 | 0.93 |
| burst p=0.02 30 dB | 0.001 | blank(bern masks)   | 13.60 [14.17] | 3.2  | -11.3 | 0.69 | 0.93 |
| burst p=0.02 30 dB | 0.001 | blank(burst masks)  | 13.43 [14.00] | 3.5  | -11.0 | 0.68 | 0.93 |
| burst p=0.05 30 dB | 0.01  | clean-thr clip6     | 3.98 [ 4.22]  | -2.0 | +0.4  | 0.82 | 0.95 |
| burst p=0.05 30 dB | 0.01  | cert(bern) clip4    | 0.16 [ 0.40]  | n.r. | n.r.  | 0.06 | 0.09 |
| burst p=0.05 30 dB | 0.01  | cert(burst) clip4   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.01  | cert(bern) clip6    | 1.25 [ 1.45]  | 4.1  | +6.5  | 0.62 | 0.72 |
| burst p=0.05 30 dB | 0.01  | cert(burst) clip6   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.01  | cert(bern) clip9    | 1.66 [ 1.83]  | 2.5  | +4.9  | 0.69 | 0.84 |
| burst p=0.05 30 dB | 0.01  | cert(burst) clip9   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.01  | cert(bern) clip12   | 2.23 [ 2.49]  | 2.4  | +4.8  | 0.70 | 0.87 |
| burst p=0.05 30 dB | 0.01  | cert(burst) clip12  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.01  | cert(bern) clip18   | 2.84 [ 3.22]  | 2.8  | +5.2  | 0.70 | 0.90 |
| burst p=0.05 30 dB | 0.01  | cert(burst) clip18  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.01  | cert(bern) trim t=5 | 3.60 [ 3.68]  | 2.8  | +5.1  | 0.70 | 0.92 |
| burst p=0.05 30 dB | 0.01  | blank(bern masks)   | 4.14 [ 4.23]  | -1.2 | +1.2  | 0.78 | 0.96 |
| burst p=0.05 30 dB | 0.01  | blank(burst masks)  | 4.10 [ 4.17]  | -1.1 | +1.3  | 0.78 | 0.96 |
| burst p=0.05 30 dB | 0.001 | clean-thr clip6     | 8.92 [12.32]  | 19.0 | +4.5  | 0.44 | 0.52 |
| burst p=0.05 30 dB | 0.001 | cert(bern) clip4    | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(burst) clip4   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(bern) clip6    | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(burst) clip6   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(bern) clip9    | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(burst) clip9   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(bern) clip12   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(burst) clip12  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(bern) clip18   | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(burst) clip18  | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | cert(bern) trim t=6 | 0.00 [ 0.00]  | n.r. | n.r.  | 0.00 | 0.00 |
| burst p=0.05 30 dB | 0.001 | blank(bern masks)   | 32.43 [33.64] | 3.5  | -11.0 | 0.68 | 0.92 |
| burst p=0.05 30 dB | 0.001 | blank(burst masks)  | 32.35 [33.49] | 3.9  | -10.6 | 0.67 | 0.92 |

Verdicts (frozen expectations): G1 met, G2 met, G3 NOT met, G4 met, G5 met, K1 met, K2 NOT met, T1 met, T2 NOT met, X1 met, X2 met, X3 met, B1 met, B2 met, B3 met  
met 12 of 15

## 22 Independent re-check of the main laws and of Theorem 1

r14/theory\_check\_independent.py re-derives the main closed forms of Section IV and Theorem 1 by Monte Carlo on a GPU (the remaining laws are checked by the study scripts, Section 20). It was written from scratch and does not import the code that implements the laws. It checks:

- Proposition 1, including repeated eigenvalues and a singular  $Q$ ;
- Proposition 4;
- the horizon of Corollary 4;
- the strong-target OS law (7), the integration laws and the known-bin P-ANMF law;
- the Beta law (by Monte Carlo) and the calibration sizes (by exact computation);
- Theorem 1 under strong, matched and cancelling interference.

It also includes a misspecified case: bursts against a certificate calibrated on independent hits, which violates the dominance condition. There the clipped certificate exceeds  $\alpha$ , as the theorem's condition implies it may.

|                                                                                 |            |             |         |
|---------------------------------------------------------------------------------|------------|-------------|---------|
| OK Prop1 AR(1) centre, Q=I                                                      | MC 0.84463 | law 0.84460 | z +0.14 |
| OK Prop1 zero centre, white Sigma (repeated eigenvalues)                        | MC 0.66741 | law 0.66788 | z -1.39 |
| OK Prop1 singular Q (rank 2)                                                    | MC 0.34575 | law 0.34580 | z -0.13 |
| OK Prop4 S=0dB opposite look 0                                                  | MC 0.53657 | law 0.53641 | z +0.32 |
| OK Prop4 S=0dB opposite look 1                                                  | MC 0.81305 | law 0.81315 | z -0.24 |
| OK Prop4 S=0dB opposite look 2                                                  | MC 0.70894 | law 0.70896 | z -0.04 |
| OK Prop4 S=0dB opposite look 4                                                  | MC 0.17562 | law 0.17575 | z -0.33 |
| OK Prop4 S=0dB opposite look 8                                                  | MC 0.00381 | law 0.00387 | z -1.05 |
| OK Prop4 S=0dB random look 0                                                    | MC 0.53664 | law 0.53641 | z +0.46 |
| OK Prop4 S=0dB random look 1                                                    | MC 0.62243 | law 0.62197 | z +0.94 |
| OK Prop4 S=0dB random look 2                                                    | MC 0.46771 | law 0.46758 | z +0.27 |
| OK Prop4 S=0dB random look 4                                                    | MC 0.10737 | law 0.10790 | z -1.72 |
| OK Prop4 S=0dB random look 8                                                    | MC 0.00602 | law 0.00611 | z -1.10 |
| OK Prop4 S=10dB opposite look 0                                                 | MC 0.93152 | law 0.93148 | z +0.15 |
| OK Prop4 S=10dB opposite look 1                                                 | MC 0.97868 | law 0.97857 | z +0.79 |
| OK Prop4 S=10dB opposite look 2                                                 | MC 0.96165 | law 0.96169 | z -0.22 |
| OK Prop4 S=10dB opposite look 4                                                 | MC 0.15337 | law 0.15389 | z -1.45 |
| OK Prop4 S=10dB opposite look 8                                                 | MC 0.00047 | law 0.00045 | z +0.99 |
| OK Prop4 S=10dB random look 0                                                   | MC 0.93142 | law 0.93148 | z -0.26 |
| OK Prop4 S=10dB random look 1                                                   | MC 0.94759 | law 0.94767 | z -0.39 |
| OK Prop4 S=10dB random look 2                                                   | MC 0.89941 | law 0.89926 | z +0.52 |
| OK Prop4 S=10dB random look 4                                                   | MC 0.08163 | law 0.08146 | z +0.63 |
| OK Prop4 S=10dB random look 8                                                   | MC 0.00077 | law 0.00079 | z -0.94 |
| OK Cor2 eq.(3) onset, 10 dB                                                     | MC 0.93176 | law 0.93148 | z +1.10 |
| horizon 1* (opposite Doppler,  rho =0.93, m=16, Pfa 0.01) = 3.730 (paper: 3.73) |            |             |         |
| law at S=60 dB, look 3: Pd = 1.0000 (expected -> 1)                             |            |             |         |
| law at S=60 dB, look 4: Pd = 0.0000 (expected -> 0)                             |            |             |         |
| OK eq.(7) OS k=8 look 2, 30 dB (strong-target approx.)                          | MC 0.99973 | law 0.99972 | z +0.05 |
| OK eq.(7) OS k=8 look 4, 30 dB (strong-target approx.)                          | MC 0.99966 | law 0.99965 | z +0.23 |
| OK eq.(7) OS look 10 > m=k at 40 dB: Pd = 0.0000 (expected -> 0)                |            |             |         |

```

OK eq.(8)/(9) integration K=8 H0 MC 0.00998 law 0.01000 z -0.16
OK eq.(8)/(9) integration K=8 0 dB MC 0.86949 law 0.86912 z +1.09
OK eq.(8)/(9) integration K=8 5 dB MC 0.95670 law 0.95595 z -1.21
OK eq.(10) P-ANMF known bin, H0 MC 0.01007 law 0.01000 z +0.74
OK eq.(10) P-ANMF known bin, -5 dB MC 0.95225 law 0.95215 z +0.45
OK eq.(10) P-ANMF known bin, 0 dB MC 0.98432 law 0.98450 z -1.45
OK calibration size alpha=0.01: first n = 149 (paper 149); violations beyond 3.15/alpha in next 2/alpha: 0
OK calibration size alpha=0.001: first n = 1497 (paper 1497); violations beyond 3.15/alpha in next 2/alpha: 0
OK calibration size alpha=0.0001: first n = 14978 (paper 14978); violations beyond 3.15/alpha in next 2/alpha: 0
P(Pfa>2a) at n=23000, a=1e-4: 0.0563 (paper 0.056)
OK conditional Pfa ~ Beta(n+1-k,k): KS p = 0.370
OK Thm1 Bernoulli 5% hits, attack=huge : certified clip Pfa 0.0004, binary 0.0003 (bound 0.01)
OK Thm1 Bernoulli 5% hits, attack=matched : certified clip Pfa 0.0004, binary 0.0003 (bound 0.01)
OK Thm1 Bernoulli 5% hits, attack=cancel : certified clip Pfa 0.0003, binary 0.0001 (bound 0.01)
Thm1 MISSPECIFIED (bursty L=6, same 5% rate, calibrated on Bernoulli): clip Pfa 0.0114, binary 0.0192 -> violates the bound, as the dominance condition fails
OK Thm1 bursty hits, calibrated on the bursty model: clip Pfa 0.0000, binary 0.0000

```

ALL CHECKS PASSED (GPU=True, N=2000000)

The guarded post-onset law (Proposition 4 with  $\ell$  replaced by  $\ell - \Delta$ ) is checked separately by `r14/theory_check_guard.py`, also written from scratch. With  $\Delta = 8$  the decay is that of  $\Delta = 0$  delayed by eight looks:

```

Guarded post-onset law vs Monte Carlo (400000 episodes per point; m=16, Pfa=0.01, rho=0.911-0.185j)
guard 0, SCR 10 dB, opposite looks 0-12: MC 0.932 0.979 0.962 0.882 0.154 0.010 0.002 0.001 0.000 0.000 0.000 0.000 0.000
law 0.931 0.979 0.962 0.882 0.154 0.010 0.002 0.001 0.000 0.000 0.000 0.000 0.000 max |z| 1.23
guard 0, SCR 0 dB, opposite looks 0-12: MC 0.536 0.814 0.710 0.482 0.176 0.047 0.015 0.007 0.004 0.002 0.002 0.001 0.001
law 0.536 0.813 0.709 0.483 0.176 0.047 0.016 0.007 0.004 0.002 0.002 0.001 0.001 max |z| 2.18
guard 0, SCR 10 dB, quadrature looks 0-12: MC 0.931 0.953 0.910 0.690 0.088 0.011 0.003 0.001 0.001 0.000 0.000 0.000 0.000
law 0.931 0.952 0.909 0.691 0.088 0.011 0.003 0.001 0.001 0.000 0.000 0.000 0.000 max |z| 1.90
guard 8, SCR 10 dB, opposite looks 0-12: MC 0.931 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.979 0.962 0.883 0.153
law 0.931 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.981 0.979 0.962 0.882 0.154 max |z| 1.65
guard 8, SCR 0 dB, opposite looks 0-12: MC 0.535 0.830 0.830 0.830 0.830 0.830 0.829 0.830 0.830 0.813 0.709 0.485 0.176
law 0.536 0.831 0.831 0.831 0.831 0.831 0.831 0.831 0.831 0.813 0.709 0.483 0.176 max |z| 2.56
guard 8, SCR 10 dB, quadrature looks 0-12: MC 0.932 0.962 0.963 0.962 0.963 0.962 0.962 0.962 0.962 0.952 0.909 0.692 0.088
law 0.931 0.962 0.962 0.962 0.962 0.962 0.962 0.962 0.962 0.952 0.909 0.691 0.088 max |z| 1.42

```

ALL CHECKS PASSED (max |z| = 2.56)

The line for  $n = 23,000$  checks the formula only. The low-false-alarm calibration sets have 17,200–23,100 episodes per unit, for which the same formula gives  $\mathbb{P}\{P_{fa} > 2\alpha\} = 0.02\text{--}0.09$  (Section IV-F).

## 23 The real IPIX target: detection versus dwell length

Table 6: Real IPIX target (primary cell, test thirds, 56 units): mean  $P_d$  at design  $P_{fa} = 10^{-3}$  versus dwell  $N$ . In parentheses: measured clutter  $P_{fa} (\times 10^{-3})$ . P-ANMF with Fisher’s  $g$  uses the analytic threshold.

| $N$  | P-ANMF     |              | Range CA    | ANMF  |
|------|------------|--------------|-------------|-------|
|      | conformal  | Fisher’s $g$ | integration | Tyler |
| 8    | 0.22 (1.1) | 0.37 (13)    | 0.003       | 0.13  |
| 16   | 0.22 (1.3) | 0.45 (32)    | 0.003       | 0.18  |
| 32   | 0.22 (1.3) | 0.56 (69)    | 0.004       | –     |
| 64   | 0.30 (1.3) | 0.67 (121)   | 0.004       | –     |
| 128  | 0.37 (1.2) | 0.75 (178)   | 0.004       | –     |
| 256  | 0.40 (1.5) | 0.79 (235)   | 0.004       | –     |
| 512  | 0.37 (2.0) | 0.83 (299)   | 0.005       | –     |
| 1024 | 0.41 (2.6) | 0.87 (404)   | 0.007       | –     |

Calibration of the self-normalized detector degrades for longer, overlapping windows. Its analytic Fisher’s  $g$  threshold fails already at  $N = 8$ , consistent with imperfect whitening by an AR(4) model fitted on neighboring cells. The value of P-ANMF here is a calibrated short-dwell threshold, not sensitivity.

## 24 77 GHz data: receivers, range bins and thermal noise

Range bins 8–247 of receivers 0, 5, 10 and 15 of both stations are used. As the CNR rises from noise-dominated cells to 10–20 dB, the measured gain of IN-ARCP over local power detection rises from  $-1.45$  dB to  $3.11$  dB, following the sign of Proposition 2.

## 25 Further detector notes

- **Power-detector laws.** At  $10^{-3}$  the white-noise laws of the power detectors err in both directions: CA16 runs at 0.56 times design and the OS detector on raw powers at 5.56 times design.
- **OS variants after onset.** The OS variants with  $k = 12$  and with AR(4) innovations collapse at the first look at which the target occupies more than  $m - k$  history innovations, as (7) states. The pre-specified count was off by one and was corrected after the data were seen (Section 2).

- **Worst case in Theorem 1.** The worst case over amplitudes is attained for the clipped dwell terms, which strong interference drives to  $\kappa$ . It is conservative only through the history scale, whose corrupted entries are set to zero.
- **Certified binary integration.** In the final IPIX study (R12, sparse calibration), the detection probability of certified binary integration for a 10 dB target was within 0.03 of that of certified clipping at both hit rates and powers (output in Section 20).

## 26 Second sea-clutter radar: NetRAD (R15)

Frozen protocol `r15/PROTOCOL_R15.md` (sha256 945c5e51...). It was written before any statistic of the NetRAD samples was computed. Before that point only the following had been read: file sizes, array shapes, and the providers' scripts and log. In addition, two of the providers' own quick-look images were viewed while the dataset was being located. Expectations and verdicts are in `r15/RESULTS_R15.md` (12 of 15 met), and deviations in `r15/DEVIATIONS_R15.md`. The deviations are a corrected site description (the Atlantic side of the Cape Peninsula, not False Bay) and run mechanics; none changes a method, setting or expectation.

*Data.* Node 3 (monostatic) of the NetRAD system: seven HH and seven VV recordings of 130,000 pulses at 1 kHz. The clutter cells are the providers' per-recording range windows (`NR_Range_Selection_June9.m`). Each cell has its complex mean removed and is scaled to unit power. The cross-polarized recordings were set aside before any data were read. All methods are the frozen R12 and R14 code. The only change is that the ANMF secondary cells are restricted to those two to four cells away, which gives 15–30 vectors (20–35 on IPIX). The AR(1) clamp  $|r| \leq 0.98$  binds in 12 of the 14 recordings.

*Unmet expectations of R15.*

- **Calibration at  $10^{-4}$  (N1).** At least 12 of 13 conformal ratios were expected within  $[0.5, 2]$  at  $10^{-4}$ ; 7 were. Outside: Clip-OS1 (0.39), D-IN4 (4.26), NA4 (2.08), P-ANMF (2.33), P-ANMF(bin) (2.52), PAMF-H(w) (2.18). At  $10^{-3}$  all 13 are within  $[0.5, 2]$ , and the 12 non-saturating statistics lie within 1.01–1.36 times design.
- **Pivotality ranking (N4).** IN-ARCP's quintile spread at  $10^{-2}$  was expected to be at most 2 and below CA16's. It is 2.40 against 1.51. In the same analysis on IPIX it is 1.31 against 1.06, so this expectation had no support on IPIX either.
- **Flagged pulses (W2).** Non-coherent integration with a clean threshold was expected to exceed  $1.5\alpha$  in segments with flagged pulses. It stays at 1.08 times design.

Expectation N5 (PAMF-H within 0.5 dB of the best dwell detector) was met, but it is uninformative: every whitened dwell detector reaches  $P_d = 0.5$  at the lowest SCR tested.

*Exploratory (post hoc, after the outcomes and the claim audit).* Two scripts look at where the conformal exceedances at  $10^{-4}$  fall: `r15/explore_r15_hotspots.py` and `r15/explore_r15_attribution.py`.

- **Integration and PAMF-H.** Most of their excess comes from two VV units (1450 and 1501, assignment 3). Their exceedances fall within one to three seconds of the test third and coincide with flagged pulses, while their calibration thirds contain no flagged pulse. Without these two units the pooled ratios are 1.56 (integration) and 1.33 (PAMF-H).
- **P-ANMF statistics.** Their excess comes from one HH unit (1128, assignment 3) with frequent flagged pulses in both thirds. Fewer than half of its exceedances coincide with flagged pulses.
- **Unexplained.** The flagged pulses do not explain the excess of the noise-aware score (2.08). Nor do they explain an excess of integration of more than five times design in one unit that has no flagged pulse at all (1508, assignment 2).

**What the flags are is not established.** `r15/explore_r15_flag_extent.py` counts, in the raw recordings, the range bins above ten times their median:

- a flagged HH pulse has a median of about 108 such bins, of 1024, and 85–91% of them lie outside the clutter window;
- unflagged HH pulses with an elevated window bin have about 80.

The flags are therefore the upper tail of a phenomenon that extends over range. They may be wideband interference: the providers remove WiFi pulses in their own processing of the bistatic nodes. They may equally be range-extended HH sea spikes. The main text therefore calls them flagged pulses, not interference.

In the clutter cells, uncertified non-coherent integration runs at 1.05 times design at  $10^{-2}$  and 1.20 at  $10^{-3}$  over all test segments of the 7 HH recordings, the only ones with at least 0.5% flagged pulses (VV: at most 0.1%). Segments without a flagged pulse give 1.04 and 1.17; segments with flagged pulses give at most 1.08 and 1.33, with wide intervals at  $10^{-3}$  (Part W below). A certificate whose hit model must cover these frequent, bursty masks is nearly vacuous ( $P_d = 0.04$ ). Blanking with mask-matched calibration keeps  $P_d = 0.92$  at 0.57 times design.

Output of `r15/analyze_r15.py` (Parts L, A, G, I, K/X/B and W; intervals resample recordings):

R15 NetRAD analysis: 28 units, 14 recordings (7 HH)

|         |     |   |         |       |      |   |        |     |           |          |         |              |      |
|---------|-----|---|---------|-------|------|---|--------|-----|-----------|----------|---------|--------------|------|
| 1113.58 | HH: | T | 130000, | cells | 101, | r | 0.975, | nu4 | 0.0205,   | hit rate | 0.1654, | mean hit run | 6.00 |
| 1118.34 | HH: | T | 130000, | cells | 101, | r | 0.980, | nu4 | 0.00194,  | hit rate | 0.2283, | mean hit run | 5.98 |
| 1123.49 | HH: | T | 130000, | cells | 101, | r | 0.980, | nu4 | 0.00893,  | hit rate | 0.0427, | mean hit run | 3.74 |
| 1128.12 | HH: | T | 130000, | cells | 71,  | r | 0.980, | nu4 | 0.00223,  | hit rate | 0.0898, | mean hit run | 4.69 |
| 1132.56 | HH: | T | 130000, | cells | 71,  | r | 0.980, | nu4 | 0.00118,  | hit rate | 0.0782, | mean hit run | 4.41 |
| 1139.49 | HH: | T | 130000, | cells | 61,  | r | 0.980, | nu4 | 0.000729, | hit rate | 0.0655, | mean hit run | 4.67 |
| 1146.48 | HH: | T | 130000, | cells | 51,  | r | 0.980, | nu4 | 0.000456, | hit rate | 0.0329, | mean hit run | 4.12 |
| 1446.21 | VV: | T | 130000, | cells | 101, | r | 0.975, | nu4 | 0.0156,   | hit rate | 0.0000, | mean hit run | 1.00 |
| 1450.16 | VV: | T | 130000, | cells | 101, | r | 0.980, | nu4 | 0.00295,  | hit rate | 0.0008, | mean hit run | 1.00 |
| 1457.22 | VV: | T | 130000, | cells | 81,  | r | 0.980, | nu4 | 0.00171,  | hit rate | 0.0000, | mean hit run | 0.00 |
| 1501.53 | VV: | T | 130000, | cells | 71,  | r | 0.980, | nu4 | 0.00319,  | hit rate | 0.0001, | mean hit run | 1.00 |
| 1508.24 | VV: | T | 130000, | cells | 61,  | r | 0.980, | nu4 | 0.00149,  | hit rate | 0.0000, | mean hit run | 0.00 |
| 1512.14 | VV: | T | 130000, | cells | 41,  | r | 0.980, | nu4 | 0.00175,  | hit rate | 0.0001, | mean hit run | 1.10 |
| 1518.58 | VV: | T | 130000, | cells | 41,  | r | 0.980, | nu4 | 0.000916, | hit rate | 0.0000, | mean hit run | 0.00 |

Part L: pooled Pfa / alpha [95% recording-cluster interval]; texture-quintile spread (max/min)

| alpha = 0.01   |      |       |               |        |       |  |  |  |  |
|----------------|------|-------|---------------|--------|-------|--|--|--|--|
| ANMF-FP(w)     | conf | 1.01  | [0.98, 1.03]  | spread | 2.10  |  |  |  |  |
| ANMF-FP(w)     | anal | 2.01  | [1.97, 2.04]  | spread | 2.02  |  |  |  |  |
| ANMF-SCM(w)    | conf | 1.00  | [0.98, 1.02]  | spread | 1.90  |  |  |  |  |
| ANMF-SCM(w)    | anal | 2.27  | [2.20, 2.33]  | spread | 1.81  |  |  |  |  |
| CA16           | conf | 1.01  | [0.99, 1.03]  | spread | 1.51  |  |  |  |  |
| CA16           | anal | 0.32  | [0.27, 0.36]  | spread | 2.07  |  |  |  |  |
| Clip-OS1       | conf | 1.05  | [1.02, 1.10]  | spread | 2.48  |  |  |  |  |
| Clip-OS1       | anal | 2.04  | [1.89, 2.23]  | spread | 1.89  |  |  |  |  |
| D-IN4          | conf | 1.06  | [1.00, 1.15]  | spread | 1.24  |  |  |  |  |
| D-IN4          | anal | 1.44  | [1.36, 1.56]  | spread | 1.19  |  |  |  |  |
| IN1            | conf | 1.04  | [0.99, 1.09]  | spread | 2.40  |  |  |  |  |
| IN1            | anal | 0.80  | [0.63, 0.94]  | spread | 2.34  |  |  |  |  |
| IN1            | boot | 0.82  | [0.55, 1.05]  | spread | 2.20  |  |  |  |  |
| IN4            | conf | 1.03  | [1.00, 1.06]  | spread | 1.11  |  |  |  |  |
| IN4            | anal | 0.98  | [0.95, 1.02]  | spread | 1.12  |  |  |  |  |
| NA4            | conf | 1.08  | [1.00, 1.19]  | spread | 1.42  |  |  |  |  |
| NA4            | anal | 1.90  | [1.75, 2.03]  | spread | 1.40  |  |  |  |  |
| OS1_8          | conf | 1.05  | [0.99, 1.14]  | spread | 1.83  |  |  |  |  |
| OS1_8          | anal | 0.91  | [0.73, 1.07]  | spread | 1.86  |  |  |  |  |
| OSraw          | conf | 1.01  | [0.99, 1.02]  | spread | 1.13  |  |  |  |  |
| OSraw          | anal | 0.65  | [0.60, 0.70]  | spread | 1.11  |  |  |  |  |
| P-ANMF         | conf | 1.04  | [0.97, 1.12]  | spread | 2.04  |  |  |  |  |
| P-ANMF         | anal | 2.46  | [2.29, 2.66]  | spread | 1.70  |  |  |  |  |
| P-ANMF(bin)    | conf | 0.98  | [0.90, 1.09]  | spread | 1.80  |  |  |  |  |
| P-ANMF(bin)    | anal | 2.28  | [1.89, 2.66]  | spread | 1.74  |  |  |  |  |
| PAMF-H(w)      | conf | 1.03  | [0.97, 1.10]  | spread | 1.71  |  |  |  |  |
| PAMF-H(w)      | anal | 2.38  | [2.05, 2.72]  | spread | 1.65  |  |  |  |  |
| alpha = 0.001  |      |       |               |        |       |  |  |  |  |
| ANMF-FP(w)     | conf | 1.01  | [0.94, 1.12]  | spread | 2.20  |  |  |  |  |
| ANMF-FP(w)     | anal | 3.00  | [2.82, 3.21]  | spread | 2.17  |  |  |  |  |
| ANMF-SCM(w)    | conf | 1.02  | [0.95, 1.11]  | spread | 2.06  |  |  |  |  |
| ANMF-SCM(w)    | anal | 3.52  | [3.29, 3.75]  | spread | 1.92  |  |  |  |  |
| CA16           | conf | 1.07  | [1.03, 1.12]  | spread | 2.35  |  |  |  |  |
| CA16           | anal | 0.18  | [0.12, 0.23]  | spread | 3.41  |  |  |  |  |
| Clip-OS1       | conf | 1.13  | [1.05, 1.22]  | spread | 6.69  |  |  |  |  |
| Clip-OS1       | anal | 4.40  | [3.63, 5.45]  | spread | 4.42  |  |  |  |  |
| D-IN4          | conf | 1.36  | [0.96, 1.99]  | spread | 1.60  |  |  |  |  |
| D-IN4          | anal | 2.42  | [1.88, 3.22]  | spread | 1.42  |  |  |  |  |
| IN1            | conf | 1.12  | [0.99, 1.26]  | spread | 2.76  |  |  |  |  |
| IN1            | anal | 0.90  | [0.63, 1.13]  | spread | 2.44  |  |  |  |  |
| IN1            | boot | 0.75  | [0.32, 1.16]  | spread | 2.05  |  |  |  |  |
| IN4            | conf | 1.08  | [0.97, 1.20]  | spread | 1.17  |  |  |  |  |
| IN4            | anal | 1.06  | [0.95, 1.24]  | spread | 1.18  |  |  |  |  |
| NA4            | conf | 1.36  | [1.02, 1.75]  | spread | 1.45  |  |  |  |  |
| NA4            | anal | 4.23  | [3.53, 4.85]  | spread | 1.50  |  |  |  |  |
| OS1_8          | conf | 1.22  | [0.98, 1.50]  | spread | 1.95  |  |  |  |  |
| OS1_8          | anal | 1.17  | [0.80, 1.53]  | spread | 1.98  |  |  |  |  |
| OSraw          | conf | 1.03  | [1.01, 1.06]  | spread | 1.16  |  |  |  |  |
| OSraw          | anal | 1.09  | [0.91, 1.28]  | spread | 1.17  |  |  |  |  |
| P-ANMF         | conf | 1.16  | [0.85, 1.61]  | spread | 2.99  |  |  |  |  |
| P-ANMF         | anal | 4.33  | [3.67, 5.15]  | spread | 2.47  |  |  |  |  |
| P-ANMF(bin)    | conf | 1.15  | [0.79, 1.75]  | spread | 1.81  |  |  |  |  |
| P-ANMF(bin)    | anal | 3.50  | [2.77, 4.24]  | spread | 1.83  |  |  |  |  |
| PAMF-H(w)      | conf | 1.15  | [0.98, 1.42]  | spread | 1.36  |  |  |  |  |
| PAMF-H(w)      | anal | 4.09  | [3.32, 5.08]  | spread | 1.62  |  |  |  |  |
| alpha = 0.0001 |      |       |               |        |       |  |  |  |  |
| ANMF-FP(w)     | conf | 1.18  | [0.91, 1.63]  | spread | 2.47  |  |  |  |  |
| ANMF-FP(w)     | anal | 4.73  | [4.08, 5.81]  | spread | 2.15  |  |  |  |  |
| ANMF-SCM(w)    | conf | 1.10  | [0.91, 1.41]  | spread | 2.08  |  |  |  |  |
| ANMF-SCM(w)    | anal | 5.64  | [4.90, 6.63]  | spread | 1.94  |  |  |  |  |
| CA16           | conf | 1.18  | [1.04, 1.33]  | spread | 3.05  |  |  |  |  |
| CA16           | anal | 0.17  | [0.07, 0.25]  | spread | 5.33  |  |  |  |  |
| Clip-OS1       | conf | 0.39  | [0.13, 0.65]  | spread | 8.78  |  |  |  |  |
| Clip-OS1       | anal | 11.86 | [8.29, 16.54] | spread | 11.98 |  |  |  |  |
| D-IN4          | conf | 4.26  | [1.46, 9.24]  | spread | 2.00  |  |  |  |  |
| D-IN4          | anal | 7.78  | [3.95, 13.54] | spread | 1.77  |  |  |  |  |
| IN1            | conf | 1.30  | [0.94, 1.72]  | spread | 1.79  |  |  |  |  |
| IN1            | anal | 1.70  | [0.91, 2.66]  | spread | 1.56  |  |  |  |  |
| IN1            | boot | 1.22  | [0.52, 1.93]  | spread | 1.13  |  |  |  |  |
| IN4            | conf | 1.31  | [1.02, 1.60]  | spread | 1.19  |  |  |  |  |
| IN4            | anal | 1.85  | [1.05, 3.07]  | spread | 1.44  |  |  |  |  |
| NA4            | conf | 2.08  | [1.14, 3.34]  | spread | 1.68  |  |  |  |  |
| NA4            | anal | 12.69 | [9.13, 16.26] | spread | 1.46  |  |  |  |  |
| OS1_8          | conf | 1.38  | [0.82, 2.05]  | spread | 1.19  |  |  |  |  |
| OS1_8          | anal | 2.76  | [1.25, 4.85]  | spread | 1.55  |  |  |  |  |
| OSraw          | conf | 1.02  | [0.91, 1.10]  | spread | 1.85  |  |  |  |  |
| OSraw          | anal | 2.21  | [1.69, 2.78]  | spread | 1.52  |  |  |  |  |
| P-ANMF         | conf | 2.33  | [0.85, 5.19]  | spread | 4.01  |  |  |  |  |
| P-ANMF         | anal | 9.25  | [6.27, 13.24] | spread | 4.02  |  |  |  |  |
| P-ANMF(bin)    | conf | 2.52  | [0.80, 6.10]  | spread | 2.13  |  |  |  |  |
| P-ANMF(bin)    | anal | 6.59  | [3.88, 11.12] | spread | 1.82  |  |  |  |  |
| PAMF-H(w)      | conf | 2.18  | [1.18, 3.78]  | spread | 1.69  |  |  |  |  |
| PAMF-H(w)      | anal | 8.49  | [5.83, 12.23] | spread | 1.22  |  |  |  |  |

W1: conformal statistics within [0.5, 2]: 1e-3 13/13, 1e-4 7/13; outside: [('Clip-OS1', 0.0001, np.float64(0.39)), ('D-IN4', 0.0001, np.float64(4.26)), ('NA4', 0.0001, np.float64(2.08)), ('P-ANMF', 0.0001, np.float64(2.33)), ('P-ANMF(bin)', 0.0001, np.float64(2.52)), ('PAMF-H(w)', 0.0001, np.float64(2.18))]

W2: whitened model-based laws > 2x at 1e-4: 7 of 9 ['OS1\_8', 'NA4', 'D-IN4', 'PAMF-H(w)', 'P-ANMF(bin)', 'P-ANMF', 'Clip-OS1']

W3: IN1 CA law at 1e-4: 1.70

W4: quintile spread at 1e-2: IN1 2.40, CA16 1.51

Part A: SCR (dB) for Pd = 0.5 / 0.9 (pooled curves, alpha = 0.01), measured Pfa

|           |       |      |       |      |         |      |     |        |
|-----------|-------|------|-------|------|---------|------|-----|--------|
| ANMF-FP   | SCR50 | <=-5 | SCR90 | 6.6  | Pd@10dB | 0.93 | Pfa | 0.0101 |
| ANMF-SCM  | SCR50 | <=-5 | SCR90 | 6.0  | Pd@10dB | 0.94 | Pfa | 0.0103 |
| Bin-OS1   | SCR50 | <=-5 | SCR90 | 4.5  | Pd@10dB | 0.95 | Pfa | 0.0078 |
| Clip-OS1  | SCR50 | <=-5 | SCR90 | 3.7  | Pd@10dB | 0.95 | Pfa | 0.0106 |
| D-IN1     | SCR50 | <=-5 | SCR90 | -0.3 | Pd@10dB | 0.99 | Pfa | 0.0107 |
| D-IN4     | SCR50 | <=-5 | SCR90 | -1.6 | Pd@10dB | 0.99 | Pfa | 0.0105 |
| Exc-D-IN1 | SCR50 | n.r. | SCR90 | n.r. | Pd@10dB | 0.11 | Pfa | 0.0106 |
| IN1look   | SCR50 | <=-5 | SCR90 | 2.8  | Pd@10dB | 0.98 | Pfa | 0.0097 |
| OSraw     | SCR50 | 7.2  | SCR90 | 17.8 | Pd@10dB | 0.64 | Pfa | 0.0101 |

P-ANMF SCR50 <=-5 SCR90 n.r. Pd@10dB 0.81 Pfa 0.0115  
 PAMF-H SCR50 <=-5 SCR90 -3.4 Pd@10dB 0.99 Pfa 0.0104  
 N5: PAMF-H SCR50 <=-5 vs best dwell detector <=-5 (['PAMF-H', 'D-IN1', 'D-IN4', 'Clip-OS1', 'Bin-OS1', 'P-ANMF', 'ANMF-SCM', 'ANMF-PP'])  
 N6: OSraw minus IN1look at Pd 0.5: inf dB

#### Part G: guarded history scale

g=0 alpha=0.01: clean Pfa/alpha 1.05 [0.98, 1.14]  
 g=0 alpha=0.001: clean Pfa/alpha 1.22 [0.94, 1.65]  
 g=8 alpha=0.01: clean Pfa/alpha 1.05 [1.00, 1.11]  
 g=8 alpha=0.001: clean Pfa/alpha 1.23 [0.98, 1.64]  
 g=16 alpha=0.01: clean Pfa/alpha 1.05 [1.00, 1.12]  
 g=16 alpha=0.001: clean Pfa/alpha 1.36 [1.05, 1.87]  
 random g=0: per-look Pd@10dB 0.97 0.79 0.74 0.64 0.24 0.01 0.00 0.00 0.00 | mean looks1-8 0.30 | SCR50 look0 <=-5  
 random g=8: per-look Pd@10dB 0.97 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 | mean looks1-8 0.92 | SCR50 look0 <=-5  
 random g=16: per-look Pd@10dB 0.97 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 | mean looks1-8 0.92 | SCR50 look0 <=-5  
 opposite g=0: per-look Pd@10dB 0.97 0.99 0.98 0.96 0.50 0.01 0.00 0.00 0.00 | mean looks1-8 0.43 | SCR50 look0 <=-5  
 opposite g=8: per-look Pd@10dB 0.97 0.99 0.99 0.99 0.99 0.99 0.99 0.99 0.99 | mean looks1-8 0.99 | SCR50 look0 <=-5  
 opposite g=16: per-look Pd@10dB 0.97 0.99 0.99 0.99 0.99 0.99 0.99 0.99 0.99 | mean looks1-8 0.99 | SCR50 look0 <=-5  
 matched g=0: per-look Pd@10dB 0.97 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 | mean looks1-8 0.00 | SCR50 look0 <=-5  
 matched g=8: per-look Pd@10dB 0.97 0.03 0.03 0.03 0.03 0.03 0.03 0.03 0.03 | mean looks1-8 0.03 | SCR50 look0 <=-5  
 matched g=16: per-look Pd@10dB 0.97 0.03 0.03 0.03 0.03 0.03 0.03 0.03 0.03 | mean looks1-8 0.03 | SCR50 look0 <=-5  
 law (exploratory, AR(1) no noise) random g=0: 0.98 0.79 0.74 0.61 0.05 0.00 0.00 0.00 0.00  
 law (exploratory, AR(1) no noise) random g=8: 0.98 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92  
 law (exploratory, AR(1) no noise) random g=16: 0.98 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92  
 ramp L=8 g=0: detection within 8 looks @10dB 0.862  
 ramp L=8 g=8: detection within 8 looks @10dB 0.964  
 ramp L=8 g=16: detection within 8 looks @10dB 0.966  
 ramp L=16 g=0: detection within 8 looks @10dB 0.016  
 ramp L=16 g=8: detection within 8 looks @10dB 0.937  
 ramp L=16 g=16: detection within 8 looks @10dB 0.959  
 N7: g=0 look0 0.97, look4 0.24; N8: g=8 mean looks1-8 0.92 vs g=0 0.30; N9: g=8 clean Pfa/a 1.05; onset cost nan dB

#### Part I (R12 code): injected interference, pooled Pfa / alpha [upper], Pd@10dB, alpha = 0.01

|                            |       |               |              |
|----------------------------|-------|---------------|--------------|
| p=0.02 10 dB D-IN1         | Pfa/a | 12.55 [12.85] | Pd@10dB 0.92 |
| p=0.02 10 dB PAMF-H        | Pfa/a | 12.77 [13.37] | Pd@10dB 0.95 |
| p=0.02 10 dB P-ANMF        | Pfa/a | 0.90 [ 1.02]  | Pd@10dB 0.76 |
| p=0.02 10 dB ANMF-PP       | Pfa/a | 1.04 [ 1.11]  | Pd@10dB 0.88 |
| p=0.02 10 dB Clip-OS1      | Pfa/a | 1.28 [ 1.37]  | Pd@10dB 0.95 |
| p=0.02 10 dB Clip-OS1-cert | Pfa/a | 0.46 [ 0.52]  | Pd@10dB 0.94 |
| p=0.02 10 dB Bin-OS1-cert  | Pfa/a | 0.27 [ 0.33]  | Pd@10dB 0.93 |
| p=0.02 30 dB D-IN1         | Pfa/a | 13.07 [13.20] | Pd@10dB 0.74 |
| p=0.02 30 dB PAMF-H        | Pfa/a | 13.52 [13.83] | Pd@10dB 0.77 |
| p=0.02 30 dB P-ANMF        | Pfa/a | 0.90 [ 1.01]  | Pd@10dB 0.65 |
| p=0.02 30 dB ANMF-PP       | Pfa/a | 1.05 [ 1.11]  | Pd@10dB 0.80 |
| p=0.02 30 dB Clip-OS1      | Pfa/a | 1.30 [ 1.39]  | Pd@10dB 0.95 |
| p=0.02 30 dB Clip-OS1-cert | Pfa/a | 0.48 [ 0.54]  | Pd@10dB 0.94 |
| p=0.02 30 dB Bin-OS1-cert  | Pfa/a | 0.27 [ 0.33]  | Pd@10dB 0.93 |
| p=0.05 10 dB D-IN1         | Pfa/a | 21.31 [21.80] | Pd@10dB 0.84 |
| p=0.05 10 dB PAMF-H        | Pfa/a | 20.84 [21.81] | Pd@10dB 0.89 |
| p=0.05 10 dB P-ANMF        | Pfa/a | 0.62 [ 0.69]  | Pd@10dB 0.70 |
| p=0.05 10 dB ANMF-PP       | Pfa/a | 1.23 [ 1.33]  | Pd@10dB 0.80 |
| p=0.05 10 dB Clip-OS1      | Pfa/a | 1.86 [ 2.04]  | Pd@10dB 0.95 |
| p=0.05 10 dB Clip-OS1-cert | Pfa/a | 0.09 [ 0.11]  | Pd@10dB 0.92 |
| p=0.05 10 dB Bin-OS1-cert  | Pfa/a | 0.05 [ 0.07]  | Pd@10dB 0.91 |
| p=0.05 30 dB D-IN1         | Pfa/a | 22.03 [22.21] | Pd@10dB 0.52 |
| p=0.05 30 dB PAMF-H        | Pfa/a | 21.85 [22.35] | Pd@10dB 0.55 |
| p=0.05 30 dB P-ANMF        | Pfa/a | 0.61 [ 0.68]  | Pd@10dB 0.47 |
| p=0.05 30 dB ANMF-PP       | Pfa/a | 1.46 [ 1.58]  | Pd@10dB 0.63 |
| p=0.05 30 dB Clip-OS1      | Pfa/a | 1.88 [ 2.08]  | Pd@10dB 0.95 |
| p=0.05 30 dB Clip-OS1-cert | Pfa/a | 0.10 [ 0.13]  | Pd@10dB 0.92 |
| p=0.05 30 dB Bin-OS1-cert  | Pfa/a | 0.05 [ 0.07]  | Pd@10dB 0.91 |

#### Parts K/X/B (R14 code): dwell integration, Pfa/alpha [upper], Pd@10dB

|                                                                |              |
|----------------------------------------------------------------|--------------|
| clean clip4 alpha=0.01: Pfa/a 1.07 [1.03, 1.12]                | Pd@10dB 0.95 |
| clean clip4 alpha=0.001: Pfa/a 0.94 [0.77, 1.08]               | Pd@10dB 0.73 |
| clean clip6 alpha=0.01: Pfa/a 1.05 [1.02, 1.10]                | Pd@10dB 0.95 |
| clean clip6 alpha=0.001: Pfa/a 1.14 [1.06, 1.23]               | Pd@10dB 0.93 |
| clean clip9 alpha=0.01: Pfa/a 1.05 [1.02, 1.08]                | Pd@10dB 0.95 |
| clean clip9 alpha=0.001: Pfa/a 1.10 [1.03, 1.19]               | Pd@10dB 0.93 |
| clean clip12 alpha=0.01: Pfa/a 1.05 [1.03, 1.08]               | Pd@10dB 0.95 |
| clean clip12 alpha=0.001: Pfa/a 1.11 [1.05, 1.19]              | Pd@10dB 0.93 |
| clean clip18 alpha=0.01: Pfa/a 1.08 [1.03, 1.14]               | Pd@10dB 0.95 |
| clean clip18 alpha=0.001: Pfa/a 1.13 [1.07, 1.20]              | Pd@10dB 0.93 |
| clean blank alpha=0.01: Pfa/a 1.05 [1.03, 1.08]                | Pd@10dB 0.94 |
| clean blank alpha=0.001: Pfa/a 1.08 [1.03, 1.15]               | Pd@10dB 0.92 |
| bern p=0.02 30 dB a=0.01 clean-thr clip6 Pfa/a 1.31 [ 1.36]    | Pd@10dB 0.95 |
| bern p=0.02 30 dB a=0.01 cert(bern) clip6 Pfa/a 0.46 [ 0.51]   | Pd@10dB 0.94 |
| bern p=0.02 30 dB a=0.01 cert(bern) clip9 Pfa/a 0.43 [ 0.47]   | Pd@10dB 0.94 |
| bern p=0.02 30 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.02 30 dB a=0.01 blank(bern masks) Pfa/a 1.10 [ 1.13]  | Pd@10dB 0.94 |
| bern p=0.02 30 dB a=0.001 clean-thr clip6 Pfa/a 1.23 [ 1.32]   | Pd@10dB 0.93 |
| bern p=0.02 30 dB a=0.001 cert(bern) clip6 Pfa/a 0.33 [ 0.40]  | Pd@10dB 0.87 |
| bern p=0.02 30 dB a=0.001 cert(bern) clip9 Pfa/a 0.26 [ 0.32]  | Pd@10dB 0.91 |
| bern p=0.02 30 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] | Pd@10dB 0.00 |
| bern p=0.02 30 dB a=0.001 blank(bern masks) Pfa/a 1.18 [ 1.26] | Pd@10dB 0.92 |
| bern p=0.05 30 dB a=0.01 clean-thr clip6 Pfa/a 1.95 [ 2.09]    | Pd@10dB 0.95 |
| bern p=0.05 30 dB a=0.01 cert(bern) clip6 Pfa/a 0.08 [ 0.10]   | Pd@10dB 0.92 |
| bern p=0.05 30 dB a=0.01 cert(bern) clip9 Pfa/a 0.07 [ 0.08]   | Pd@10dB 0.91 |
| bern p=0.05 30 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 30 dB a=0.01 blank(bern masks) Pfa/a 1.25 [ 1.30]  | Pd@10dB 0.94 |
| bern p=0.05 30 dB a=0.001 clean-thr clip6 Pfa/a 1.61 [ 1.88]   | Pd@10dB 0.92 |
| bern p=0.05 30 dB a=0.001 cert(bern) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 30 dB a=0.001 cert(bern) clip9 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 30 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] | Pd@10dB 0.00 |
| bern p=0.05 30 dB a=0.001 blank(bern masks) Pfa/a 2.36 [ 2.45] | Pd@10dB 0.91 |
| bern p=0.02 10 dB a=0.01 clean-thr clip6 Pfa/a 1.31 [ 1.36]    | Pd@10dB 0.95 |
| bern p=0.02 10 dB a=0.01 cert(bern) clip6 Pfa/a 0.46 [ 0.51]   | Pd@10dB 0.94 |
| bern p=0.02 10 dB a=0.01 cert(bern) clip9 Pfa/a 0.43 [ 0.47]   | Pd@10dB 0.94 |
| bern p=0.02 10 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.02 10 dB a=0.01 blank(bern masks) Pfa/a 1.10 [ 1.13]  | Pd@10dB 0.94 |
| bern p=0.02 10 dB a=0.001 clean-thr clip6 Pfa/a 1.23 [ 1.32]   | Pd@10dB 0.93 |
| bern p=0.02 10 dB a=0.001 cert(bern) clip6 Pfa/a 0.31 [ 0.38]  | Pd@10dB 0.87 |
| bern p=0.02 10 dB a=0.001 cert(bern) clip9 Pfa/a 0.25 [ 0.31]  | Pd@10dB 0.91 |
| bern p=0.02 10 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] | Pd@10dB 0.00 |
| bern p=0.02 10 dB a=0.001 blank(bern masks) Pfa/a 1.16 [ 1.24] | Pd@10dB 0.92 |
| bern p=0.05 10 dB a=0.01 clean-thr clip6 Pfa/a 1.93 [ 2.05]    | Pd@10dB 0.95 |
| bern p=0.05 10 dB a=0.01 cert(bern) clip6 Pfa/a 0.08 [ 0.10]   | Pd@10dB 0.91 |
| bern p=0.05 10 dB a=0.01 cert(bern) clip9 Pfa/a 0.07 [ 0.09]   | Pd@10dB 0.91 |
| bern p=0.05 10 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 10 dB a=0.01 blank(bern masks) Pfa/a 1.24 [ 1.28]  | Pd@10dB 0.94 |
| bern p=0.05 10 dB a=0.001 clean-thr clip6 Pfa/a 1.64 [ 1.88]   | Pd@10dB 0.92 |
| bern p=0.05 10 dB a=0.001 cert(bern) clip6 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 10 dB a=0.001 cert(bern) clip9 Pfa/a 0.00 [ 0.00]  | Pd@10dB 0.00 |
| bern p=0.05 10 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] | Pd@10dB 0.00 |
| bern p=0.05 10 dB a=0.001 blank(bern masks) Pfa/a 2.20 [ 2.28] | Pd@10dB 0.91 |
| burst p=0.02 30 dB a=0.01 clean-thr clip6 Pfa/a 2.73 [ 2.79]   | Pd@10dB 0.93 |
| burst p=0.02 30 dB a=0.01 cert(bern) clip6 Pfa/a 1.69 [ 1.74]  | Pd@10dB 0.92 |

```

burst p=0.02 30 dB a=0.01 cert(bern) clip9 Pfa/a 2.00 [ 2.05] Pd010dB 0.92
burst p=0.02 30 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.02 30 dB a=0.01 blank(bern masks) Pfa/a 2.29 [ 2.33] Pd010dB 0.94
burst p=0.02 30 dB a=0.001 clean-thr clip6 Pfa/a 9.73 [10.77] Pd010dB 0.91
burst p=0.02 30 dB a=0.001 cert(bern) clip6 Pfa/a 5.59 [ 5.87] Pd010dB 0.85
burst p=0.02 30 dB a=0.001 cert(bern) clip9 Pfa/a 8.27 [ 8.93] Pd010dB 0.89
burst p=0.02 30 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.02 30 dB a=0.001 blank(bern masks) Pfa/a 13.89 [14.00] Pd010dB 0.91
burst p=0.05 30 dB a=0.01 clean-thr clip6 Pfa/a 5.14 [ 5.26] Pd010dB 0.90
burst p=0.05 30 dB a=0.01 cert(bern) clip6 Pfa/a 1.66 [ 1.70] Pd010dB 0.87
burst p=0.05 30 dB a=0.01 cert(bern) clip9 Pfa/a 2.53 [ 2.56] Pd010dB 0.87
burst p=0.05 30 dB a=0.01 cert(burst) clip6 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.05 30 dB a=0.01 blank(bern masks) Pfa/a 4.11 [ 4.16] Pd010dB 0.93
burst p=0.05 30 dB a=0.001 clean-thr clip6 Pfa/a 22.52 [25.03] Pd010dB 0.88
burst p=0.05 30 dB a=0.001 cert(bern) clip6 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.05 30 dB a=0.001 cert(bern) clip9 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.05 30 dB a=0.001 cert(burst) clip6 Pfa/a 0.00 [ 0.00] Pd010dB 0.00
burst p=0.05 30 dB a=0.001 blank(bern masks) Pfa/a 33.54 [33.72] Pd010dB 0.90

```

Part W: real interference (hit rule: >10% of range bins above 10x the bin median)

```

recordings with >= 0.5% hit pulses: 7 of 14
a=0.01 D-IN1|clean all 1.05 [ 1.12] Pd10 0.98 | s0 1.04 [ 1.11] Pd10 0.98 | s12 1.08 [ 1.26] Pd10 0.97 | s3 1.08 [ 1.20] Pd10 0.97
a=0.01 clip6|clean all 1.04 [ 1.12] Pd10 0.94 | s0 1.04 [ 1.11] Pd10 0.94 | s12 1.08 [ 1.20] Pd10 0.93 | s3 1.03 [ 1.23] Pd10 0.93
a=0.01 clip9|clean all 1.03 [ 1.09] Pd10 0.94 | s0 1.03 [ 1.08] Pd10 0.94 | s12 1.06 [ 1.16] Pd10 0.93 | s3 1.00 [ 1.19] Pd10 0.93
a=0.01 clip6|cert all 0.04 [ 0.15] Pd10 0.04 | s0 0.04 [ 0.14] Pd10 0.05 | s12 0.05 [ 0.25] Pd10 0.04 | s3 0.04 [ 0.21] Pd10 0.04
a=0.01 clip9|cert all 0.04 [ 0.14] Pd10 0.04 | s0 0.04 [ 0.13] Pd10 0.05 | s12 0.05 [ 0.22] Pd10 0.04 | s3 0.03 [ 0.19] Pd10 0.04
a=0.01 blank|mask all 0.57 [ 0.82] Pd10 0.92 | s0 0.61 [ 0.80] Pd10 0.92 | s12 0.55 [ 0.94] Pd10 0.91 | s3 0.45 [ 0.94] Pd10 0.91
a=0.001 D-IN1|clean all 1.20 [ 1.53] Pd10 0.96 | s0 1.17 [ 1.43] Pd10 0.96 | s12 1.09 [ 1.88] Pd10 0.96 | s3 1.33 [ 1.96] Pd10 0.96
a=0.001 clip6|clean all 1.10 [ 1.18] Pd10 0.91 | s0 1.11 [ 1.22] Pd10 0.91 | s12 1.00 [ 1.18] Pd10 0.91 | s3 1.08 [ 1.24] Pd10 0.91
a=0.001 clip9|clean all 1.08 [ 1.19] Pd10 0.91 | s0 1.10 [ 1.23] Pd10 0.92 | s12 1.00 [ 1.24] Pd10 0.91 | s3 1.02 [ 1.19] Pd10 0.91
a=0.001 clip6|cert all 0.00 [ 0.00] Pd10 0.00 | s0 0.00 [ 0.00] Pd10 0.00 | s12 0.00 [ 0.00] Pd10 0.00 | s3 0.00 [ 0.00] Pd10 0.00
a=0.001 clip9|cert all 0.00 [ 0.00] Pd10 0.00 | s0 0.00 [ 0.00] Pd10 0.00 | s12 0.00 [ 0.00] Pd10 0.00 | s3 0.00 [ 0.00] Pd10 0.00
a=0.001 blank|mask all 0.24 [ 0.49] Pd10 0.83 | s0 0.25 [ 0.48] Pd10 0.84 | s12 0.24 [ 0.57] Pd10 0.81 | s3 0.18 [ 0.62] Pd10 0.79

```

Verdicts (frozen expectations): N1 NOT met, N2 met, N3 met, N4 NOT met, N5 met, N6 met, N7 met, N8 met, N9 met, N10 met, N11 met, N12 met, N13 met, W1 met, W2 NOT met  
met 12 of 15

## Exploratory hotspot analysis (r15/explore\_r15\_hotspots.py):

```

1450.16 rot3 (VV, hit rate 0.001), test pulses (43333, 84666), cells 101
D-IN4 923 exceedances of 173518 (Pfa/a 53.19); cells with most: 62:15, 63:14, 30:13 (share 0.05); 1-s blocks with most: 40s:604, 39s:288, 41s:18 (share 0.99); with a
    flagged hit 0.99
PAMF-H(w) 318 exceedances of 173518 (Pfa/a 18.33); cells with most: 49:7, 46:7, 48:6 (share 0.06); 1-s blocks with most: 40s:261, 39s:45, 41s:2 (share 0.97); with a
    flagged hit 0.97
P-ANMF 10 exceedances of 173518 (Pfa/a 0.58); cells with most: 81:1, 66:1, 53:1 (share 0.30); 1-s blocks with most: 13s:2, 9s:2, 35s:1 (share 0.50); with a flagged
    hit 0.00
IN1 0 exceedances of 245531 (Pfa/a 0.00); cells with most: 100:0, 99:0, 98:0 (share 0.00); 1-s blocks with most: 41s:0, 40s:0, 39s:0 (share 0.00); with a
    flagged hit nan

1128.12 rot3 (HH, hit rate 0.090), test pulses (43333, 84666), cells 71
D-IN4 0 exceedances of 121978 (Pfa/a 0.00); cells with most: 70:0, 69:0, 68:0 (share 0.00); 1-s blocks with most: 41s:0, 40s:0, 39s:0 (share 0.00); with a flagged
    hit nan
PAMF-H(w) 0 exceedances of 121978 (Pfa/a 0.00); cells with most: 70:0, 69:0, 68:0 (share 0.00); 1-s blocks with most: 41s:0, 40s:0, 39s:0 (share 0.00); with a flagged
    hit nan
P-ANMF 467 exceedances of 121978 (Pfa/a 38.29); cells with most: 38:25, 67:24, 66:23 (share 0.15); 1-s blocks with most: 28s:102, 37s:88, 34s:85 (share 0.59); with a
    flagged hit 0.48
IN1 16 exceedances of 172601 (Pfa/a 0.93); cells with most: 60:3, 50:1, 67:1 (share 0.31); 1-s blocks with most: 4s:3, 21s:3, 36s:2 (share 0.50); with a flagged
    hit 0.25

1501.53 rot3 (VV, hit rate 0.000), test pulses (43333, 84666), cells 71
D-IN4 101 exceedances of 121978 (Pfa/a 8.28); cells with most: 6:3, 1:3, 2:3 (share 0.09); 1-s blocks with most: 1s:88, 10s:2, 4s:2 (share 0.91); with a flagged hit
    0.86
PAMF-H(w) 27 exceedances of 121978 (Pfa/a 2.21); cells with most: 33:2, 1:2, 38:2 (share 0.22); 1-s blocks with most: 1s:17, 39s:1, 22s:1 (share 0.70); with a flagged
    hit 0.63
P-ANMF 17 exceedances of 121978 (Pfa/a 1.39); cells with most: 68:1, 34:1, 37:1 (share 0.18); 1-s blocks with most: 17s:3, 24s:2, 29s:1 (share 0.35); with a flagged
    hit 0.00
IN1 110 exceedances of 172601 (Pfa/a 6.37); cells with most: 66:3, 45:3, 54:3 (share 0.08); 1-s blocks with most: 16s:59, 2s:37, 36s:2 (share 0.89); with a
    flagged hit 0.34

1123.49 rot3 (HH, hit rate 0.043), test pulses (43333, 84666), cells 101
D-IN4 18 exceedances of 173518 (Pfa/a 1.04); cells with most: 92:1, 91:1, 90:1 (share 0.17); 1-s blocks with most: 0s:18, 41s:0, 39s:0 (share 1.00); with a flagged
    hit 0.00
PAMF-H(w) 22 exceedances of 173518 (Pfa/a 1.27); cells with most: 92:1, 91:1, 90:1 (share 0.14); 1-s blocks with most: 0s:20, 40s:1, 16s:1 (share 1.00); with a flagged
    hit 0.00
P-ANMF 19 exceedances of 173518 (Pfa/a 1.09); cells with most: 9:4, 8:3, 11:2 (share 0.47); 1-s blocks with most: 5s:12, 10s:2, 27s:1 (share 0.79); with a flagged
    hit 0.16
IN1 26 exceedances of 245531 (Pfa/a 1.06); cells with most: 79:3, 1:2, 17:2 (share 0.27); 1-s blocks with most: 7s:3, 28s:3, 31s:2 (share 0.31); with a flagged
    hit 0.46

```

## Exploratory attribution (r15/explore\_r15\_attribution.py): flagged-pulse fraction in each unit's calibration and test thirds, the per-unit ratios at $10^{-4}$ , and the pooled ratios without the units named above:

```

EXPLORATORY. Per unit: flagged fraction in calibration / test third; Pfa/alpha at 1e-4 (conformal)
unit pol cal% test% IN1 IN4 OS1.8 NA4 CA16 OSraw D-IN4 PAMF-H(w) P-ANMF(bin) P-ANMF ANMF-SCM(w ANMF-FP(w) Clip-OS1
1113_58_rot2 HH 16.71 12.86 0.81 0.53 0.86 1.10 0.94 1.63 2.07 2.07 0.46 1.96 2.07 2.19 1.27
1113_58_rot3 HH 12.86 18.36 2.12 2.44 1.83 2.40 1.14 0.73 0.00 0.17 2.48 0.86 0.52 0.52 0.46
1118_34_rot2 HH 19.53 29.43 2.20 2.73 0.98 2.93 0.69 0.90 1.50 1.90 0.00 0.00 0.29 0.23 0.00
1118_34_rot3 HH 29.43 18.63 0.00 0.00 0.00 0.00 1.38 1.38 1.56 1.33 0.75 0.00 1.50 1.33 1.44
1123_49_rot2 HH 3.50 1.56 0.81 0.86 0.90 1.43 1.06 0.57 3.98 2.36 1.56 2.36 0.86 0.92 1.33
1123_49_rot3 HH 1.56 8.09 1.06 1.59 0.73 0.65 0.98 1.43 1.04 1.27 0.63 1.09 1.09 1.27 1.21
1128_12_rot2 HH 16.79 3.65 1.27 0.98 1.33 2.20 1.74 1.68 2.13 2.05 1.07 1.15 0.74 0.66 0.00
1128_12_rot3 HH 3.65 5.89 0.93 1.33 1.27 0.64 0.93 0.87 0.00 0.00 47.55 38.29 5.00 6.80 0.00
1132_56_rot2 HH 14.46 2.32 0.64 0.70 1.22 0.41 1.27 1.39 3.69 2.46 0.00 0.00 0.49 0.33 0.00
1132_56_rot3 HH 2.32 6.95 0.87 1.10 0.93 2.26 0.70 0.64 1.80 1.97 0.57 1.15 0.98 0.90 0.00
1139_49_rot2 HH 4.07 2.21 0.00 0.07 0.00 0.00 1.01 1.21 0.29 0.76 1.62 0.67 0.48 0.67 0.00
1139_49_rot3 HH 2.21 13.81 0.20 0.27 0.61 0.81 1.48 1.28 1.62 0.86 0.19 1.53 1.24 0.86 0.00
1146_48_rot2 HH 2.23 0.22 1.94 1.29 0.65 0.24 1.37 0.32 5.25 5.02 1.83 2.97 1.37 1.37 0.00
1146_48_rot3 HH 0.22 7.75 0.00 0.40 0.00 0.08 0.73 1.45 0.00 0.00 0.23 0.00 0.23 0.46 0.00
1446_21_rot2 VV 0.00 0.00 2.08 1.22 2.65 1.87 1.14 0.81 1.67 0.98 0.63 2.54 0.69 0.75 1.21
1446_21_rot3 VV 0.00 0.00 0.61 1.51 0.00 12.54 2.36 1.26 0.86 0.98 1.15 0.40 1.04 0.98 0.58
1450_16_rot2 VV 0.00 0.00 4.32 4.40 3.95 4.32 0.94 0.98 0.00 0.00 1.50 0.86 0.69 0.92 0.00
1450_16_rot3 VV 0.00 0.24 0.00 0.00 2.40 0.04 2.08 1.22 53.19 18.33 0.63 0.58 0.98 0.81 0.00
1457_22_rot2 VV 0.00 0.00 0.15 0.91 0.71 0.71 0.66 0.81 1.87 0.50 0.65 1.22 0.93 1.51 0.00
1457_22_rot3 VV 0.00 0.00 1.78 1.57 1.12 0.66 1.22 0.81 0.29 0.50 1.65 0.57 1.51 0.93 0.00
1501_53_rot2 VV 0.00 0.00 0.70 0.52 1.04 0.35 0.52 0.46 1.07 1.07 0.90 1.48 0.49 0.66 0.00
1501_53_rot3 VV 0.00 0.02 6.37 4.52 8.23 9.73 1.62 1.22 8.28 2.21 1.07 1.39 1.80 1.72 1.07
1508_24_rot2 VV 0.00 0.00 1.75 1.35 1.48 2.23 1.75 1.75 6.20 5.63 0.95 0.95 1.34 1.34 0.00
1508_24_rot3 VV 0.00 0.00 0.88 1.28 0.81 0.67 0.81 0.47 1.24 0.48 0.67 1.62 0.48 0.95 0.00
1512_14_rot2 VV 0.03 0.00 0.00 0.00 0.00 0.00 0.80 1.10 0.00 0.00 1.28 1.28 0.14 0.28 0.00
1512_14_rot3 VV 0.00 0.00 1.71 1.00 1.00 1.20 0.70 0.30 0.57 1.14 1.56 0.71 1.70 2.13 0.00
1518_58_rot2 VV 0.00 0.00 1.00 0.80 0.90 2.81 1.61 0.30 1.14 0.43 0.28 1.42 1.28 0.85 0.00
1518_58_rot3 VV 0.00 0.00 0.50 0.80 0.70 0.30 0.50 0.50 0.85 1.42 1.42 0.43 1.14 1.28 0.00

```

```

units whose calibration third has no flagged pulse but whose test third has some: ['1446_21_rot3', '1450_16_rot3', '1501_53_rot3']
pooled Pfa/alpha at 1e-4, all units vs without ['1450_16_rot3', '1501_53_rot3']:
IN1      1.30 -> 1.18
IN4      1.31 -> 1.26
OS1_8    1.38 -> 1.07
NA4      2.08 -> 1.91
CA16     1.18 -> 1.11
OSraw    1.02 -> 1.00
D-IN4    4.26 -> 1.56
PAMF-H(w) 2.18 -> 1.33
P-ANMF(bin) 2.52 -> 2.67
P-ANMF    2.33 -> 2.45
ANMF-SCM(w) 1.10 -> 1.09
ANMF-FP(w) 1.18 -> 1.18
Clip-OS1 0.39 -> 0.39
pooled Pfa/alpha at 1e-4, all units vs without ['1450_16_rot3', '1501_53_rot3', '1128_12_rot3']:
IN1      1.30 -> 1.19
IN4      1.31 -> 1.26
OS1_8    1.38 -> 1.07
NA4      2.08 -> 1.96
CA16     1.18 -> 1.12
OSraw    1.02 -> 1.01
D-IN4    4.26 -> 1.62
PAMF-H(w) 2.18 -> 1.39
P-ANMF(bin) 2.52 -> 0.96
P-ANMF    2.33 -> 1.09
ANMF-SCM(w) 1.10 -> 0.94
ANMF-FP(w) 1.18 -> 0.97
Clip-OS1 0.39 -> 0.41

```

Exploratory range extent of the flagged pulses (r15/explore\_r15\_flag\_extent.py):

```

EXPLORATORY. Flagged pulses: number of range bins above 10x their median, and share of them outside the clutter window
1113_58 HH: flagged 0.1654 of pulses; elevated bins per flagged pulse median 109 (5-95%: 103-126) of 1024, outside the 101-bin window 0.87 (median share); unflagged pulses
with an elevated window bin: elevated bins median 86
1118_34 HH: flagged 0.2283 of pulses; elevated bins per flagged pulse median 109 (5-95%: 103-123) of 1024, outside the 101-bin window 0.85 (median share); unflagged pulses
with an elevated window bin: elevated bins median 89
1123_49 HH: flagged 0.0427 of pulses; elevated bins per flagged pulse median 108 (5-95%: 103-124) of 1024, outside the 101-bin window 0.87 (median share); unflagged pulses
with an elevated window bin: elevated bins median 76
1128_12 HH: flagged 0.0898 of pulses; elevated bins per flagged pulse median 107 (5-95%: 103-121) of 1024, outside the 71-bin window 0.90 (median share); unflagged pulses with
an elevated window bin: elevated bins median 82
1132_56 HH: flagged 0.0782 of pulses; elevated bins per flagged pulse median 107 (5-95%: 103-119) of 1024, outside the 71-bin window 0.90 (median share); unflagged pulses with
an elevated window bin: elevated bins median 81
1139_49 HH: flagged 0.0655 of pulses; elevated bins per flagged pulse median 108 (5-95%: 103-124) of 1024, outside the 61-bin window 0.91 (median share); unflagged pulses with
an elevated window bin: elevated bins median 77
1146_48 HH: flagged 0.0329 of pulses; elevated bins per flagged pulse median 107 (5-95%: 103-119) of 1024, outside the 51-bin window 0.91 (median share); unflagged pulses with
an elevated window bin: elevated bins median 76

```

## 27 Closed forms against Monte Carlo: all six panels

## 28 Dwell detection on IPIX

Table 7: SCR (dB) for  $P_d = 0.5$  and  $0.9$  over an 8-pulse dwell: persistent Swerling-1 target with random Doppler from the first dwell pulse (IPIX, conformal thresholds).  $\leq -5$ : reached at the lowest SCR tested; n.r.: not reached at 25 dB.

| Detector             | $\alpha = 0.01$ |      | 0.001 |
|----------------------|-----------------|------|-------|
|                      | $P_d = 0.5$     | 0.9  | 0.5   |
| PAMF-H               | $\leq -5$       | 5.1  | -4.3  |
| Integration, AR(4)   | $\leq -5$       | 7.0  | -1.6  |
| Integration, AR(1)   | -4.4            | 8.3  | -0.1  |
| Clipped, OS scale    | -2.3            | 13.0 | n.r.  |
| Binary, OS scale     | -1.2            | 14.5 | n.r.  |
| ANMF-SCM             | -0.2            | 20.3 | 9.5   |
| ANMF-Tyler           | 0.4             | 20.3 | 9.8   |
| P-ANMF               | 0.2             | n.r. | n.r.  |
| IN-ARCP, first look  | -1.2            | 11.0 | 1.9   |
| Power OS, first look | 9.4             | 19.5 | 14.3  |

## 29 77 GHz radar: full results

```

units 24 (station x Rx x rotation); test episodes per CNR class {'<-5': 721095, '-5..0': 265416, '0..5': 229845, '5..10': 68556, '10..20': 31870, '>=20': 338}
mean CNR per class (dB): -12.9 -2.2 2.4 7.2 13.4 20.1

=== 1-alpha = 0.9 ===
J1 marginal coverage: IN1 0.9000 | IN4 0.9000 | NA1 0.8999 | NA4 0.9000 | G1 0.8974 | NA4G 0.8933 | U 0.8998 | RMS 0.9000 | CA16 0.9000 | CA1oc 0.9001
J2 coverage by CNR class (mean over units):
IN1 0.897 0.897 0.900 0.917 0.962 1.000 | mean max class deviation 0.066
NA1 0.939 0.850 0.840 0.863 0.882 0.945 | mean max class deviation 0.067
NA4 0.917 0.864 0.881 0.912 0.946 1.000 | mean max class deviation 0.055
U 0.972 0.896 0.824 0.678 0.305 0.017 | mean max class deviation 0.629
CA16 0.893 0.895 0.904 0.942 0.985 1.000 | mean max class deviation 0.086
IN1 exact law 0.896 0.900 0.905 0.914 0.932 0.966 | MAE per unit x class 0.0104

=== 1-alpha = 0.99 ===
J1 marginal coverage: IN1 0.9900 | IN4 0.9900 | NA1 0.9900 | NA4 0.9901 | G1 0.9897 | NA4G 0.9846 | U 0.9899 | RMS 0.9900 | CA16 0.9900 | CA1oc 0.9900

```

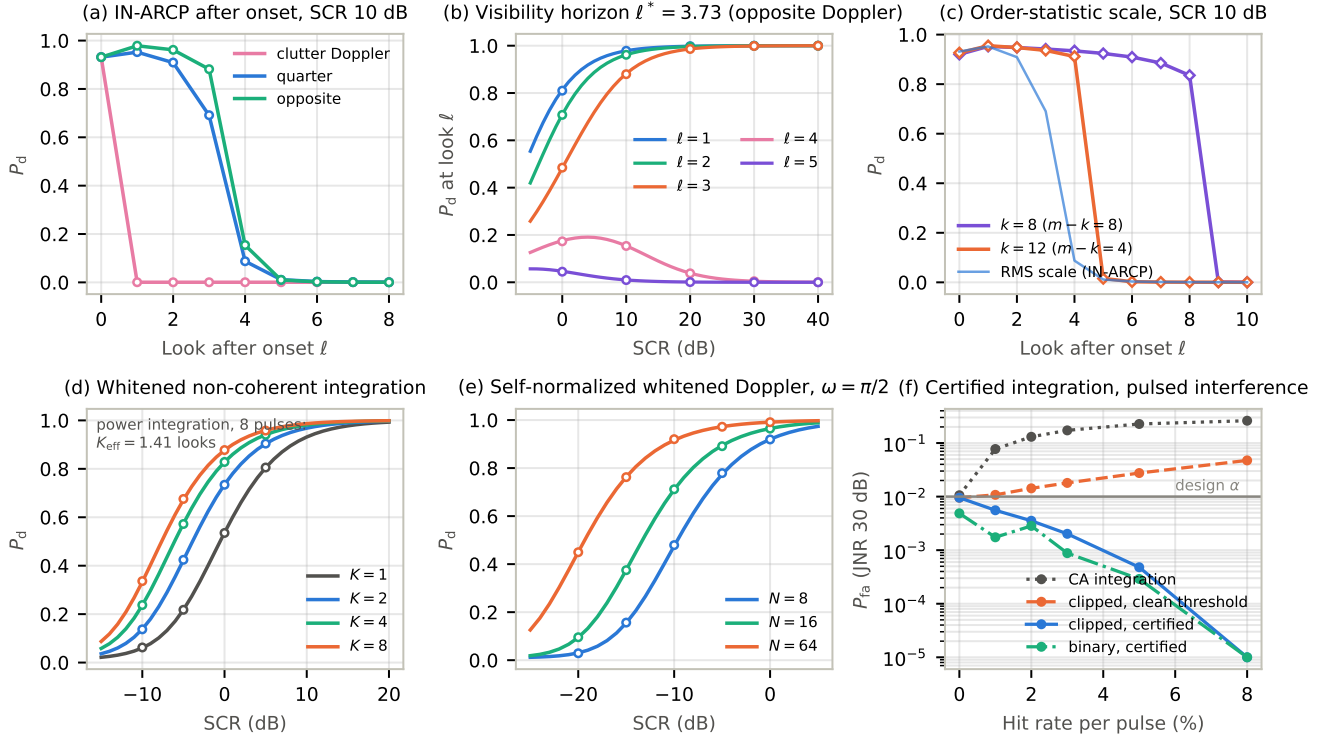


Figure 6: Closed forms (lines) against Monte Carlo (markers;  $10^5$  episodes per point in (a), (c) and (f),  $5 \times 10^4$  in (b) and (d),  $10^4$  to  $2.5 \times 10^4$  in (e)) at the true coefficient, AR(1) compound-Gaussian clutter with  $\rho = 0.93e^{-0.2j}$ , lognormal texture,  $m = 16$ ,  $P_{fa} = 0.01$ . Panels (a)–(c) and (f) are those of Fig. 2 of the paper, where (f) appears as its panel (d) and (c) here adds the RMS-scale law for reference. The two panels shown only here are (d), the integration law (9), and (e), P-ANMF with unknown Doppler. JNR in (f): interference-to-clutter-plus-noise power ratio.

J2 coverage by CNR class (mean over units):

|       | IN1   | NA1   | NA4   | U     | CA16  | IN1 exact law | mean max class deviation    |
|-------|-------|-------|-------|-------|-------|---------------|-----------------------------|
| 0.989 | 0.989 | 0.990 | 0.995 | 0.999 | 1.000 | 0.999         | 0.009                       |
| 0.996 | 0.983 | 0.979 | 0.984 | 0.989 | 1.000 | 0.989         | 0.013                       |
| 0.993 | 0.983 | 0.987 | 0.993 | 0.998 | 1.000 | 0.998         | 0.009                       |
| 1.000 | 0.997 | 0.991 | 0.995 | 0.734 | 0.245 | 0.995         | 0.305                       |
| 0.989 | 0.989 | 0.992 | 0.998 | 1.000 | 1.000 | 0.998         | 0.010                       |
| 0.989 | 0.990 | 0.991 | 0.993 | 0.995 | 0.998 | 0.998         | MAE per unit x class 0.0019 |

J3 detection (Swirling 1, Pfa 0.01): SCR gain vs CAlloc at Pd=0.5 per CNR class, mean [95% CI]; closed form G\_IN

| G_IN(c) dB | -0.79                | -0.25                | 0.28                 | 0.77                 | 1.11                 | 1.92                 |
|------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| IN1        | -1.45 [-1.79, -1.13] | -1.00 [-1.19, -0.83] | 0.02 [-0.23, 0.23]   | 1.57 [0.89, 2.20]    | 3.11 [1.97, 4.26]    | 5.78 [5.78, 5.78]    |
| IN4        | -1.42 [-1.57, -1.28] | -0.85 [-0.99, -0.72] | 0.52 [0.03, 0.96]    | 3.19 [2.27, 3.98]    | 6.78 [5.08, 8.34]    | 12.33 [12.33, 12.33] |
| NA1        | -2.10 [-2.65, -1.61] | 0.65 [0.31, 0.93]    | 1.99 [1.06, 2.70]    | 4.21 [2.71, 5.51]    | 7.81 [5.26, 10.08]   | 13.07 [13.07, 13.07] |
| NA4        | -1.26 [-1.38, -1.14] | 0.35 [0.08, 0.64]    | 1.33 [0.78, 1.88]    | 3.61 [2.70, 4.44]    | 7.03 [5.50, 8.59]    | 12.78 [12.78, 12.78] |
| U          | -5.48 [-6.02, -5.02] | -2.19 [-2.65, -1.80] | 0.22 [-0.21, 0.61]   | 4.11 [3.54, 4.65]    | 11.44 [10.93, 11.95] | 13.07 [13.07, 13.07] |
| CA16       | -0.62 [-0.64, -0.59] | -0.49 [-0.55, -0.44] | -0.47 [-0.50, -0.46] | -0.54 [-0.57, -0.51] | -0.61 [-0.67, -0.54] | -0.85 [-0.85, -0.85] |

J4 interference scenario C (Y-chirp hit rate 0.020; hit episodes 1920, clean 53520)

|      | coverage 0.90: | clean 0.901 hit 0.792 | 0.99: clean 0.990 hit 0.979 |
|------|----------------|-----------------------|-----------------------------|
| IN1  | coverage 0.90: | clean 0.828 hit 0.584 | 0.99: clean 0.964 hit 0.854 |
| NA1  | coverage 0.90: | clean 0.870 hit 0.644 | 0.99: clean 0.980 hit 0.926 |
| NA4  | coverage 0.90: | clean 0.761 hit 0.449 | 0.99: clean 0.950 hit 0.785 |
| U    | coverage 0.90: | clean 0.900 hit 0.719 | 0.99: clean 0.989 hit 0.965 |
| CA16 | coverage 0.90: | clean 0.900 hit 0.719 | 0.99: clean 0.989 hit 0.965 |

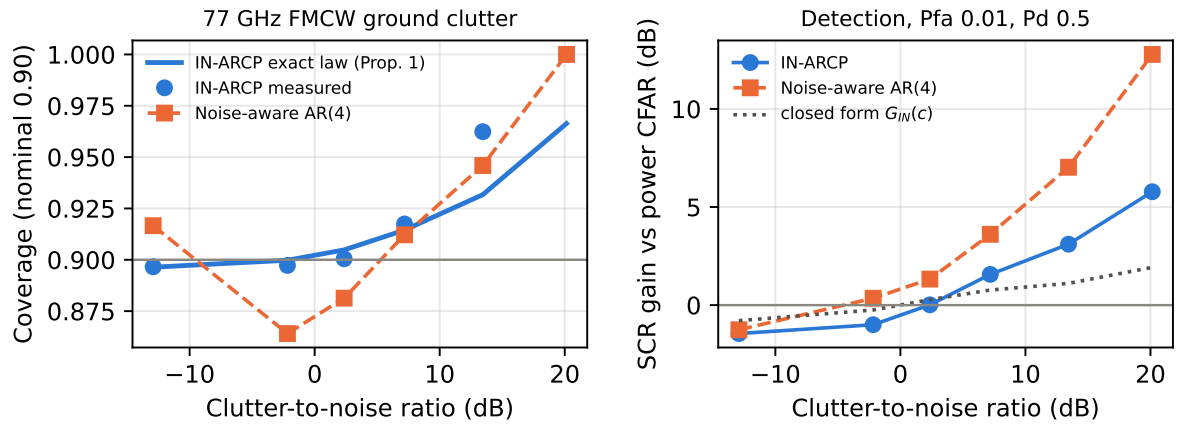


Figure 7: 77 GHz FMCW ground clutter: coverage by CNR class with the exact law, and SCR gain over the local power detector.